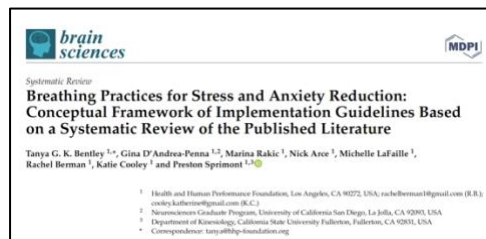


Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Breathing Practices for Stress and Anxiety Reduction: Conceptual Framework of Implementation Guidelines Based on a Systematic Review of the Published Literature

411 Released: June 2, 2024



Quick Summary

This comprehensive analysis examined 58 studies with 72 different breathing interventions to discover that an effective breathing practice for reducing stress and anxiety includes: slow-only or a mix of slow and fast breathing; a duration of at least 5 minutes; at least one session of human guidance; and at least six sessions spread out over one week.

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Article Citation

Bentley TKG, D'Andrea-Penna G, Rakic M, Arce N, LaFaille M, Berman R, Cooley K, Sprimont P. Breathing Practices for Stress and Anxiety Reduction: Conceptual Framework of Implementation Guidelines Based on a Systematic Review of the Published Literature. Brain Sci. 2023 Nov 21;13(12):1612. doi: 10.3390/brainsci13121612. PMID: 38137060; PMCID: PMC10741869.

Reading Time: 3 min 44 seconds

I hope the next 56'ish breaths are the most nourishing of your day.



4 Fundamentals

1. Essential Background Material

As you are likely aware, chronic stress and anxiety are a global problem. They adversely affect physical health, for example, increasing our risk of heart disease, and negatively impact mental health, increasing our risk of mental illness. Put simply, chronic stress and anxiety wear down our bodies and minds, reducing resilience, health, and hope.

Breathing practices are an attractive option for treating stress and anxiety because, as the authors of this paper state, they **“have the advantage of being universally accessible, scalable, and cost-free.”** Moreover, millennia of practice and decades of scientific research have shown that breathing exercises are an effective antidote to stress and anxiety.

Surprisingly, no study has systematically examined what characteristics make a breath practice effective. Why do some studies show considerable benefits while others find none? Why do so many disparate techniques seem to work? Do they share any commonalities? That’s precisely what this paper answers: it provides a practical framework to help breath practitioners, coaches, and program developers create effective protocols.

2. What Did this Research Do?

The authors found clinical trials that examined the impact of breathing exercises on stress and anxiety (which were assessed via questionnaires, inventories, observations, or self-reports). It was especially unique because they included any type of controlled breathing (slow breathing, fast breathing, alternate nostril breathing, etc.) as long as they were used in isolation (for example, it couldn’t be slow breathing *and* meditation).

Moreover, this study included people with both clinically diagnosed and subclinical levels of stress or anxiety. They included healthy adults, youth, and people with chronic health conditions, making the results applicable to the broader population.

Overall, they ended up 58 studies using 72 different breathing interventions with 5407 participants. The studies included slow breathing, deep diaphragmatic breathing, fast breathing, alternate and uninostil breathing, breathing with holds/pauses, extended exhale, paced or pursed-lips breathing, and normal rate breathing.



3 & 4. What Were the Major Findings and Why Do They Matter?

This was an incredibly comprehensive study with TONS of neat results. However, they consolidated everything into a simple framework involving five items, which I'll focus on below. But definitely check out the paper for all the goodness.

Breathing improved stress and anxiety symptoms in 54 of the 72 interventions.

That's 75% of the breathing interventions examined. The rest of this section focuses on what characteristics separated effective (n=54) interventions from ineffective (n=18).

Item 1: Pace of Breathing – Slow or a combination of slow and fast is best.

Interventions using slow-only, or a mix of slow and fast breathing, were most effective. Fast-only breathing interventions were not effective. (Caveat: Only two fast-only breathing studies were included, so perhaps more studies with fast breathing need to be examined.) Crucially, among slow breathing interventions, the specific pace, pauses, and ratio were not significant. *What mattered was that the breath was slowed, not the precise pace.*

Item 2: Session Duration – At least five minutes or longer is the most critical.

Effective breathing interventions were generally five minutes or longer. Surprisingly, there were no statistically significant improvements beyond five minutes, meaning a 20-minute session was no more effective than a 5-minute session for stress and anxiety. This doesn't mean longer is worse; it just means it isn't vital for reducing stress and anxiety.

Item 3: Human Guidance – At least one session of human guidance is best.

Effective interventions generally included at least one session of human guidance, defined as prerecorded audio or video instruction for at least the first session. Notably, it didn't matter if the practices were done individually, within a group, or both; this finding provides lots of flexibility for coaches designing a program.

Item 4: Multiple Sessions – Do the breathing practice more than once.

There are no surprises here: Effective interventions required people to perform the breathing practice more than once. Consistency is crucial to teaching our bodies and minds to relax.

Item 5: Long-Term Practice – At least six sessions over at least one week.



Again, no surprises here. Like with multiple sessions, continued practice over time leads to sustained benefits for reducing stress and anxiety.

The Five Factors Altogether

An effective breath practice for reducing stress and anxiety includes:

- Slow-only or a mix of slow and fast breathing.
- At least 5 minutes of practice.
- At least one session of human guidance.
- Multiple sessions; and
- Long-term practice of at least six sessions over one week

Limitations: Low Study Quality

There were several limitations, but study quality is one of the biggest. Of the 58 studies, 7 had “good” quality, 29 had “fair” quality, and 22 had “poor” quality. This has always been a problem with breathing research. Hopefully, papers like this one will pave the way for more rigorous trials in the future.

1 Big Takeaway

This comprehensive analysis examined 58 studies and 72 different breathing interventions to discover that an effective breathing practice for reducing stress and anxiety includes: slow-only or a mix of slow and fast breathing, a duration of at least 5 minutes, at least one session of human guidance; and at least six sessions spread out over one week.

1 Practical Application

If you're a coach:

This is an open-access paper, so go print out Figure 6 or Table 4 and pin it to your wall or place it next to your computer. Use it as guidance anytime you're helping a client, designing a class, or building a program.

If you're a practitioner:

Use and adapt these results to your situation to ensure you're balancing your unique needs with the latest science to obtain maximum effectiveness for reducing stress and anxiety.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Acute Fall and Long-Term Rise in Oxygen Saturation in Response to Meditation

411 Released: May 26, 2023



[Get the Paper Here](#)

Article Citation

Bernardi NF, Bordino M, Bianchi L, Bernardi L. Acute fall and long-term rise in oxygen saturation in response to meditation. *Psychophysiology*. 2017 Dec;54(12):1951-1966. doi: 10.1111/psyp.12972. Epub 2017 Aug 25. PMID: 28840941.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2017 study in *Psychophysiology* found that mantra-based mindfulness reduces metabolism independently of breathing rate, whereas slow breathing increases heart rate variability independently of meditation. Combining them into one practice may provide the best of both techniques.



4 Fundamentals

1. Essential Background Material

- Some common physiological effects of meditation are a reduced heart rate, decreased breathing rate, reduced blood pressure (all of which comprise a “hypometabolic state”), and improved heart rate variability in some cases.
- A confounding factor for understanding these benefits is that meditation naturally slows down breathing rate. Hundreds of studies have found that slow breathing by itself can provide many of these same benefits: it lowers blood pressure, improves HRV, and decreases breathing rate.
- Thus, it’s unclear if meditation’s physiological benefits are primarily due to slower breathing or if there are other changes elicited by meditation independent of breathing.
- This study aimed to isolate meditation’s unique, acute physiological effects by controlling for breathing rate.
- Additionally, it examined if long-term meditation practice elicits favorable cardiorespiratory changes.
- Nick’s Note: Although unintentional and not explicitly stated, they uncover a significant benefit of combining mindfulness with slow breathing to amplify both of their benefits.

2. What Did this Research Do?

There were 80 participants in this study. Forty-one participants had prior meditation experience (“meditators”), and 39 did not (“controls”).

The participants performed seven different protocols while lying on their backs with their eyes closed:

1. Initial Baseline [4 min]
 2. Paced Breathing at 15 Breaths a Minute (equal inhale:exhale) [5 min]
 3. Paced Breathing at 6 Breaths a Minute (equal inhale:exhale) [5 min]
 4. Mantra Meditation while Breathing at 15 Breaths a Minute [5 min]
-



5. Mantra Meditation while Breathing at 6 Breaths a Minute [5 min]
6. Body Scan Meditation (observing right hand & index finger specifically) [5 min]
7. Final Baseline [4 min]

The paced breathing exercise (#2 and #3) were randomized. The meditation exercises (#4, #5, #6) were also randomized.

The mantra involved silently focusing on and repeating a single word that was meaningful to them. This is almost identical to Herbert Benson's suggestion for eliciting the relaxation response ([see Book 411 here](#)).

3 & 4. What Were the Major Findings and Why Do They Matter?

The addition of the mantra-based mindfulness reduced breathing volume and heart rate independent of breathing rate (this is the most important finding from the study, in my opinion).

- The addition of meditation reduced oxygen saturation, tidal volume, minute ventilation, and an indirect measurement of chemosensitivity (urge to breathe) compared to breathing alone. *For example, breathing at 6 breaths/min with meditation significantly reduced oxygen saturation compared to 6 breaths/min without meditation.*
- These results are significant for two reasons. First, they uniquely show that the hypometabolic effects of meditation occur independently of breathing rate.
- Second (this is my personal addition, not mentioned in the paper), the only contraindication of slow breathing is that people tend to hyperventilate when learning it (the slow breathing rate is challenging, so they overcompensate by breathing more).

These results show that, in both experienced meditators and new practitioners, adding a mantra to the slow breathing practice reduces the tendency to overbreathe. Thus, this could be a significant complement to a slow breathing practice.

Slow breathing increased heart rate variability independent of meditation.

- Perhaps unsurprising given all the research on slow breathing and HRV, but it's still important to note that slow breathing provided benefits independent of meditation.
-



Long-term meditation practice may produce (in the authors' words) a "rather optimal cardiorespiratory function."

- At baseline, the meditators generally had lower blood pressure, a lower breathing rate, and greater brain and tissue oxygenation than the controls. As stated above, the practice of meditation acutely dropped oxygenation. This indicates that, over time, the acute drop in oxygenation during meditation trains the body to run more efficiently. This would explain why meditators generally had a lower breathing rate and blood pressure but still had greater oxygenation.

Altogether (this is my interpretation), we could say: Mindfulness amplifies the benefits of slow breathing. Slow breathing amplifies the benefits of mindfulness.

- In my opinion, the two practices (slow breathing and mantra-based mindfulness) complement each other perfectly. Combining the two into one practice (like they did here) seems like an excellent way to get slow breathing's HRV and nervous system benefits and meditation's metabolism benefits.

1 Big Takeaway

Mantra-based mindfulness reduces metabolism independently of breathing rate, whereas slow breathing increases heart rate variability independently of meditation. Combining them into one practice may provide the best of both techniques.

1 Practical Application

Choose a meaningful focus word or phrase (see [Thought # 2 in this Book 411](#) for more instructions). Then, place your attention on your focus word/phrase for the duration of your slow breathing practice to naturally reduce breathing volume and improve chemosensitivity. If you're an instructor, this approach may be helpful with clients learning slow breathing.



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Citation

Bernardi L, Sleight P, Bandinelli G, Cencetti S, Fattorini L, Wdowczyk-Szulc J, Lagi A. Effect of rosary prayer and yoga mantras on autonomic cardiovascular rhythms: comparative study. BMJ. 2001 Dec 22-29;323(7327):1446-9. doi: 10.1136/bmj.323.7327.1446. PMID: 11751348; PMCID: PMC61046.

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November 18, 2022

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Effect of rosary prayer and yoga mantras on autonomic cardiovascular rhythms: comparative study

4 FUNDAMENTALS

1. Essential Background Material

This study analyzed how breathing and cardio-autonomic function changed during normal breathing, slow breathing, talking, and prayer or mantra recitation.

Slow breathing at around 5 to 6 breaths a minute has a wide range of benefits for the cardiovascular and autonomic nervous systems. (See 411s on [chronic disease](#) and [general benefits](#)).

Briefly, it synchronizes rhythms of the respiratory and cardiovascular systems. This lowers blood pressure, increases oxygenation, improves baroreflex sensitivity, increases heart rate variability, and induces a state of calm and a feeling of well-being.

Interestingly, the authors unintentionally discovered that the recitation of the Ave Maria (Hail Mary) prayer led to a breathing rate of nearly 6 breaths/min in a separate experiment, suggesting that this religious practice may provide some of these same health benefits.

Thus, this study aimed to analyze in detail how breathing rate and cardio-autonomic function changes under recitation of the prayer (and several other conditions), which we'll describe next.



2. What Did this Research Do?

The study had 23 subjects, 16 men and 7 women, with an average age of 34.

Here was the protocol the participants followed:

- Baseline measurements of spontaneous breathing (3 min)
- Free talking (6 min)
- Slow breathing at 6 breaths/min (6 min)
- Recitation of the Ave Maria (Hail Mary) in Latin (part of the rosary prayer) (6 min)
- Repetition of yoga mantra (om-mani-padme-om) (6 min)

Critically, they were not given instructions on how fast/slow to recite the prayer or yoga mantra. Moreover, for the prayer, the participants worked in groups of two: one spoke the priest's part, and the other the response from the congregation.

The measurements recorded were breathing rate, heart rate via electrocardiogram, blood pressure, and "[midcerebral arterial flow velocity](#)," which supplies specific brain tissue with oxygenated blood and is the brain's most commonly affected blood vessel.

3. What Were the Major Findings?

Recitation of the Ave Maria (Hail Mary) prayer and yoga mantra both slowed breathing rate to approximately 5.5 to 6 breaths a minute.

Moreover, breathing and heart measurements synchronized during prayer and mantra, indicating coherence between these systems. Simultaneously, both led to rhythmic fluctuations in brain blood flow. Baroreflex sensitivity (an overall marker of cardio-autonomic health) also increased during prayer and mantra.

Interestingly, heart rate variability increased (a good thing) while respiratory variability decreased.

For example, during the 3 minutes of normal breathing, participants' breathing variability was about 22%. During slow controlled breathing using a pacer, it was only 5.4% (less variability makes sense because they were using a timer).

Remarkably, it was 8.3% during the Hail Mary prayer and 6.2% during the mantra. **Thus, without instruction, these methods regulated breathing to almost the same extent as controlled breathing using a pacer.**



4. Why Do These Results Matter?

These results show that reciting a common religious prayer or yoga mantra leads to cardio-autonomic-respiratory coherence. Moreover, we learn that these methods stabilize breathing rates to almost the same extent as using a controlled breathing timer.

Thus, this study shows that these spiritual methods are also healthy for our hearts and nervous systems.

The authors hypothesize that these methods may have developed to help people breathe slowly when stopwatches and metronomes were unavailable.

They also hypothesize that the rosary, which involves many repetitions of the Ave Maria, might have evolved to induce a state of calm and a feeling of well-being, which may have made people more receptive to the message.

1 BIG TAKEAWAY

Recitation of the Ave Maria (Hail Mary) prayer or a common yoga mantra slows breathing rate to 5.5-6 breaths per minute and improves markers of cardiovascular and autonomic health. They can be thought of as both health and religious practices.

1 PRACTICAL APPLICATION

If you don't like using a breathing app, try finding a mantra or prayer that resonates with you (it can be anything) that will slow your breathing to around 5-6 breaths/min. This may be a more fun and engaging way to perform the practice.

Either way, this study is another reminder of the power of slow breathing, regardless of how it's practiced. So make sure you practice a little more today



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Citation

Bernardi L, Gordin D, Bordino M, Rosengård-Bärlund M, Sandelin A, Forsblom C, Groop PH. Oxygen-induced impairment in arterial function is corrected by slow breathing in patients with type 1 diabetes. Sci Rep. 2017 Jul 20;7(1):6001. doi: 10.1038/s41598-017-04947-4. PMID: 28729675; PMCID: PMC5519543.

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April 25, 2022

Additional Note

This is one of the first papers I read that made me genuinely appreciate just how powerful all this “breathing stuff” is. In fact, this paper was a critical catalyst for creating “The Breathing Diabetic.”

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Oxygen-induced impairment in arterial function is corrected by slow breathing in patients with type 1 diabetes

4 FUNDAMENTALS

1. Essential Background Material

This study evaluated the difference between slow breathing and oxygen administration on autonomic and arterial function in type-1 diabetics.

People with diabetes suffer from higher rates of cardiovascular disease and nervous system issues. Many of the problems are thought to be permanent damage. However, the author and colleagues (they have several papers we’ll eventually get to) have found that some of these issues are partially reversible in diabetes.

For example, they found that the parasympathetic nervous system could be stimulated using either slow breathing or oxygen administration. This led to a transient improvement in autonomic function in people with diabetes, **and a temporary reversal of complications.**

Although slow breathing and oxygen administration both help, it’s unclear how they work. For instance, as mentioned, both lead to improvements in autonomic function. However, oxygen administration generally increased blood pressure, whereas slow breathing naturally lowered it. This suggests they’re working through different mechanisms that could either be complementary or offsetting.

The goal of this paper was to find out.



2. What Did this Research Do?

They recruited 91 type-1 diabetics and 40 age-matched controls, none of which showed evidence of cardiovascular disease. They had the participants lay down, and measurements were recorded during 5 minutes of spontaneous breathing followed by 2 minutes of slow breathing at 6 breaths/min (5-sec inhale, 5-sec exhale). Next, they had them repeat this breathing procedure with additional inhaled oxygen (“hyperoxia”).

(Note: Inhaled oxygen is potentially valuable for diabetes because many complications are thought to arise from resting tissue hypoxia. Thus, the idea is that the hypoxia will be alleviated by supplementing with additional oxygen, and complications will be reduced.)

So, they ended up with four conditions:

1. Baseline
2. Slow Breathing
3. Oxygen Administration
4. Slow Breathing + Oxygen Administration

Measurements they took:

- Blood pressure
- Oxygen saturation
- Baroreflex sensitivity (BRS), which generally reflects your body’s ability to quickly adjust your blood pressure to match the current circumstances (higher is better)
- Heart rate variability (HRV)
- Arterial stiffness

3. What Were the Major Findings?

At baseline, the type-1 diabetics basically had worse measurements of everything compared to the controls: They had lower BRS and HRV, higher blood pressure, lower oxygen saturation, and more arterial stiffness.

During oxygen administration, HRV and BRS increased. However, blood pressure and arterial stiffness also increased (likely due to more oxidative stress).

During slow breathing, on the other hand, HRV and BRS increased while diastolic blood pressure and arterial stiffness decreased.



During slow breathing + oxygen, the increases in blood pressure and arterial stiffness were counteracted by the slow breathing.

4. Why Do These Results Matter?

There are several significant takeaways from these findings. **First, slow breathing appears to be more protective for the autonomic nervous system and arterial function than oxygen administration in people with diabetes.**

Second, slow breathing might also act as an antioxidant. Recall from above that oxygen administration increased arterial stiffness and blood pressure, likely due to excess oxidative stress. When oxygen administration was paired with slow breathing, those effects didn't occur. **This suggests that slow breathing acts as a natural antioxidant.**

How might this be happening? The authors hypothesize that it's through vagal nerve stimulation. Several experiments have shown that vagal stimulation produces antioxidant effects. However, this paper is the first to show it occurs using slow breathing alone in people with diabetes. **This is especially significant because oxidative stress is a central driver of diabetic complications.**

Lastly, another important finding was that the parasympathetic nervous system is not destroyed, just suppressed, in people with diabetes. A simple slow breathing exercise “wakes it up” and acutely reverses components of diabetic autonomic dysfunction.

1 BIG TAKEAWAY

Slow breathing improves autonomic and arterial function while also acting as an antioxidant in people with diabetes. The last line of their abstract summarizes it perfectly: **“Slow breathing could be a simple beneficial intervention in diabetes.”**

1 PRACTICAL APPLICATION

If there is any disappointment at all in this paper, it's this: I emailed the authors and discovered that they performed the breathing protocol through the mouth. It's unclear if it was for measurement purposes or not.



However, the fact that it was through the mouth might actually be even more powerful. **This suggests the slow breathing pace is most important for autonomic and arterial health**, independent of all the other adverse effects of mouth breathing.

Thus, we can improve it further by doing it through the nose. So, the 1 Application is to simply practice slow breathing at around 6 breaths per minute to improve autonomic function and arterial function and reduce oxidative stress.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Yoga Breathing, Meditation, and Longevity

411 Released: September 1, 2023

LONGEVITY, REGENERATION, AND OPTIMAL HEALTH

Yoga Breathing, Meditation, and Longevity

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Article Citation

Brown RP, Gerbarg PL. Yoga breathing, meditation, and longevity. Ann N Y Acad Sci. 2009 Aug;1172:54-62. doi: 10.1111/j.1749-6632.2009.04394.x. PMID: 19735239.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

Yogic breathing and meditation are complementary practices that may enhance longevity by improving nervous system health and stress resiliency, boosting emotional health, enhancing brain health, and reducing suffering associated with disasters.



4 Fundamentals

1. Essential Background Material

- Tibetan Buddhist meditation and yogic breathing techniques (pranayama) have been used for thousands of years to combat the effects of stress and aging, such as mental decline, loss of energy, and loss of independence.
- Moreover, both meditation and yogic breathing have a common goal of understanding the mind's nature and developing compassion, awareness, and perhaps even enlightenment.
- Buddhist meditation and yogic breathing also agree that breathing and emotions are reciprocal: emotional changes affect breathing, and changes in breathing affect emotions.
- Across many schools of meditation and yoga, breathing exercises are used to clear distractions and reduce negative emotions, thus preparing practitioners for reflection and allowing them to reach deeper meditative states.
- Yogic breathing and meditation are therefore very complementary and share common goals. They can each be used independently or combined into one practice, with breathing most commonly serving as preparation for meditation.
- With this in mind, this paper examined the similarities and differences between these two practices and reviewed scientific evidence for how their benefits may be related to stress reduction and longevity.

2. What Did this Research Do?

This article was a **narrative review** of research supporting the use of yogic breathing and meditation for stress reduction and longevity. There were tons of neat studies cited, but when they got deeper into breathing, they relied heavily on research performed with the Sudarshan Kriya Yoga (SKY) program. However, the underlying mechanisms they discuss are relevant to virtually all breathing techniques.

(But for your reference, the SKY program has four breathing methods: slow resistance breathing (ujjayi), bellows breath (fast breathing), om chanting, and cyclical breathing.)



3 & 4. What Were the Major Findings and Why Do They Matter?

In general, slow breathing exercises and meditation activate the parasympathetic nervous system and induce altered states of consciousness.

- These big-picture neurophysiological effects positively impact stress and longevity by improving the balance of the autonomic nervous system, which enhances stress resilience, a critical factor in many age-related diseases (cardiovascular disease being a prominent one).
- Moreover, by enhancing quality of life, these practices get at a primary goal of all ancient contemplative practices: reducing suffering. Brown and Gerbarg summarize it perfectly in their abstract: **“By inducing stress resilience, breath work enables us to rapidly and compassionately relieve many forms of suffering.”**

For people who find meditation difficult, breathing exercises can provide almost immediate benefits while laying the foundation for meditation practice.

- This was a short remark from Brown and Gerbarg based on their clinical experience. However, it’s a critical finding, one that many breathing practitioners and coaches experience in their own lives, so it’s worth emphasizing here.

Breathing and emotions are reciprocal.

- We mentioned that this reciprocal relationship is a core tenant of yogic breathing and Buddhist meditation in Section 1. It’s also backed by science. Brown and Gerbarg cite a paper [we recently covered](#) as evidence of this relationship.
- Healthy emotional responses are critical to longevity, so this research supports using breathing exercises and meditation to improve this aspect of aging.

The combination of slow and fast breathing exercises may provide a “neurophysiological workout,” which would improve the flexibility and resiliency of the nervous system.

- Interleaving calming, slow breathing methods with stress-inducing, fast breathing methods may give the nervous system a workout, building its adaptability and resiliency while enhancing longevity.

Breathing and meditation positively influence the brain in ways that may enhance longevity.



- Fast and cyclical breathing techniques disrupt typical brain wave patterns, reorganizing them into a meditative-like state upon completion. These approaches also seem to improve overall integrative brain functioning and improve brain wave coherence.
- At the same time, meditation itself can increase the physical thickness of specific brain tissue. Together, these changes may have an anti-aging effect on the brain.

SKY breathing, in particular, helps relieve the suffering associated with natural and man-made disasters.

- They have implemented SKY breathing in both civilian and military populations following disasters and have found it extremely helpful in reducing depression and PTSD.

1 Big Takeaway

Yogic breathing and meditation are complementary practices that may enhance longevity by improving nervous system health and stress resiliency, boosting emotional health, enhancing brain health, and reducing suffering associated with disasters.

1 Practical Application

You could sign up for a SKY class (on my bucket list) to experience the exact method on which many of these results are based. However, a slightly modified version of a breathing protocol [we covered previously](#) also uses many similar mechanisms:

1. Ujjayi breathing (~4-5 min)
 2. Kapalabhati (10-30 cycles; check with pranayama instructor to assess contraindications before practicing)
 3. Alternate nostril breathing (4-5 min)
 4. Mindful breathing (~4-5 min)
 5. Humming (~1 min)
-

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Citation

Buijze GA, De Jong HMY, Kox M, van de Sande MG, Van Schaardenburg D, Van Vugt RM, Popa CD, Pickkers P, Baeten DLP. An add-on training program involving breathing exercises, cold exposure, and meditation attenuates inflammation and disease activity in axial spondyloarthritis - A proof of concept trial. PLoS One. 2019 Dec 2;14(12):e0225749. doi: 10.1371/journal.pone.0225749. PMID: 31790484; PMCID: PMC6886760.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

An add-on training program involving breathing exercises, cold exposure, and meditation attenuates inflammation and disease activity in axial spondyloarthritis - A proof of concept trial.

4 FUNDAMENTALS

1. Essential Background Material

[The landmark PNAS study](#) of Wim Hof's method showed that it could reduce inflammation associated with an endotoxin injection. Although groundbreaking, it was unclear if long-term practice could reduce chronic inflammation, which would be valuable in autoimmune diseases.

This study aimed to fill that gap. They implemented Wim's method (WHM) in a group of people with axial spondyloarthritis (axSpA), a type of arthritis involving chronic inflammation of the spine and the joints at the bottom of the spine.

This disease is a good first test case because people with it have an altered immune response, are generally younger, have fewer comorbidities, and the condition can remain somewhat stable for many years.

Let's see what they did.

2. What Did this Research Do?

Twenty-four patients with axSpA were selected for this study. They were



randomized into early intervention (n=13) and late intervention groups (n=11). The early group trained in the WHM for 8 weeks. The late group started WHM training when the early group finished. Thus, the 8-week waiting period for the late group served as the control period to compare the intervention against.

The WHM Training

The WHM group trained with Wim and four instructors for 8 weeks. The core training included the 3 pillars of Wim's method: breathing, cold, and meditation. (They also did exercises such as push-ups, yoga poses, horse stance, etc.)

- The breathing consisted of the two exercises described in the 2014 PNAS study. See that [Science 411](#) for the full description.
- Cold exposure was achieved through cold showers at home. On group training days, participants were exposed to ice-cold water for several minutes (up to 5 min max).
- The meditation involved 15-20 minutes of visualization, typically used at the end of the training session.

During the first 4 weeks, there was group training twice a week; it was reduced to once a week for the second four weeks. They were also expected to perform the method at home daily.



Measurements

The study's primary outcome was safety: They simply wanted to test if the WHM was safe in autoimmune patients.

However, they also measured a host of blood and quality-of-life markers, which are probably most interesting to us breathing nerds:

- [C-reactive protein](#) (CRP): Measure of inflammation
- [Erythrocyte sedimentation rate](#) (ESR): Measure of inflammation
- [Calprotectin](#): Measure of inflammation
- ASDAS-CRP and BASDAI: Markers of disease activity
- Quality of Life Questionnaire



They assessed how these measurements changed from baseline to the 8-week mark when the training was completed. (The 24-week assessment shown in the graphic above was to determine longer-term safety.)

3. What Were the Major Findings?

- No serious adverse events occurred during the training period, meaning it “passed” the primary objective of the study (listen to the audio version of this 411 for more on this).
- **ESR significantly declined during the intervention but not during the control period.**
- Calprotectin and CRP both decreased, but the changes did not reach statistical significance.
- **The two global measures of axSpA disease activity, ASDAS-CRP and BASDAI, decreased significantly.**
- Quality of life improved following the intervention.

4. Why Do These Results Matter?

This study shows for the first time that voluntary activation of the immune system using the WHM is not only effective in short-term inflammation, but also in chronic inflammation associated with autoimmune disease.

And it’s more than just inflammation. The WHM also improved quality of life and two overall markers of disease activity in axSpA. These would be significant results for any intervention, much less a free mind-body one.

1 BIG TAKEAWAY

This study showed that the daily practice of Wim Hof’s method can help people with autoimmune disease reduce chronic inflammation, improve markers of disease status, and improve overall quality of life.



1 PRACTICAL APPLICATION

NOTE: To use the method, it's critical that you work with a certified Wim Hof instructor to ensure you perform it safely (as was done in this study).

If you have an autoimmune disease, these results suggest that daily practice of Wim's method is generally safe and might reduce inflammation associated with the overactivation of the innate immune response. Moreover (and maybe more importantly), it may also improve your quality of life with the disease.

If you're a regular WHM practitioner, these results can simply motivate you to continue your practice.

Lastly, remember that all three pillars were used. So, we must assume that all three are needed to get these positive outcomes.



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Citation

Chaddha A, Modaff D, Hooper-Lane C, Feldstein DA. Device and non-device-guided slow breathing to reduce blood pressure: A systematic review and meta-analysis. *Complement Ther Med*. 2019 Aug;45:179-184. doi: 10.1016/j.ctim.2019.03.005. Epub 2019 Mar 15. PMID: 31331557.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Device and non-device-guided slow breathing to reduce blood pressure: A systematic review and meta-analysis

4 FUNDAMENTALS

1. Essential Background Material

This meta-analysis aimed to analyze if slow breathing reduces blood pressure in hypertensive and prehypertensive patients based on randomized controlled trials.

Hypertension is a significant risk factor for cardiovascular disease, which is the leading cause of death in the U.S. For instance, if systolic blood pressure rises from 115 mmHg to 135 mmHg, your risk of cardiovascular disease doubles.

Slow breathing has consistently been shown to reduce blood pressure. In particular, an FDA-approved device called RESPeRATE, which slowly reduces breathing rate down to below ten breaths per minute, has been studied extensively. It consistently lowers blood pressure in hypertensive patients. Due to the ample evidence, the American Heart Association has even given device-guided slow breathing a “class IIA” rating for reducing blood pressure.

However, although many individual studies provide support for slow breathing, meta-analyses and systematic reviews have been inconclusive. For instance, one showed that no positive effects are found when industry-funded studies are excluded. Others have had small sample sizes or excluded non-device guided slow breathing.

Thus, this meta-analysis aimed to fill that gap by providing a concise yet comprehensive summary of studies that have examined slow breathing (device



and non-device guided) and hypertension using strict inclusion criteria and randomized controlled trials (RCTs).

2. What Did this Research Do?

The authors searched several public databases (e.g., Web of Science, MEDLINE) from their inception until mid-2015. They used a combination of search terms like “hypertension OR prehypertension” and “slow breathing OR device-guided breathing” to identify papers relevant to the meta-analysis.

In the identified papers, slow breathing was defined as anything below ten breaths per minute. They included studies of people with both hypertension and prehypertension. The subjects had to perform slow breathing at least three times a week for at least 5 minutes each session. The follow-up period had to be at least 4 weeks, and changes in blood pressure had to be reported.

The authors started with 1,984 studies, but only 17 met their criteria for inclusion in the meta-analysis. Although this was a reasonably strict search, it certainly wasn't perfect, and there are some important caveats.

Of the 17 studies selected, five were abstracts only. Additionally, only two had slow breathing without a device. The other fifteen were device-guided slow breathing using the RESPeRATE, and the maker of the device sponsored six of these.

Thus, there was still a high risk of publication bias in these studies.

(Note that this current meta-analysis was self-funded. So, that gives us some hope that the authors had no conflict of interest with this review.)

3. What Were the Major Findings?

Despite these limitations, the pooled results were impressive. The average decrease in systolic blood pressure (SBP) across all seventeen studies was 5.62 mmHg. The two non-device slow breathing studies had an even more significant drop of 7.69 mmHg. For diastolic blood pressure, the mean decrease was 2.67 mmHg for the device-guided slow breathing.

They also examined how the intensity of the slow breathing practice affected results—the conclusion: the longer subjects practiced, the more significant their reduction in blood pressure. For example, for slow breathing <100 min a week,



the decrease in SBP was 3.01 mmHg, for 100-200 min, it was 6.44 mmHg, and for >200 min, it was 14.00 mmHg.

4. Why Do These Results Matter?

The significance of these findings is that modest reductions in blood pressure lower the chances of strokes, coronary events, heart failure, cardiovascular deaths, and total deaths. This is especially important for diabetics who are at higher risk of developing hypertension and heart disease.

Moreover, the improvements from slow breathing were similar to those seen with antihypertensive medications. Those medications have been shown to improve long-term outcomes in hypertensive and prehypertensive patients. Therefore, slow breathing could potentially provide similar positive results if practiced consistently over a long period.

1 BIG TAKEAWAY

Slow breathing reduces systolic blood pressure by 5.62 mmHg and diastolic blood pressure by 2.67 mmHg. **The more time you practice per week, the greater the blood pressure reductions.** This is a go-to reference for anything slow breathing and hypertension-related.

1 PRACTICAL APPLICATION

Many of the studies cited used a device that gradually guided people's breath to slower rates.

We can mimic this with The Breathing Zone App. This app lets you set a target breathing rate (I suggest 5-6 breaths per minute) and a starting breathing rate (say 7-10 breaths per minute). Then, it will gradually guide your breathing from your starting rate to your target rate so that it feels more manageable and effortless.

Moreover, their "Breathing Zone" setting also starts with a 1:1 ratio and gradually makes your exhales longer until you reach a 1:2 ratio (exhales twice as long as inhales). This was also the approach used by most of the studies cited in this meta-analysis.



You could use this app for ~20 minutes a day to be in the sweet spot of blood pressure reduction found in this meta-analysis.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Exploring the Health Benefits of Bhramari Pranayama (Humming Bee Breathing): A Comprehensive Literature Review

Article Published: March 2024

411 Released: February 7, 2025

Quick Summary

This comprehensive review published in 2024 found that bhramari pranayama (humming bee breathing) is a simple and effective exercise for reducing stress, enhancing cognitive function, improving heart and lung health, and potentially easing symptoms of several chronic diseases—most notably tinnitus and hypertension.

Article Citation

Chetry, D., Chhetri, A., Rajak, D. K., Rathore, V., & Gupta, A. (2024). Exploring the health benefits of bhramari pranayama (humming bee breathing): A comprehensive literature review. *Indian Journal of Physiology and Pharmacology*, 68(1), 71-85.

[Link to Get the Paper](#)



4 Fundamentals

1. Essential Background Material

Bhramari is a widely researched and practiced pranayama exercise. It involves inhaling comfortably through the nose, then exhaling with a high-pitched humming noise, which sounds like a buzzing bee (which is why it's also called "humming bee breath"). The ears are often plugged during practice to amplify its resonating effects.

According to the authors of this paper, ancient yogic texts suggest that bhramari pranayama enhances mental bliss and supports the attainment of Samadhi, the final stage in Patanjali's eight limbs of yoga. Moreover, there are many scientific and anecdotal claims around the benefits of the practice.

However, there are no recent literature reviews synthesizing these effects that researchers and practitioners can draw upon for guidance. Therefore, this study sought to integrate all available literature on bhramari and its effects on health outcomes to provide such a resource.

2. What Did this Research Do?

This was a comprehensive review of peer-reviewed literature on bhramari, not a systematic review or meta-analysis, which are typically more rigorous. The authors searched multiple databases for studies analyzing the isolated effects of bhramari, meaning it could not be combined with other practices. They included studies on both healthy and patient populations, assessing a broad range of psychological and physiological health outcomes.

A total of 46 studies with 3,097 participants were included in the final review. The studies were single-arm clinical trials, clinically controlled trials, and randomized controlled trials. Study durations ranged from less than a week to several months, but none extended beyond a year.

Importantly, this review did not assess the quality of the included studies. In general, breath-focused research tends to be of low to moderate quality, but without a formal quality assessment, it is possible that some included studies were of very low quality.



3 & 4. What Were the Major Findings and Why Do They Matter?

General Characteristics of the Bhramari Technique.

- The most common technique used a 1:3 inhale:exhale ratio, with an inhale of ~5 seconds and an exhale of ~15 seconds, resulting in a breathing rate of 3–6 breaths per minute.
- Most participants covered their ears with their index fingers or thumbs during practice.
- Across all the papers, participants practiced 16 minutes per day, with most studies focusing on short-term effects (<3 months).

Brain and Psychological Effects: Better Attention and Less Stress.

- Bhramari pranayama increased gamma and alpha brain waves and improved brain wave synchrony, particularly in the prefrontal and temporal lobes.
- It improved attention performance and shortening reaction times in both auditory and visual tasks, suggesting it can enhance cognitive function.
- Bhramari reduced symptoms of stress, anxiety, and depression while improving quality of life, sleep, and mental health.

Bhramari Improved Cardiovascular Health.

- During the practice, it caused temporary increases in heart rate and blood pressure.
- After the practice, it consistently led to lowered heart rate and blood pressure, indicating that it elicited the relaxation response.
- It also improved several different measures of heart rate variability (HRV), all of which suggested a shift toward parasympathetic (relaxation) dominance.

Bhramari Improved Lung Health and Lessened Breathlessness.

- It improved several markers of lung function, including forced vital capacity, forced expiratory volume in 1 second, peak expiratory flow rate, and maximal voluntary ventilation.
- It also reduced feelings of breathlessness in people with chronic rhinosinusitis and COPD.

Bhramari Helped Patient Populations.

- The greatest benefits were in people suffering from tinnitus and hypertension.
 - However, benefits were also seen in patients with chronic rhinosinusitis, COPD, insomnia, and anxiety disorders.
-



1 Big Takeaway

Bhramari pranayama (humming bee breathing) is a simple and effective exercise for reducing stress, enhancing cognitive function, improving heart and lung health, and potentially easing symptoms of several chronic diseases—most notably tinnitus and hypertension.

1 Practical Application

The studies reviewed suggest an average practice of ~15 minutes per day for about 10 weeks, although the individual durations varied widely. If that feels like too much, even three humming bee breaths can be a great way to start or end a practice. But for full benefits, consider making it a longer practice—and don't forget to plug your ears for maximum effect.

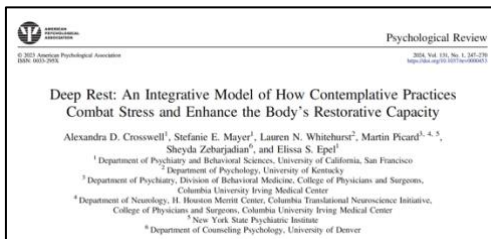
[See this video for a demonstration.](#) Note that don't have to pulsate as this video suggests—simply closing the ears is the most important part.

Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Deep Rest: An Integrative Model of How Contemplative Practices Combat Stress and Enhance the Body's Restorative Capacity

411 Released: May 5, 2024



Quick Summary

This 2024 review/hypothesis defines deep rest as a state of mental & physical safety marked by parasympathetic dominance via slow rhythmic breathing. This state reduces threat arousal, promoting peace, safety, and contentment. Using contemplative practice to spend more time in deep rest may be a powerful way to promote better physical, mental, and cellular health.

Get the Paper Here

Article Citation

Crosswell, A. D., Mayer, S. E., Whitehurst, L. N., Picard, M., Zebbarjadian, S., & Epel, E. S. (2024). Deep rest: An integrative model of how contemplative practices combat stress and enhance the body's restorative capacity. *Psychological Review*, 131(1), 247–270.
<https://doi.org/10.1037/rev0000453>

Reading Time: 2 min 49 seconds

I hope the next 42'ish breaths are the most nourishing of your day.



4 Fundamentals

1. Essential Background Material

“Contemplative practice” is a broad term encompassing most mind-body exercises used to improve well-being, promote flourishing, and enhance our connection with our inner self and outer world. It includes things like meditation, breathing, chanting, mindful movement, and reading. These practices have been used for millennia to improve wellness.

Modern research has shown that these ancient practices have many physiological and psychological benefits. One of the most robust is that they help reduce stress. But they also enhance the functioning of our cardiovascular, immune, and nervous systems. Many studies try to isolate such effects to find an “active ingredient” explaining the benefits. Although informative, this procedure may miss the overarching holistic impact of contemplative practice.

This review/hypothesis paper takes a different approach. It defines a unifying philosophy explaining the benefits of all contemplative practices, termed “deep rest.” This “deep rest” framework is robust because it provides an avenue for understanding how mind-body practices enhance health, even if their specific benefits differ on a case-by-case basis.

2. What Did this Research Do?

Because this is a hypothesis paper/review, there were no strict methods, such as in controlled trials, systematic reviews, and meta-analyses. Instead, this is a literature synthesis to support an overarching theory. (As an aside, these are some of my favorites to read.)

They explored five broad categories:

1. Stress
2. Safety
3. Parasympathetic Dominance
4. Cellular Restoration
5. Sleep

In the next section, I will summarize my favorite findings and hypotheses on deep rest.

3 & 4. What Were the Major Findings and Why Do They Matter?

What is deep rest?

Deep rest is a state of physiological and psychological safety marked by parasympathetic dominance (via slow rhythmic breathing). This state reduces threat arousal, promoting peace, safety, and contentment while enhancing mental, physical, and cellular health.

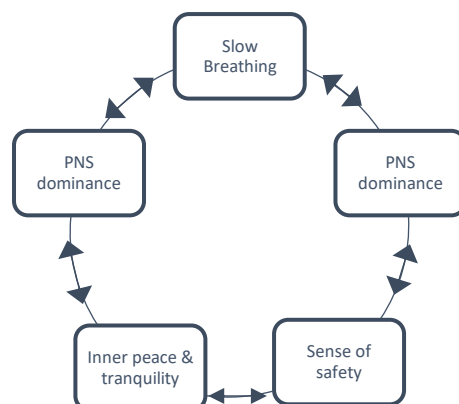
To get into deep rest, we need three main ingredients: physical safety, psychological safety, and parasympathetic dominance.

1. *Physical safety* occurs when our environment doesn't threaten us. Here, it's vital to acknowledge that many people don't live in safe homes, cities, or countries, so they may be unable to achieve this feeling of physical safety on a regular basis.
2. *Psychological safety* occurs when worrisome thoughts and rumination no longer exist.
3. *Parasympathetic (PNS) dominance* occurs via slow or rhythmic breathing, which sends safety signals to the brain via the vagus nerve, allowing it to turn down its threat arousal.

Slow rhythmic breathing is a central component of achieving deep rest.

As stated above, PNS dominance is necessary for deep rest. And (rightfully so), this paper emphasized the role of slowed and/or rhythmic breathing in achieving this PNS dominance. *In short, they hypothesized that slow breathing is the common denominator to all contemplative practices that allows PNS dominance to occur.*

Here's a diagram I made to visualize deep rest and slow breathing. I used double-sided arrows to indicate how, although slow breathing is critical, all these processes are interdependent:





Deep rest may promote cellular restoration and healing.

Excess stress takes energy away from healing processes and directs it toward stress processes, weakening the function of the mitochondria (where cells produce energy). Thus, deep rest may optimize cellular functioning by reducing stress, allowing that energy to be used toward healing. This may improve mitochondrial function and promote cellular restoration.

1 Big Takeaway

Deep rest is a state of mental and physical safety marked by parasympathetic dominance via slow rhythmic breathing. This state reduces threat arousal, promoting peace, safety, and contentment. Using contemplative practices to spend more time in deep rest may be a powerful way to promote better physical, mental, and cellular health.

1 Practical Application

First, prioritize creating a nurturing environment. If you're a coach, go the extra mile to create a safe and comforting space for those you work with. Keep in mind that this sense of safety is essential. Without it, any breathing or meditation practice may not yield its intended benefits.

Then, considering everyone has different preferences and needs, take time to find contemplative practices that put you into a state of deep rest, thus promoting cellular healing and enhancing physical and mental health. If you'd like assistance, you can try out any of the practices [on this page](#) to help you find the right one for you.



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Citation

Dhruva A, Miasowski C, Abrams D, Acree M, Cooper B, Goodman S, Hecht FM. Yoga breathing for cancer chemotherapy-associated symptoms and quality of life: results of a pilot randomized controlled trial. J Altern Complement Med. 2012 May;18(5):473-9. doi: 10.1089/acm.2011.0555. Epub 2012 Apr 23. PMID: 22525009; PMCID: PMC3353818.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Yoga breathing for cancer chemotherapy-associated symptoms and quality of life: results of a pilot randomized controlled trial

4 FUNDAMENTALS

1. Essential Background Material

- Cancer chemotherapy usually has harmful effects, such as fatigue, sleep disturbances, stress and anxiety, and a general worsening of quality of life.
- However, chemotherapy patients have very few options—additional drugs often come with more side effects.
- Pranayama breathing exercises (defined as “*regulation and expansion of breath*”) are appealing because **they are free, relatively easy to perform, and have virtually no adverse side effects.**
- Moreover, physiologically, **some pranayama exercises have been shown to increase parasympathetic tone**, which could be helpful with chemotherapy side effects.
- Previous studies using pranayama in cancer patients were encouraging. However, they used multidimensional yoga approaches, making it hard to know how much pranayama alone contributed to positive outcomes.
- **This study aimed to see if pranayama (without other yoga components) could help cancer patients undergoing chemotherapy.**



Side Note:

Based on the info provided, my quick calculations say that the total duration of the experiment was about 42 weeks.

2. What Did this Research Do?

A total of 16 participants enrolled in the study throughout two cycles of chemotherapy (labeled “Cycle A” and “Cycle B”) over a 1-year period.

The treatment group (n=8) received pranayama training throughout both cycles. The control group (n=8) did not receive pranayama during Cycle A but did receive it during Cycle B.

Pranayama Protocol

Once a week, the participants received a 60-minute group pranayama class. They were also instructed to practice at home twice a day for 10-15 minutes (totaling 20-30 minutes daily).

The protocol used the following exercises:

1. Breath Awareness
2. Ujjayi with slow, deep, extended exhales and slightly closed glottis
3. Kapalabhati (the number of cycles depended on the participant’s health and comfort during this practice)
4. Alternate Nostril Breathing (ANB)

These exercises were chosen because they were easy to learn and had an effective history with cancer patients.

Measurements

Self-assessments of **fatigue, sleep disturbances, stress, anxiety, depression, and overall quality of life** were given at the start of the experiment, after the first cycle of chemotherapy (Cycle A), and at the end of the investigation (after Cycle B). They used established, clinically-accepted questionnaires for each of these self-assessments.

3. What Were the Major Findings?

- **Weekly pranayama class attendance was almost 100%**, indicating that pranayama is feasible for cancer patients undergoing chemotherapy.
- Patients averaged 20-27 daily minutes of at-home pranayama practice, signifying excellent adherence to the protocol.



A Limitation + Nick's Note:

All participants were also using other forms of complementary and alternative medicine. Who wouldn't? And I personally think that makes these results even more powerful because the breathing practice was applied in the patients' real life. However, scientifically, this confounds the results because there's no way of knowing how much those other therapies played a role.

- All of the control group's symptoms got worse during the first round of chemotherapy (when they weren't practicing pranayama). Then, they got better after they began pranayama.
- **The patients who practiced pranayama for both cycles of chemotherapy improved in all symptom and quality of life measurements.**
 - **Caveat:** In this group, fatigue and depression initially worsened from baseline during the first chemo treatment (meaning the pranayama couldn't "overpower" the chemotherapy). Then, it got better during the second treatment with continued pranayama practice.
- Increases in practice time were correlated with decreases in symptoms. **This means the more time people practiced pranayama, the better their chemo symptoms got, and the more their quality of life improved.**

4. Why Do These Results Matter?

This was the first study to show that pranayama (by itself) improves chemotherapy symptoms and enhances cancer patients' quality of life. This is extremely meaningful because the breathing exercises used here are safe and relatively easy to perform, regardless of health status. **This makes them an attractive supplemental treatment option for chemotherapy patients.**

Moreover, the pranayama classes had almost 100% attendance, indicating that this treatment option is feasible and perhaps even enjoyable to the patients (which is the most important, in my opinion).

Lastly, this study is significant because it discovered a dose-response correlation: **The more pranayama practice a person did, the fewer symptoms they felt and the better their quality of life.**

1 BIG TAKEAWAY

A simple pranayama practice (without additional yoga components) was safe and improved stress, anxiety, depression, sleep disturbances, and overall quality of life in cancer patients undergoing chemotherapy, suggesting that pranayama may be a helpful supplemental treatment for these patients.



1 PRACTICAL APPLICATION

Although our personal life stressors seem insignificant compared to what cancer patients endure during chemotherapy, we can still apply this pranayama protocol to help improve our sleep, stress, anxiety, and overall quality of life.

Moreover, the protocol is simple and safe for most people (**Kapalabhati can be dangerous in some cardiovascular conditions or if you are pregnant, so check with a pranayama instructor beforehand...or just skip it and add an extra minute to each of the other practices below**).

Give it a try for a few days and see how it makes you feel:

1. Mindful breathing (~4-5 min)
2. Ujjayi breathing (~4-5 min)
3. Kapalabhati (10-30 cycles; check with pranayama instructor to assess contraindications before practicing)
4. Alternate nostril breathing (4-5 min)

Share the Research

Did any of these findings stand out based on your prior knowledge and experience? If so, grab one, add your personal take on its importance, and share it, so people learn the importance of these findings from your unique perspective.



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Citation

Esch T, Fricchione GL, Stefano GB. The therapeutic use of the relaxation response in stress-related diseases. Med Sci Monit. 2003 Feb;9(2):RA23-34. PMID: 12601303.

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January 5, 2023

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

The therapeutic use of the relaxation response in stress-related diseases

4 FUNDAMENTALS

1. Essential Background Material

- Negative stress can be defined as mental or environmental challenges that trigger the nervous system's sympathetic (fight-or-flight) branch.
- If chronic, this stress can be harmful in immunological, cardiovascular, and neurodegenerative diseases and mental disorders.
- **The relaxation response (RR) is the opposite of the stress response.** It serves as a protective mechanism against excessive stress and might help in the prevention or treatment of stress-related diseases.
- The RR is triggered when someone performs a repetitive mental or physical activity and passively ignores intruding thoughts (like meditation, etc.)
- Primary characteristics of the RR include decreased oxygen consumption (hypometabolism), lower heart rate and blood pressure, and slower breathing rates.
- This paper aimed to look at how the RR helps in many stress-related diseases and see if there are any common mechanisms that explain its benefits in these diseases.



2. What Did this Research Do?

This was a “speculative review,” not a systematic review or meta-analysis. Thus, there were no specific inclusion criteria. Instead, they synthesized peer-reviewed studies on all aspects of the RR and stress-related diseases.

They ended up with over 150 peer-reviewed studies to support their analysis of the RR and stress-related diseases, which they grouped into three categories:

1. Immunological Diseases
2. Cardiovascular Diseases
3. Neurodegenerative Diseases/Mental Disorders

Then, they provided speculation as to why the RR might be helping these chronic conditions to better understand the common mechanisms underlying the benefits of the RR in stress-related diseases.

3. What Were the Major Findings?

- Overall, the RR benefits many diseases (see below) for a few common reasons. In general, **it counteracts the harmful effects of stress**, which play a central role in many diseases.
- More specifically, **the RR reduces the release of stress hormones** such as norepinephrine and cortisol. These hormones can cause damage if they remain elevated in the bloodstream for prolonged periods.
- Related to this lowering of stress hormones, **the RR also reduces inflammation and oxidative stress**, which play a crucial role in many diseases.
- The RR also activates brain areas associated with memory, attention, and motivation, which can benefit neurodegenerative diseases and mental disorders.
- Lastly, the authors speculate that **the RR might result in therapeutic levels of nitric oxide (NO) production**, which is beneficial in many diseases.

For those reasons, this study found that the RR is beneficial in:

Immunological Diseases

- Infectious diseases
- Autoimmune diseases



- Cancer

Cardiovascular Diseases

- Hypertension: shown to be as effective as some drugs
- Endothelial dysfunction
- Atherosclerosis
- Coronary artery disease
- Heart attacks

Neurodegenerative Disease and Mental Disorders

- Alzheimer's disease
- Anxiety and depression: RR might qualify for clinical use most readily here

The authors note that the RR will be most helpful **in the prevention or early stages of all these diseases**. Of course, it will never be harmful to elicit the RR, but to get the maximum benefits, it's best to start using it early.

4. Why Do These Results Matter?

Stress is a significant contributor to many chronic diseases and complications. Moreover, chronic stress prevents our bodies from performing their natural healing processes.

Here, we learn that a simple mind-body approach acts to counter the adverse effects of stress and can help in many disease states.

Thus, the RR may be a therapeutic (and free) tool for combatting stress-related diseases, specifically immunological, cardiovascular, and neurodegenerative diseases and mental disorders. However, the RR's benefits likely extend beyond these categories because stress harms virtually every health condition.

1 BIG TAKEAWAY

Regularly eliciting the relaxation response can counter the adverse effects of stress and serve as a free therapeutic tool in many chronic diseases, especially immunological, cardiovascular, and neurodegenerative/mental disorders.



1 PRACTICAL APPLICATION

The RR is generally elicited when you perform a repetitive mental or physical activity and passively ignore intrusive thoughts. Thus, it can be elicited during breathing, meditation, Tai Chi, yoga, walking, and so on.

Although the paper doesn't give specific instructions, the easiest way I've found to elicit the RR comes from Dr. Herbert Benson's book, [*Timeless Healing*](#):

"To evoke the relaxation response, you need to follow only two basic steps. You need to repeat a word, sound, prayer, phrase, or muscular activity, and when common everyday thoughts intrude on your focus, you need to passively disregard them and return to your repetition.

...

The choice of a focused repetition is up to the individual. ... If you are a religious person, you can choose a prayer; if you are a nonreligious person, choose a secular focus."

So, if you chose the word "peace," here's what you can do:

1. Sit comfortably with a straight back or lie down.
2. Focus on your breath, and say "peace" in your head with each exhale. Extend it to match the length of your exhale, for example, "peaaaacccceeee."
3. When thoughts intrude, passively disregard them: say to yourself, "oh well," and return to your word/phrase.
4. Do this for 10-20 minutes daily.

For more benefits, combine it with your slow breathing practice (that's what I've been doing for about 6 months). For example, my practice currently uses a 6-second inhale, and an 8.6-second exhale. So, I inhale for 6 and then silently draw out my focus phrase in my head for the 8.6-second exhale.

It's one of my favorite ways to perform slow breathing.

Give it a try and enjoy getting all the stress-busting benefits of the relaxation response.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Voluntary Control of Respiration Patterns

A Book Chapter in: *A Clinical Guide to the Treatment of the Human Stress Response*

411 Released: April 14, 2023

[Home](#) > [A Clinical Guide to the Treatment of the Human Stress Response](#) > Chapter

Voluntary Control of Respiration Patterns

[George S. Everly Jr.](#) & [Jeffrey M. Lating](#)

Chapter | [First Online: 06 March 2019](#)

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Article Citation

Everly, G.S., Lating, J.M. (2019). Voluntary Control of Respiration Patterns. In: *A Clinical Guide to the Treatment of the Human Stress Response*. Springer, New York, NY.

https://doi.org/10.1007/978-1-4939-9098-6_15

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A chapter in “A Clinical Guide to the Treatment of the Human Stress Response” found that controlled diaphragmatic breathing is one of the most effective methods for reducing excessive stress. **They concluded:** “Any clinician treating patients who manifest excessive stress syndromes should consider controlled respiration as a potentially suitable intervention for virtually all patients.”



4 Fundamentals

1. Essential Background Material

- Breath control is one of the oldest and most effective methods of reducing stress, having been used for thousands of years to induce a state of calm.
- From a clinical standpoint, they present three very general types of breathing:
 - **Clavicle:** Shallow “vertical” breathing where the shoulders move up and down, and the chest moves slightly.
 - **Thoracic:** Greater expansion of the rib cage followed by vertical movement of the shoulders.
 - **Diaphragmatic:** The deepest form of breathing, relying primarily on the diaphragm, resulting in the greatest blood oxygenation. A full diaphragmatic breath fills the belly, expands the ribcage, and finally, the shoulders move slightly.
- Although all three patterns are essential to understand, diaphragmatic breathing is generally regarded as the most effective for reducing stress, so that’s what this book chapter focuses on.
- In their words: **“Any clinician treating patients who manifest excessive stress syndromes should consider controlled respiration as a potentially suitable intervention for virtually all patients.”** The rest of this will look at why that’s so.

2. What Did this Research Do?

This is a book chapter, so it’s not a formal research study. It’s more of a broad overview of how diaphragmatic breathing can help reduce stress.

It has three main parts: “Mechanisms of Action,” “Clinical Research,” and “How to Implement.” Let’s look at the key points from each of these.



3. What Were the Major Findings?

Mechanisms of Action

- In general, diaphragmatic breathing works by reducing the activity of the sympathetic (fight-or-flight) branch of the nervous system and increasing the action of the parasympathetic (rest-and-digest) branch.
- This is accompanied by **reduced activity in the amygdala**, which helps calm the mind.
- Moreover, **longer exhalations** typically associated with diaphragmatic breathing **play a key role in eliciting the relaxation response** and reducing activity in the amygdala and hippocampus (the latter portion potentially being significant in organizing emotional responses and for learning and memory).
- Additionally, diaphragmatic breathing may activate stretch receptors in the lungs that reduce metabolism and increase parasympathetic dominance.
- Lastly, aside from pure physiological mechanisms, breath control may give people a sense of agency and provide a means of controlling obsessive thought patterns.

That all sounds pretty awesome. Let's now look at what clinical research has shown.

Clinical Research

They describe around 10 general findings on slow diaphragmatic breathing. Some key themes and results include:

- Slow diaphragmatic breathing **can reduce arousal and skin conductance** (meaning less sympathetic activity) in subjects about to receive mild electric shocks.
 - It can reduce both psychological and physiological (as measured by cortisol) stress.
 - It can reduce performance stress and anxiety, such as that associated with test taking.
 - It has been shown to help with panic disorder and PTSD.
 - It can help with reducing asthma symptoms in both children and adults.
 - It can decrease the severity of motion sickness symptoms.
-



4. Why Do These Results Matter?

These findings are significant because they show that **diaphragmatic breathing is versatile**, helping many conditions related to excessive stress. Moreover, **it's simple to learn** and can be applied to a wide range of conditions.

Here, it's important to point out the only potential side effect of deep breathing: **unintentional hyperventilation**. When instructed to "take a deep breath," people often interpret this as taking a big breath and overbreathe. Additionally, they often compensate by taking much bigger breaths if they are breathing slower than usual.

Thus, they suggest that careful attention be taken to ensure people breathe deeply and slowly without hyperventilation. Symptoms of hyperventilation to look out for include dizziness, chest pain, feelings of suffocation, and a racing heart, among many others.

However, this is relatively easy to overcome, making diaphragmatic breathing an excellent first-line treatment for stress. This brings us back to the quote from earlier: **"Any clinician treating patients who manifest excessive stress syndromes should consider controlled respiration as a potentially suitable intervention for virtually all patients."**

1 Big Takeaway

Controlled diaphragmatic breathing is one of the simplest and most effective methods for reducing excessive stress that can be easily taught and safely used in essentially any condition or setting.

1 Practical Application

The authors provide several easy exercises that any clinician can implement. **Most reduce breathing to 6.5-10 breaths a minute**, faster than we usually discuss here. However, when dealing with people experiencing excessive stress, starting at these slightly faster rates may be helping because slower speeds might feel too challenging. This would then cause additional stress associated with the exercise itself, which is not what we want.

The general instructions they provide are:



1. Inhale through the nose to the point of fullness that doesn't cause strain (generally 2-3 seconds)
2. Pause the breath (typically 1-2 seconds). If the air feels like it "bursts out" after the pause, then it was too long.
3. Exhale steadily and smoothly through the mouth or nose (generally 2-4 seconds).
4. Pause slightly, ensuring the next inhale occurs quietly and smoothly (1-2 seconds max).

If you're a breathing coach, consider using these straightforward guidelines when you work with people with excessive stress 🙏



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Nocturnal Mouth-Taping and Social Media: A Scoping Review of the Evidence

Article Published: December 2024

411 Released: December 19, 2024

Quick Summary

This 2024 review found that nighttime mouth-taping may help people with obstructive sleep apnea, especially in moderate to severe OSA when combined with other therapies. It may also improve snoring and might improve subjective sleep quality in some cases. However, evidence for its benefits as a standalone treatment is limited and many claims made on social media do not currently have scientific support.

Article Citation

Fangmeyer SK, Badger CD, Thakkar PG. Nocturnal mouth-taping and social media: A scoping review of the evidence. Am J Otolaryngol. 2024 Dec 4;46(1):104545. doi: 10.1016/j.amjoto.2024.104545. Epub ahead of print. PMID: 39662104.

[Link to Get the Paper](#)



4 Fundamentals

1. Essential Background Material

This paper addresses the growing trend of mouth-taping promoted on social media platforms like TikTok. Many influencers claim that mouth-taping can improve sleep, increase immune function, improve energy, and lead to better skin, among others.

The basis of these claims is primarily on the well-documented benefits of nasal breathing over mouth breathing in general, with the assumption that those benefits should persist during sleep. Yet very few studies have actually been carried out to test this hypothesis and most benefits are extrapolated from this general knowledge.

Thus, the goal of this review was to summarize existing research on mouth-taping and give healthcare providers evidence-based information to respond to patient questions about the topic. This is also helpful for breath and meditation coaches who are increasingly asked about mouth taping.

Let's look at what they did...

2. What Did this Research Do?

This was a scoping review of published literature on mouth taping. As such, they only included studies that specifically used mouth tape during sleep (with or without other devices). **Overall, nine papers were included**, each having different methods, participants, and study sizes, highlighting the variability in the available research.

In addition to this literature review, the authors analyzed 50 TikTok videos discussing mouth-taping that featured both influencers and medical professionals. These videos were used to compare social media claims with the scientific findings.

As an aside, this study is a testament to how social movements and platforms like TikTok can influence healthcare conversations and even scientific inquiry. While in some ways it's concerning that TikTok trends drive medical consultations, it's also inspiring that public interest can spark meaningful research.



3 & 4. What Were the Major Findings and Why Do They Matter?

Mouth-taping generally helps people with obstructive sleep apnea.

- Overall, mouth-taping can sometimes help reduce snoring, apnea index, and breath holding, especially when combined with other treatments.
- People with more severe OSA seem to benefit more from mouth taping, but it's unclear if mouth-taping alone can significantly reduce apnea index.
- Mouth tape plus a mandibular advancement device (MAD) or a CPAP was more effective than either device by itself.
- Some studies were inconclusive, often due to small sample sizes, lack of controls, or unclear comparisons.

Mouth taping might help reduce snoring.

- One study examined the effect of mouth tape as part of a combined airway and positional therapy that included a nasal dilator, nasal wash, and side-sleeping measures. Importantly the participants did not have sleep apnea. Snoring did show a 58% improvement according to participants' partners. (It's worth noting that MAD-only users in this study showed a 91% improvement.) Further, because mouth tape wasn't used by itself, it's hard to conclude how much of a role it played in the 58% improvements.

Mouth taping showed mixed results in people with respiratory disease.

- A four-week study with 50 participants found no significant improvements in morning peak expiratory flow and overall symptom scores in people with asthma who mouth taped.
- In another study of people with nighttime respiratory failure using a device to support breathing, it was found that mouth taping increased REM sleep, reduced mouth leak, and reduced arousal levels.

Some claims from social media were supported, but there is a long way to go (audio version has a fun mini rant on this at 8:51 mark).

- Their social media review found the following claims for mouth taping:
 - Better sleep (partially supported).
 - Less cavities (not studied yet).
 - Better morning breath (not studied yet).
 - Less morning mouth dryness (not studied yet).
 - Prevent snoring (partially supported).
 - Improve or cure sleep apnea (partially supported).
 - Improve asthma (not supported)
-



1 Big Takeaway

Nighttime mouth-taping may help people with obstructive sleep apnea, especially in moderate to severe OSA when combined with other therapies. It may also improve snoring and might improve subjective sleep quality in some cases. However, evidence for its benefits as a standalone treatment is limited and many claims made on social media do not currently have scientific support.

1 Practical Application

This review did not address safety considerations. So, always start by addressing whether mouth taping is safe for yourself or your clients. If someone has nasal obstruction, severe OSA, or a respiratory disease, consult a physician first. Once deemed safe, consider using mouth tape alone or in conjunction with other devices (such as CPAP or MAD) to potentially help improve symptoms of OSA, reduce snoring, and improve subjective sleep quality.

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PAPER](#)

Citation

Fincham, G.W., Strauss, C., Montero-Marin, J. et al. Effect of breathwork on stress and mental health: A meta-analysis of randomised-controlled trials. *Sci Rep* 13, 432 (2023). <https://doi.org/10.1038/s41598-022-27247-y>

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January 18, 2023

Additional Note

This was published in the prestigious *Nature Scientific Reports*. I never thought I'd see the word "breathwork" there. And although I dislike the word "breathwork" for some reason, it's an amazing step forward for the scientific acceptance of breathing for better health. **To date, this is probably the most significant study ever published on breathing.**

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Effect of breathwork on stress and mental health: A meta-analysis of randomised-controlled trials

4 FUNDAMENTALS

1. Essential Background Material

- Although "breathwork" has been practiced for thousands of years, it has recently gained popularity due to COVID-19 and James Nestor's book—yes, this *Nature* publication even referenced his book as part of the trend :)
- Concurrently, **chronic stress has also increased markedly over the past few years**, negatively contributing to physical and mental health conditions like anxiety, hypertension, cardiovascular disease, and depression.
- Breathwork can be defined as various practices that regulate the way we breathe to achieve better mental, emotional, and physical health. **It can be easily taught in a variety of formats**, making it an attractive option for mitigating the recent rise in stress and its associated adverse health effects.
- **In general, slow breathing techniques have the most evidence for reducing stress.** They work by increasing parasympathetic activity in the nervous system, increasing HRV, influencing brain regions associated with emotion, and ultimately making the stress-response system more resilient.
- Although slow breathing clearly works, more broadly, it's unclear how much of breathwork's popularity for stress reduction is "hype" and how much is grounded in science. **That's what this review aimed to investigate.**



2. What Did this Research Do?

This is the first [meta-analysis](#) of randomized controlled trials (RCTs) examining the effect of breathing exercises on self-reported stress, anxiety, and depression. Nothing is perfect, but **a meta-analysis of RCTs is essentially the most robust evidence you can get from published research**, so this is perhaps the most rigorous review of breathing published to date.

Stress

The primary question they sought to answer scientifically was: **Does breathwork help people feel less stressed?**

They used robust search criteria to find published studies that measured how any type of breathwork (slow or fast) affected self-reported stress. Note that it had to be “breathwork,” not meditation, singing, chanting, or any other practice that indirectly changes breathing patterns.

They ended up with 12 studies having 785 total participants. Nine papers were scored as having “some concerns” for bias, while the other three scored as having a high risk of bias—not surprising given the low quality of most breathing studies. Two papers used fast breathing, and the other ten used slow.

Anxiety and Depression

The second (very related) question they sought to answer was: **Does breathwork help people experience fewer symptoms of anxiety and depression?**

Excess stress can cause or exacerbate symptoms of anxiety and depression, so there is considerable overlap between these conditions and stress.

For this secondary question, they analyzed 20 studies for anxiety and 18 for depression.

3. What Were the Major Findings?

Stress

- Overall, breathwork significantly improved self-reported stress (statistically, it was a “small-medium effect”).



- **Statistically, the overall stress-reducing effects of breathwork are the same as those found for cognitive behavioral therapy (CBT), meditation, mindfulness, and acceptance of emotions therapy.**
- No one individual study accounted for these significant effects. If they removed one study at a time, the results did not change significantly, indicating they were robust.
- **This is really neat: The stress-reducing results found in the 12 studies were so powerful that there would have to be 69 additional studies published showing zero effects on stress to negate their statistical significance.**
- To restate that last point, there would need to be 69 papers published finding that “breathwork does not reduce stress” to overturn the results of this meta-analysis. So, if you see a headline claiming “new study finds breathwork doesn’t reduce stress,” it would take 68 more until it matters :)
- **The results remained significant regardless of if the breathwork was delivered remotely without an instructor (self-help), in-person, in a group setting, or in combinations of all the above.**
- A sub-analysis found that the results were only significant for the slow breathing studies. There were just two fast breathing papers, but if those two were analyzed alone, the stress-reducing effects were insignificant.
- Nonclinical patients (i.e., those without a physical or mental health condition) saw the greatest improvements. If the studies with patients with physical or mental health conditions were analyzed separately, the stress-reducing effects were insignificant (like with fast breathing, the sample size may be too small).

Anxiety and Depression

- Breathwork had a significant effect on reducing anxiety and depression (statistically, a “small-medium effect”).
- Heterogeneity among the studies was significant, making the results less robust. This suggests that variables other than breathwork could explain the positive results.
- **Nonetheless, breathwork’s effects on anxiety and depression are statistically similar to those seen for physical activity.** (They’re not saying



breathwork can replace exercise—just that the impact on anxiety and depression is similar).

- Breathwork's effects were also statistically the same as those found for CBT, mindfulness, meditation, and acceptance of emotions therapy.

4. Why Do These Results Matter?

Stress significantly contributes to many chronic diseases and complications, like cardiovascular disease, hypertension, and depression. Additionally, stress is pervasive—over 60% of adults report financial and occupational stress.

Here, for the first time, it is shown with the most rigorous scientific analysis (a meta-analysis of RCTs) that the stress-reducing effects of slow breathing exercises are legitimate and significant. Breathing yielded results comparable to the most widely used treatments like CBT, mindfulness, and meditation.

Slow breathing exercises also significantly reduced symptoms of anxiety and depression, similar to effects seen for commonly-used therapies.

Most importantly, this meta-analysis showed that the mode of delivery did not significantly impact the results. Most treatments for stress rely on therapists with extensive training and require ongoing supervision. Breathing exercises, on the other hand, can be self-learned remotely, in-person, or in group settings, and the benefits for stress remain significant.

1 BIG TAKEAWAY

The most rigorous scientific analysis to date shows that slow breathing significantly improves self-reported stress, anxiety, and depression to the same extent as many standard therapies, such as CBT and mindfulness. This holds true regardless of how the breathwork training is delivered (remote, in-person, or group settings).



1 PRACTICAL APPLICATION

For Yourself

Regularly practice essentially any slow breathing technique to reduce stress and symptoms of anxiety and depression.

If You Coach Others

This paper shows that you can teach your students slow breathing practices in-person, remotely, 1-on-1, or in a group setting and be confident that you will significantly reduce their stress levels.

For Your Business

Apply this knowledge by sharing it with others. Grab one of the results that resonates with you, re-write it in your own words, add your personal take on its importance, and share it so people learn the incredible power of the breath.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Effect of Coherent Breathing on Mental Health and Wellbeing: A Randomised Placebo-Controlled Trial

411 Released: January 5, 2023

[nature](#) > [scientific reports](#) > [articles](#) > [article](#)

Article | [Open access](#) | [Published: 13 December 2023](#)

Effect of coherent breathing on mental health and wellbeing: a randomised placebo-controlled trial

[Guy W. Fincham](#) , [Clara Strauss](#) & [Kate Cavanagh](#)

[Scientific Reports](#) **13**, Article number: 22141 (2023) | [Cite this article](#)

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Article Citation

Fincham GW, Strauss C, Cavanagh K. Effect of coherent breathing on mental health and wellbeing: a randomised placebo-controlled trial. Sci Rep. 2023 Dec 13;13(1):22141. doi: 10.1038/s41598-023-49279-8. PMID: 38092805; PMCID: PMC10719279.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

Controlled, nasal, diaphragmatic breathing (even at a relatively fast pace) for 10 minutes daily over 4 weeks significantly reduces stress, anxiety, and depression while increasing wellbeing.



4 Fundamentals

1. Essential Background Material

- Of all the breathing techniques available, slow breathing has received the most attention from the scientific community, and its effects on the body and mental and emotional health have been documented extensively.
- “Coherent” or “resonant” breathing is a subset of slow breathing that uses a pace of 5 to 6 breaths a minute without pauses. It has been studied extensively and found to be helpful for physical and mental health. This rate has also been found in various cultures over time.
- Coherent breathing is beneficial because it is believed to align signals between the respiratory, cardiovascular, and autonomic nervous systems, which leads to a wide variety of benefits for the body and mind.
- However, there haven’t been any well-designed, placebo-controlled studies on slow and coherent breathing, which is necessary to tease out the precise benefits of breathing at 5-6 breaths a minute versus things like intention, belief, and breath control in general.
- Moreover, most studies published on slow breathing have a moderate risk of bias, which reduces the robustness of any findings.
- This study aimed to provide the most rigorous randomized controlled trial (RCT) on slow breathing to date, including a “well-deigned placebo” and blinding of participants.

2. What Did this Research Do?

This study had 379 participants. They were randomly allocated into a slow breathing group (N=189) and a placebo group (N=190).

Slow Breathing Group: practiced at 5.5 breaths a minute with an equal inhale-to-exhale ratio.

Placebo Group: practiced at 12 breaths a minute with an equal inhale-to-exhale ratio.

This study was unique because the participants were new to breathing and blinded to the study’s intention. They received the audio breathing instructions remotely, and regardless of which group they were in, it was simply referred to as “rhythmic breathing.” There was no loaded vocabulary around the benefits of breath-focused practices.



Notably, instructions for both groups were identical: sit up straight or lie down comfortably, breathe consciously through your nose, breathe diaphragmatically, and take a moment to tune into how you feel afterward. The only variable that changed was the pace of the breathing. The audio was recorded by a qualified breathing instructor, and the participants were asked to follow the guidance for 10 minutes daily for 4 weeks.

The primary outcome they were interested in was stress. But they also measured anxiety and depression, sleep disturbances, wellbeing, and the participants' take on the "credibility and expectancy" of the breathing protocol. All were measured using established questionnaires. The questionnaires were given 1) before the intervention, 2) right after the four weeks, and 3) four weeks after the completion of the study.

3 & 4. What Were the Major Findings and Why Do They Matter?

Slow breathing (5.5 breaths a minute) and placebo breathing (12 breaths a minute) both reduced stress significantly. Surprisingly, slow breathing did not reduce stress more than controlled breathing at 12 breaths a minute.

- This seems unexpected at first...shouldn't slow breathing reduce stress more than 12 breaths a minute? I certainly would've guessed so. But remember that, regardless of which group they were in, participants were going from no breathing experience to conscious, nasal, diaphragmatic, controlled breathing. In fact, it's hard to believe they called this a "placebo," given how much literature there is on the power of breath awareness, nasal breathing, diaphragmatic breathing, and controlled breathing in general. Each of these factors individually would seem enough to reduce stress.
 - Thus, using the word "placebo" is a bit of a distraction from what the study actually showed: That breathing at 12 breaths a minute for 10 minutes daily is as effective at reducing stress as breathing at 5.5 breaths a minute for 10 minutes daily.
 - **This is still very significant** because it highlights that, when it comes to stress reduction, the pace of breathing may not be as crucial as simply controlling your breath, breathing nasally, and breathing diaphragmatically.
 - Also, note that no physiological variables were measured. So, although they both improved stress to the same extent, it could be that slow breathing reduced blood pressure more, had more improvements in heart rate variability, lowered oxidative stress more, and so on. Conversely, it could also mean that controlled breathing (even
-



at 12 breaths a minute) would have these same physiological effects.

- Additionally, a study published in 2017 (that [I reviewed 5 years ago...](#) 🤖) found that controlled breathing at 12 breaths a minute significantly lowered sympathetic outflow. This highlights that controlled breathing at 12 breaths a minute has positive physiological effects, negating it as a “placebo.” Listen to the audio version of this 411 to hear me discuss why it’s important to the current findings.
- **Finally, this result is vital:** It shows that without education, no loaded vocabulary, no explanation of the benefits, no nothing, the participants still significantly reduced their stress levels in just 10 minutes a day for 4 weeks. That’s incredible.

There were no differences in credibility or outcome expectancy between the groups.

- This is meaningful because it shows that the groups did not have some inherent bias toward one of the breathing protocols.

Depression and anxiety significantly decreased while wellbeing increased in both groups.

- Again, slow breathing was not better than controlled breathing at 12 breaths/min.

Sleep disturbances were unchanged.

- Controlled breathing for 10 minutes daily did not affect their subjective sleep quality.

1 Big Takeaway

Controlled, nasal, diaphragmatic breathing (even at a relatively fast pace) for 10 minutes daily over 4 weeks significantly reduces stress, anxiety, and depression while increasing wellbeing.

1 Practical Application

Pick almost any breathing rate that is comfortable to you and focus on conscious, nasal, and diaphragmatic breathing for 10 minutes daily to improve mental and emotional health.

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Citation

Fournié C, Chouchou F, Dalleau G, Caderby T, Cabrera Q, Verkindt C. Heart rate variability biofeedback in chronic disease management: A systematic review. Complement Ther Med. 2021 Aug;60:102750. doi: 10.1016/j.ctim.2021.102750. Epub 2021 Jun 10. PMID: 34118390.

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April 16, 2022

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Heart rate variability biofeedback in chronic disease management: A systematic review

4 FUNDAMENTALS

1. Essential Background Material

This systematic review aimed to examine if heart rate variability biofeedback (HRVB) could be an effective supplemental treatment for people with chronic diseases.

Chronic diseases like diabetes, cancer, and mental health conditions are proliferating. Aside from the diseases themselves, these conditions also cause hormone imbalances, sleep problems, inflammation, stress, anxiety, etc. This leads to a significant reduction in the quality of life for those afflicted and adverse societal and economic outcomes.

One supplemental therapy that may be helpful is HRVB. Increases in HRV are thought to be mediated by increases in vagal tone and therefore indicate better overall cardio-autonomic health. **This suggests that HRVB could benefit several body, brain, and emotional outcomes associated with chronic diseases.**

However, although several systematic reviews exist for HRVB for specific conditions (e.g., depression or stress and anxiety), one does not exist for chronic disease management in general, which was the motivation for this current review.



2. What Did this Research Do?

The authors searched several databases for articles published up until March 31, 2020. They included studies that used HRVB in patients with various chronic diseases that had a control group to compare against. Moreover, studies that used HRVB plus another supplementary therapy (e.g., exercise) were also included (which you'll see below was very helpful for putting the results into perspective) as long as there was a control group to get specific effects of HRVB.

They ended up including 29 studies with 1127 total patients. Each study had a sample size of 10 to 210 participants with various chronic diseases measuring diverse outcomes.

3. What Were the Major Findings?

HRVB (via slow breathing) meaningfully improved several chronic disease outcomes, and the protocol was easy to follow. Over 70% of the patients completed their daily practice, and there were very few side effects.

Here is the list of the issues they found help with:

- Stress reduction and positive emotions
- Asthma symptoms
- Inflammation
- Blood pressure
- Depression and anxiety
- Post-traumatic stress disorder
- Cognitive function
- Sleep
- Overall quality of life (*most important, in my opinion*)

Note that these weren't improvements across the board. Some studies showed positive results, and others were less conclusive. However, based on the available research, they concluded that the overall evidence suggests that the above outcomes are improved from HRVB in people with chronic diseases.

A critical finding was that HRVB improved all these conditions compared to controls, **but it did not improve things more than other non-pharmacological interventions such as physical exercise and progressive muscle relaxation.** *This is so vital.* HRVB works, but so do other things like exercise and mindfulness meditation. Choose which one is right for you :)



Lastly, there were some super practical findings on minimum effective dose and how to breathe during HRVB, which I've saved for the "1 Practical Application" section below.

4. Why Do These Results Matter?

These results are meaningful because they show that HRVB via slow breathing provides a practical approach to managing a wide array of chronic diseases. Moreover, they provide evidence that HRVB could be a powerful supplementary therapy when combined with standard care (this is always the best approach, in my opinion—it's not going to cure anything, but it can be a powerful addition to everything else you do to manage your chronic condition).

1 BIG TAKEAWAY

HRVB (and, hence, slow breathing) is a powerful adjunctive therapy for managing a wide variety of chronic diseases. To be effective, it should be practiced for a minimum of 10 min/day (but probably 20+ for best results—see next section). More broadly, this study supports using slow breathing at around 5-6 breaths per minute in people with chronic diseases.

1 PRACTICAL APPLICATION

Wonderfully, this paper summarized all of the research and outlined the best practices for applying their findings. So, my 1 Practical Application will come directly from their advice (which you can find in Sections 4.4 and 4.5 of the paper).

Here's what gave the best results based on the 29 studies included in this review:

- **Best Results:** 4-12 supervised sessions plus daily home practice for ~20 min/day
- **Minimum Effective Dose:** One supervised practice followed by home practice for 10 min/day for 4 weeks
- During supervised practice, trainees should learn to breathe slowly without stress & w/o overbreathing
- Use abdominal breathing w/ nasal in & pursed-lips out



- The breathing ratio should have a slightly longer exhale ([I like the 40/60 approach](#): 40% inhaling, 60% exhaling)

That's about as good as it gets from a scientific paper! Follow those guidelines for yourself or give them to your clients and loved ones.

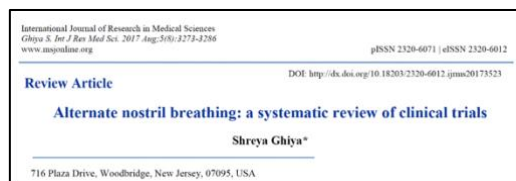


Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Alternate Nostril Breathing: A Systematic Review of Clinical Trials

411 Released: August 5, 2024



Quick Summary

Across studies using different protocols, alternate nostril breathing consistently improved nervous system and cardiovascular health, lung function, and cognitive ability, suggesting that it is an effective technique for holistically boosting health & wellness.

[Click Here to Get the Paper](#)

Article Citation

Ghiya, S. (2017). Alternate nostril breathing: a systematic review of clinical trials. *Int J Res Med Sci*, 5(8), 3273-86.

Reading Time: 3 min 23 seconds

I hope the next 50'ish breaths are the most nourishing of your day.



4 Fundamentals

1. Essential Background Material

Pranayama is one of the eight limbs of yoga and is the precursor to most modern breathing techniques. Pranayama includes many different forms of breath regulation, one popular approach being alternate nostril breathing (ANB). ANB involves breathing through one nostril at a time. For example, gently pressing your right nostril closed while breathing in through the left, then gently closing your left nostril and breathing out through the right. The process is then reversed on the other side.

Research has shown that ANB has powerful effects. Because the left nostril is tied to the parasympathetic and the right to the sympathetic, ANB generally balances the nervous system. (It's like working out your left arm and then your right to find harmony.) ANB has also been shown to reduce blood pressure, improve the retention of motor skills, enhance focus, change brain activation, and reduce stress.

Despite these meaningful effects, a comprehensive review consolidating these findings was lacking. This paper, therefore, sought to fill this gap by providing a systematic review of the impacts of ANB on physiological and cognitive functions.

2. What Did this Research Do?

This was a systematic review of randomized controlled studies that used ANB as an independent intervention (that is, they didn't review studies that used ANB and yoga, etc.).

They found 44 studies that met this criterion. The participants' ages in these studies ranged from 8 to 70 years old. Most of the participants were healthy, but they also included two studies with hypertensive participants, one with pre-hypertensive, one with females suffering from premenstrual syndrome, and one study with asymptomatic alcoholics.

Twenty-nine of the studies had a low risk of bias, ten had a high risk, and the other five had an unknown risk due to a lack of sufficient data. (As an aside, having 29 studies with a low risk of bias is considerably better than most breath-focused systematic reviews I've encountered, suggesting that ANB studies generally have a better design.)



3 & 4. What Were the Major Findings and Why Do They Matter?

There was a large variety of ANB techniques used across the studies.

- Eighteen studies used the standard approach of in one side, out the other, reverse, and repeat. Six studies had people breathe in through the left, out the right for 1-3 cycles, and then repeat in the other direction. Still, another used a mix of ANB and single-nostril breathing. Lastly, some studies used small breath holds while others did not. Keeping this diversity in mind is critical for the practical application of its findings.

Length of individual practice and duration of the study varied substantially.

- Some studies examined a single session of ANB lasting between 5 and 40 minutes. Others involved multiple sessions spread over time, lasting up to four months. Lastly, the length of practice was also sometimes reported in number of breathing cycles (ranging from 9 to 27) rather than minutes, which is common in pranayama instructions. Unfortunately, they did not report on whether the length of practice influenced positive outcomes.

ANB generally improved autonomic nervous system functioning.

- HRV results were mixed: two found improved HRV, one found no change, and two more found increased sympathetic activity. However, the nervous system was also measured by heart rate responses to several tests, like deep breathing and standing, and blood pressure response to hand grip and cold pressure tests, which revealed improved functioning.

ANB helped the cardiovascular system by decreasing heart rate and systolic and diastolic blood pressure.

- A majority of studies found that ANB reduced these parameters. However, it was noted that when ANB was initiated via the right nostril, blood pressure and heart rate both increased, which is consistent with the predominant theory that the left nostril is parasympathetic and the right sympathetic.

ANB improved lung function based on a wide range of tests.

- ANB enhanced various lung function tests, such as Peak Expiratory Flow Rate (maximum speed of exhalation) and Forced Vital Capacity (a measure of lung capacity), indicating that it may be helpful for general lung health and people suffering from chronic lung conditions.

ANB improved several markers of cognitive function.

- These included enhanced [retention of motor skills](#), improved memory, and improved verbal and spatial memory in children. One study also found improved anxiety and depression symptoms.
-



1 Big Takeaway

Across studies using a range of different protocols, alternate nostril breathing consistently improved nervous system and cardiovascular health, lung function, and cognitive ability, suggesting that it is an effective technique for holistically boosting health and wellness.

1 Practical Application

This review supports using ANB for a minimum of 5 minutes or 9 cycles to see meaningful improvements. However, I've found that even 1-3 minutes at the start of a breathing or meditation session is helpful for improving focus and enhancing the benefits of the practice.

Check out [the 26-minute mark](#) of the Breathing for Better Brain Health Workshop for video instructions on how to perform ANB. Consider using the 12-minute practice provided there, which combines ANB and slow mindful breathing, to get the benefits of both practices.



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Citation

Gholamrezaei, A., Van Diest, I., Aziz, Q., Vlaeyen, J. W. S., & Van Oudenhove, L. (2021). Psychophysiological responses to various slow, deep breathing techniques. *Psychophysiology*, 58(2), [e13712].
<https://doi.org/10.1111/psyp.13712>

411 Created On:
October 14, 2022

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Psychophysiological responses to various slow, deep breathing techniques

4 FUNDAMENTALS

1. Essential Background Material

This study examined differences in cardiovascular and emotional outcomes between slow breathing using pursed-lips (PLB), right-nostril-only (RNB), left-nostril-only (LNB), and resistance (RB).

There are many ways to perform slow breathing. We can change how we breathe (like listed above), the rate at which we breathe (e.g., 5 vs. 6 breaths/min), or breathe with different inhale-to-exhale ratios.

In general, they all seem to help cardiovascular function, reduce pain, and lessen stress and anxiety.

However, their specific effects might differ, and some methods might be better for some conditions (like hypertension) than others.

Therefore, this study aimed to examine how four common approaches (PLB, RNB, LNB, and RB) uniquely impact cardiovascular and emotional outcomes, focusing on which might be best for hypertension and pain.

2. What Did this Research Do?

Thirty-five volunteers partook in the study, aged 18 to 32. They were taught the four breathing techniques:



- **Pursed-Lips Breathing:** Inhale through the nose and exhale audibly through pursed lips.
- **Left-or-Right Nostril-Only Breathing:** Use the thumb to block either the left or right nostril and breathe exclusively through the other.
- **Resistance Breathing:** Use a resistance device to breathe with a load of 10 cmH₂O

The participants rested for 5 minutes for baseline measurements. **Then, they performed each breathing technique for 3 minutes**, with a 5-minute washout between each exercise.

For all methods, they breathed at a rate of 6 breaths/min using a 1:2 ratio, meaning the exhale was twice as long as the inhale (this comes out to a 3.3-sec inhale with a 6.6-sec exhale).

Because none of the participants had experience with breathing exercises, they only did each exercise for 3 min to make it reasonable to maintain the RB.

Physiological measurements were taken during the breathing (heart rate variability [HRV], blood pressure, etc.). Emotional outcomes were scored via questionnaires given after each breathing exercise.

3. What Were the Major Findings?

Important Overall Caveat

The findings only apply to comparisons amongst the different breathing conditions—not compared to the baseline. Usually, we'd like to see something like 'blood pressure was "X" at baseline and "Y" after the breathing.' But here, it's simply between "X" during RB and "Y" during PLB or RNB, etc.

Here's why that's critical: systolic blood pressure actually went up during all of the breathing conditions compared to baseline—not the result we want, right? But, because the breathing conditions were given in random order, it's impossible to know what caused it. For example, maybe RB caused blood pressure to go up, which carried over into the other exercises.

Of course, the results here are still significant. But, it would be nice to see them performed on separate days or with different groups of people to evaluate the changes from baseline, which are most important, in my opinion. But, alas, no study is perfect.



Overall, pursed-lips and resistance breathing had the best outcomes. Here are the specific results:

- All four slow breathing techniques increased tidal volume and minute ventilation
- Blood pressure was lowest during PLB
- Blood pressure variability was highest during RB, followed by PLB
- HRV was highest during RB, followed by PLB
- Blood pressure and HRV were similar between LNB and RNB, which might be due to the short breathing time (3 min)
- RB caused the most feelings of breathlessness and was the least enjoyable
- PLB was the most enjoyable and provided the best sense of control
- Some participants reported dizziness, sleepiness, cough, and nausea (it appears the cough and nausea were from the RB device)

A limitation of these results is that capnometry was not used. Since this was the participants' first time performing slow breathing, they were likely overbreathing (this effect generally goes away after a few training sessions). This helps explain the feelings of dizziness in 6 participants.

Another limitation is that nasal-only breathing was not tested. Thus, it's unclear if PLB would be better or worse than nasal-only slow breathing.

4. Why Do These Results Matter?

These results are helpful because there are many ways to perform slow breathing. Here we learn that pursed-lip exhales provide the most physiological and emotional benefits with no additional equipment or cost (keeping in mind the above caveat about nasal-only breathing).

However, we also learned that resistance breathing caused the most significant fluctuations in blood pressure and the highest HRV. Thus, RB could be a better overall “workout” for our cardiovascular systems, even though it causes the most unpleasant feelings.

1 BIG TAKEAWAY

Slow breathing using pursed-lip exhales or a resistance device both potentially improve cardiovascular health, but pursed-lips breathing is free, more pleasant, and provides a better sense of control.



1 PRACTICAL APPLICATION

The most straightforward application of this research is to use pursed-lips exhales during our slow breathing practice to improve our heart and emotional health.

If you have a resistance device, this research also shows the added benefit for HRV and blood pressure variation when slow breathing is performed with resistance.

Personally, I have used a resistance device for inspiratory muscle training. However, I used forceful, rapid breaths, which I did not particularly enjoy. Since reading this paper, I have started doing 3 minutes of slow breathing with resistance and find it more enjoyable. It's too soon to know if it's helping, but good to know there is an alternative way to perform resistance breathing.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

From Lung to Brain: Respiration Modulates Neural and Mental Activity

Article Published: June 2023

411 Released: November 19, 2024

Quick Summary

This review found that breathing, our physiology, brain function, and mental states are all linked through speed (e.g., slow breathing slows brain waves and physiology) and location (e.g., breathing and physiological changes impact common areas of the brain). This holistic insight offers a promising pathway for developing breathing-based mental health interventions.

Article Citation

Goheen J, Anderson JAE, Zhang J, Northoff G. From Lung to Brain: Respiration Modulates Neural and Mental Activity. *Neurosci Bull.* 2023 Oct;39(10):1577-1590. doi: 10.1007/s12264-023-01070-5. Epub 2023 Jun 7. PMID: 37285017; PMCID: PMC10533478.

[Link to Get the Paper](#)



4 Fundamentals

1. Essential Background Material

When we consider that breathing provides the primary fuel for cellular energy (oxygen), it makes sense that the brain (and all organs) would be coupled to the act of respiration. Yet the breath's impact specifically on the brain is often dismissed as “noise” in scientific studies.

This paper takes a different perspective. The authors hypothesize that brain signals linked to breathing are critical to understanding cognitive functions such as emotion and behavior. There are three primary ways through which this may occur:

1. **Brain Rhythms:** Breathing rates may couple with and amplify brain rhythms—slow breathing may amplify slower brain waves and vice versa for fast breathing.
2. **Physiology and Biochemistry:** By simultaneously changing heart rate variability (HRV) and carbon dioxide (CO₂), breathing can initiate signals that impact brain function. Conversely, changes in heart rhythms and CO₂ can lead the brain to adjust the breath to maintain balance.
3. **Mental Activity:** Anxiety, stress, and panic are generally associated with faster breathing rates, while calm states promote slower breathing. Thus, through breathing, these mental states also impact HRV and CO₂, which can then impact brain function.

Altogether, this suggests that breathing, brain activity, physiology, and mental states are all tightly interconnected. This review aimed to deepen our understanding of this relationship as a stepping stone for developing therapeutic breathing interventions for mental disorders.

2. What Did this Research Do?

This paper was a narrative review of published literature. The authors found studies that examined breathing (spontaneous, slow, or fast breathing) alongside measurements of CO₂, brain blood flow (fMRI), brain waves, HRV, or mental activity. They ended up with 75 studies with the following groupings:

- Brain Blood Flow (n=34)
 - Brain Waves (n=26)
 - CO₂ and HRV (n=27)
 - Mental Activity (n=9)
-



Their aim was to synthesize these studies to provide a better understanding of how changes in breathing, heart activity, and CO₂ are linked to changes in brain and mental activity.

3 & 4. What Were the Major Findings and Why Do They Matter?

Breathing impacts widespread regions of the brain.

- fMRI studies revealed that spontaneous breathing globally impacts the brain. Moreover, there was significant overlap in regions where breathing affected the brain and heart activity and CO₂ also affected the brain.
- Interestingly, the regions of brain overlap with breathing, heart activity, and CO₂ have been previously associated with our sense of self, **suggesting (but not proving) a relationship between the act of breathing and our sense of consciousness.**

Breathing is coupled to brain activity at multiple frequencies.

- Breathing appears to impact brain waves at similar frequencies (0.01 to 0.3 Hz, or 0.3-18 breaths a minute) and at much faster oscillations (1 to 30+ Hz). This suggests that breathing not only entrains brain wave activity at our breathing rate but may also indirectly influence broader neural activity.
- The amount of breath-brain coupling was dependent on region, with some areas being tightly linked at all frequencies while other brain regions were only coupled as specific frequencies, highlighting a complex relationship between breathing and brain waves.

There are clear brain and physiological differences between slow and fast breathing protocols (no surprises here).

- Across 27 studies examining slow breathing, they found that it generally increased HRV, slightly increased CO₂, slowed brain waves, and increased inter-regional brain wave coherence. In general, slow breathing increased alpha and theta power, but the magnitude and coupling were dependent on region.
- Across 5 studies examining fast breathing, they found the opposite: a decrease in HRV and CO₂ and an increase in brain frequencies. Although these findings are in line with our general understanding the breath-brain-body continuum, the small sample of studies may limit the generalizability of these findings.

Slow and fast breathing impact mental activity in distinctly different ways.

- Slow breathing (four studies) tends reduce negative feelings, stress, and anxiety.
-



- Fast breathing (two studies) tends to increase anxiety and stress during the practice but reduces these feelings after several weeks of practice. It could be because it acts as a positive stressor or due to adaptation to the practice over time.

Breathing and physiological changes are closely linked to mental states.

- In eight studies examining anxiety and panic disorders, they found that patients often had chronic hyperventilation, low CO₂, and reduced HRV. The studies highlighted how hyperventilation may self-perpetuate—hypersensitivity to CO₂ triggers chronic overbreathing, which leads to increased feelings of tension. Moreover, misinterpretation of these bodily sensations (“Am I dying?”) may also lead to feelings of panic, exacerbating stress and subsequent hyperventilation.

Breathing, physiology, brain, and mental states are all linked through frequency and location.

- **Linked via frequency:** This refers to the idea that slowing the breath can also slow brain waves and alter heart rhythms, making speed a “common currency” across these systems.
- **Linked via location:** This refers to the idea that breathing, physiological, and biochemical changes impact overlapping brain regions, suggesting that these systems work together to shape brain and mental states. This understanding offers a potential framework for developing breathing-based approaches to mental health.

1 Big Takeaway

Breathing, our physiology, brain function, and mental states are all linked through speed (e.g., slow breathing slows brain waves and physiology) and location (e.g., breathing and physiological changes impact common areas of the brain). This more holistic understanding offers a promising pathway for developing breathing-based mental health interventions.

1 Practical Application

By understanding that your breath is intricately related your brain and physiology, you can use (or help your clients use) breath-based interventions as one way to improve mental health. This research suggests that slow breathing, in particular, is helpful for slowing brain waves, improving physiological and biochemical markers, and reducing stress and anxiety.



[DOWNLOAD
PAPER](#)

Citation

Gustafson, J.R., Foster, S., & Swarthout, K. (2022). Oral and systemic effects of breathing patterns: Nasal breathing vs. mouth breathing. Dental Academy of Continuing Education. <https://dentalacademyofce.com/courses/oral-and-systemic-effects-of-breathing-patterns-nasal-breathing-vs-mouth-breathing/>

411 Created On:
October 10, 2022

Additional Note:

This peer-reviewed paper was given as part of an online course designed for dentists, dental hygienists, and dental assistants. So, it was slightly different from standard papers, but still peer-reviewed and packed with excellent information.

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Oral and Systemic Effects of Breathing Patterns: Nasal Breathing vs. Mouth Breathing

4 FUNDAMENTALS

1. Essential Background Material

This review examined the differences between nose and mouth breathing and its implications for our overall health and wellness.

You likely know by now that nasal breathing is healthy breathing.

But what defines a nasal breather from a mouth breather? They say this is determined by how a person breathes at rest.

If you breathe nasally at rest, you're a nasal breather; otherwise, you're a mouth breather. (I had never heard an actual clinical definition of it.)

Moreover (**and this is critical**), combining nasal and mouth breathing during exercise still keeps us in the "nasal breather" category.

But if you're a mouth breather at rest, there are several detrimental effects on your face, mouth, and overall health. Fortunately, there are ways to fix them, too.

This article reviews all of this and highlights just how powerful the route of each breath we take is for our health.



2. What Did this Research Do?

This narrative review pulled together several disparate studies and fields to examine the impacts of nose vs. mouth breathing.

For example, it explores everything from craniofacial development to plaque buildup to inflammation to Buteyko.

Although not as rigorous as a systematic review or meta-analysis, narrative reviews are some of the most fun to read because of the breadth of topics covered and the practicality of the findings.

3. What Were the Major Findings?

Causes of Mouth Breathing

Question: If nasal breathing is so good, why do so many of us mouth breathe?

Here are some of the common causes they found:

- Allergies or getting a cold can create a habit of mouth breathing.
- Asthma
- Narrow airways
- Reduced nasal function of any kind

Adverse Effects of Mouth Breathing

- Changes in force on the lips, tongue, and cheeks significantly modify craniofacial development. This leads to adaptations that reduce nasal airway function, further promoting mouth breathing.
- Airway collapse and sleep apnea.
- Tooth decay and loss.
- Loss of lip tone and the development of tongue thrusting (where the tongue sticks through the front teeth).
- Increased and/or chronic activation of the sympathetic nervous system
- Increased risk of cardiovascular disease, chronic fatigue, hypertension, and systemic inflammation.
- Increased chances of anxiety.
- Increased plaque buildup.
- Increased risk of systemic infections.

How to Fix It (and the benefits of nasal breathing)

Breathing retraining can be highly beneficial to reverse, mitigate, or prevent these issues altogether. This can include using various pranayama techniques, Buteyko breathing, or capnometry-assisted breathing training (the latter being the monitoring of exhaled carbon dioxide to ensure proper breathing volume).



Moreover, humming techniques may help children correct dysfunctional mouth breathing patterns.

Using one (or all) of these methods **to restore nasal breathing can:**

- Engage the parasympathetic nervous system more.
- Reduces stress and anxiety.
- Decrease heart rate and blood pressure (better cardiovascular health).
- Improve overall lung function.
- Decrease the risk of asthma attacks.
- Reduce oxidative stress and inflammation.
- Improve sleep.

4. Why Do These Results Matter?

Mouth breathing is pervasive in our modern world. As we learn in this study, this causes many adverse effects. On the flip side, there are **many positive outcomes from breathing through our noses**, and training ourselves to breathe this way is safe, effective, and essentially free.

Thus, all the results outlined here are meaningful because they provide scientific evidence for why we should be breathing through our noses (most of the time), along with straightforward ways we can retrain our breathing.

1 BIG TAKEAWAY

Nasal breathing is healthy breathing, promoting better facial structure, oral health, and cardiovascular function, as well as deeper sleep and less stress and anxiety. Mouth breathing, on the other hand, contributes to many facial abnormalities and oral and systemic health complications.

1 PRACTICAL APPLICATION

The most straightforward application of this research is to begin nasal breathing as much as possible.



If you have a structural issue that prevents nasal breathing (for example, a deviated septum), it's essential to seek professional help to restore it and mitigate the harmful effects of mouth breathing.

Finally, we can apply the results of this review by seeking out pranayama techniques, Buteyko, or capnometry-assisted breathing retraining to improve our ability to breathe nasally and enhance our overall health and well-being.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

The Brain's Resonance with Breathing—Decelerated Breathing Synchronizes Heart Rate and Slow Cortical Potentials

411 Released: October 20, 2024

Quick Summary

Slow breathing, particularly at 6 breaths a minute, rapidly synchronizes breath, heart, and brain rhythms, which may provide insight into how slow breathing exercises enhance mental and emotional well-being.

Article Citation

Hinterberger T, Walter N, Doliwa C, Loew T. The brain's resonance with breathing-decelerated breathing synchronizes heart rate and slow cortical potentials. J Breath Res. 2019 Jun 27;13(4):046003. doi: 10.1088/1752-7163/ab20b2. PMID: 31071704.

[Link to Get the Paper](#)



4 Fundamentals

1. Essential Background Material

Breathing techniques have been used for millennia to influence our mental state, and in recent decades, science has begun revealing why. Research has discovered that breathing is tied to the nervous system through its connections with the heart and vagus nerve. Moreover, slow breathing has been shown to influence brainwave activity, slowing down brainwaves and contributing to its calming effects. These connections provide helpful clues as to how breathing has such powerful impacts on our mental and emotional state.

Yet, many unknowns exist regarding the breath-brain-nervous system connection. One critical area of exploration is how breathing impacts slow cortical potentials (SCPs). SCPs are important because they are directly related to mental effort and excitability: A negative SCP shift means more brain excitability, while a positive one means less brain excitability.

This study sought to understand how controlled breathing influences SCPs, potentially showing another way by which breathing influences mental and emotional health. By concurrently measuring SCPs with breathing and heart rate variability, they assessed how breathing at different rates impacted the resonance between brain, heart, and breath, providing clues into what rates might be most beneficial.

2. What Did this Research Do?

This study involved 37 participants (20 female, 17 male) between the ages of 19 and 71. They participated in a one-day experiment where they performed six different breathing rates for 7 minutes each: 10, 7.5, 6, 5, and 4.3 breaths per minute, followed by a return to 10 breaths per minute. The breathing was performed in a comfortable seated position with their eyes open. Each rate was performed with a 40% inhale and 60% exhale (for example, for 6 breaths a minute, this would equate to a 4-second inhale and 6-second exhale). After each 7-minute breathing session, they rested for three minutes.

In addition to using EEG to measure SCPs, they also measured HRV using ECG and respiratory amplitude using a chest strap. The analysis focused on determining the relationship between breathing, SCPs, and HRV, and whether the changes observed in these signals were significant.



3 & 4. What Were the Major Findings and Why Do They Matter?

Slow breathing was rated as the most calming.

- Questionnaires revealed that slower breathing rates (7.5 or slower) were most relaxing. At the same time, breathing rates of 6 breaths a minute or less were rated as most unnatural, which is not surprising given how much slower they are than an average breathing rate.

All breathing rates elicited a response in HRV and in the brain.

- Regardless of the rate of breathing, there was a clear signal in both HRV and the brain (via SCP) at the frequency of breathing. This shows that controlled breathing, irrespective of rate, influences the rhythms of both the heart and brain.

Breathing at six breaths a minute led to the most significant responses in the heart and brain.

- Although all rates elicited a response, the peaks in both HRV and SCP stood out most when breathing at 6 breaths a minute.

Slow breathing rapidly alters brain rhythms, and these changes last throughout the session.

- The results revealed that, after just a few cycles of slow breathing at 6 breaths a minute, brain activity shifted and remained altered for the duration of the breathing session.

Breathing at 6 breaths a minute (a 10-second breathing cycle) led to the greatest synchronicity between breath, heart, and brain.

- By analyzing responses over the breath cycle, they found that all three systems became most synchronized during the 10-second breathing rhythm. In their words: *"We term this phenomenon the great 10 second synchrony between breath, heart and the brain."*

1 Big Takeaway

Slow breathing, particularly at 6 breaths a minute, rapidly synchronizes breath, heart, and brain rhythms, which might provide insight into how slow breathing exercises enhance mental and emotional health.



1 Practical Application

This one is straightforward: Consider practicing slow breathing—especially at 6 breaths a minute—to quickly bring your breath, heart, and brain into resonance. You can use the tones available on the [Breath Practices](#) page to get started.

[DOWNLOAD
PAPER](#)

Citation

Hopper SI, Murray SL, Ferrara LR, Singleton JK. Effectiveness of diaphragmatic breathing for reducing physiological and psychological stress in adults: a quantitative systematic review. JBI Database System Rev Implement Rep. 2019 Sep;17(9):1855-1876. doi: 10.11124/JBISRI-2017-003848. PMID: 31436595.

411 Created On:
May 21, 2022

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Effectiveness of diaphragmatic breathing for reducing physiological and psychological stress in adults: a quantitative systematic review

4 FUNDAMENTALS

1. Essential Background Material

This systematic review looked at the effects of slow diaphragmatic breathing on emotional and bodily measures of stress.

Stress is widespread. However, “stress” is a vague term, and there is good stress (eustress) and bad stress (distress). This review is focused on the bad stuff.

They define stress as a situation where environmental demands lead to harmful outcomes to a person’s physical or mental health.

And we know the consequences are alarming. For example, [more than 60%](#) of primary care physician visits are due to stress, and stress is a significant risk factor for many diseases.

Thus, there is an urgent need to develop effective, low-cost interventions to help with stress.

Slow deep breathing increases activity in the parasympathetic (rest-and-digest) nervous system. This is not only mentally calming, but it also reduces blood pressure and improves blood flow.



Moreover, **slow diaphragmatic breathing is practical, accessible, and can be performed anywhere**, making it an attractive option for managing stress.

2. What Did this Research Do?

This review looked at studies with adults that were 18 or older, and they included people with or without mental or physical illness.

Studies could only use diaphragmatic breathing by itself (for example, studies with mindfulness or yoga were excluded). The diaphragmatic breathing exercise needed to be taught to participants by a qualified trainee, or via audio recording.

The outcomes they considered were physiological and psychological stress. Physiological (bodily) stress was measured through blood pressure, breath rate, and cortisol. Psychological (mental) stress is measured by questionnaires, the main one used here being the Depression Anxiety Stress Scales-21 (DASS-21).

They used detailed exclusion criteria, employing independent reviewers and searching for both published and unpublished papers. Lastly, they also used software to “grade” the quality of the evidence found within the studies.

After starting with 766 studies, they only found 3 that met their criteria.

3. What Were the Major Findings?

The 3 studies included didn’t have enough participants and consistency to generalize the results to the broader public (this is often the case with breathing).

The studies also weren’t top tier. They ranged from low confidence: “*Our confidence in the effect estimate is limited: The true effect may be substantially different from the estimate of the effect,*” to moderate: “*We are moderately confident in the effect estimate: The true effect is likely to be close to the estimate of the effect, but there is a possibility that it is substantially different.*”

Nonetheless, the results were encouraging.

One study found that 8 weeks of diaphragmatic breathing significantly reduced cortisol, signifying a reduction in stress and inflammation.

Another study found significant decreases in blood pressure following one session of slow breathing, demonstrating a reduction in bodily stress.



Lastly, the third study found that diaphragmatic breathing significantly decreased DASS-21 scores over 9 months compared to a control group. This indicates that breathing improved the participants mental levels of stress and anxiety.

4. Why Do These Results Matter?

Negative bodily and emotional stress is detrimental in basically every situation. This review shows that diaphragmatic breathing significantly improves both of these measures of stress.

This is significant because diaphragmatic is free, easy to learn, and can be implemented anywhere (school, workplace, and so on). With practice, it can be self-administered, or we can use a simple phone app to guide us.

1 BIG TAKEAWAY

Slow diaphragmatic breathing helps reduce hard metrics (cortisol, blood pressure) and soft metrics (questionnaires) of stress. Because it was a systematic review, this paper is an excellent reference for any statements related to breathing and stress.

1 PRACTICAL APPLICATION

Because there were only 3 studies with different protocols, we cannot give explicit practical guidance from this paper. But, in general, they found that diaphragmatic breathing helped both in the short and long term.

So, do it any time you feel anxious or, better yet, use it regularly to help lower your baseline levels of stress.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Self-regulation of Breathing as a Primary Treatment for Anxiety

411 Released: March 17, 2023

Review > [Appl Psychophysiol Biofeedback](#). 2015 Jun;40(2):107-15.

doi: 10.1007/s10484-015-9279-8.

Self-regulation of breathing as a primary treatment for anxiety

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Affiliations + expand

PMID: 25869930 DOI: [10.1007/s10484-015-9279-8](#)

[Get the Paper Here](#)

Article Citation

Jerath R, Crawford MW, Barnes VA, Harden K. Self-regulation of breathing as a primary treatment for anxiety. *Appl Psychophysiol Biofeedback*. 2015 Jun;40(2):107-15. doi: 10.1007/s10484-015-9279-8. PMID: 25869930.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

This 2015 paper proposes that slow deep breathing is one of the simplest and most effective ways to reduce negative emotions. By quickly activating the parasympathetic nervous system, it may reduce cell excitability and thus reduce the body's physiological capability to experience negative emotions.



4 Fundamentals

1. Essential Background Material

- Although emotions come from the brain, they are intricately tied to our entire body via our autonomic nervous system (ANS) and breathing.
- Emotional states of stress, anxiety, and depression are characterized by the activation of the sympathetic nervous system and faster breathing. In contrast, relaxed states activate the parasympathetic nervous system and result in lower heart and breath rates.
- This is also a two-way street: the way we breathe can activate our sympathetic or parasympathetic branches, influencing our emotions.
- However, the underlying mechanism behind these physiological aspects of emotions is not well understood.
- This study proposes that changes in ANS state (e.g., sympathetic vs. parasympathetic) change cell membrane potentials.
- This is a fancy way of saying that changes in ANS change the “excitability” of cells throughout the whole body. Stressed states and sympathetic activation make cells easily excitable. Relaxed states and parasympathetic activation make cells less excitable.
- From this perspective, we see that breathing (or any relaxation technique) improves emotional states by activating the parasympathetic nervous system, thus changing whole-body membrane potential and reducing emotional excitability.
- Because pharmacological interventions typically target only one neurotransmitter system, they miss these whole-body changes accompanying emotional distress. For this reason, these authors propose that mind-body methods (like slow deep breathing) might be the best option for dealing with negative emotions.

2. What Did this Research Do?

This was a hypothesis paper. They reviewed several ideas about breathing, emotions, the nervous system, and membrane potentials to support the abovementioned hypothesis.



The two main topics they covered were:

1. Possible Role of Membrane Potential in Cardiorespiratory Modulation of ANS and Emotions
2. Breathing and Meditation Techniques for Reducing Negative Emotions

Let's look at some of the neat things they discovered and hypothesized.

3. What Were the Major Findings?

Cardiorespiratory Synchronization

One way breathing and emotions are linked is through cardiorespiratory synchronization, where messages between the heart and breathing harmonize. This occurs during meditation and voluntary slow breathing (usually less than 10 breaths a minute).

They hypothesize that this synchronization leads to “widespread hyperpolarization,” which makes cells in emotional brain regions (like the amygdala) less excitable. This is one way in which meditation and slow breathing reduce negative emotions.

GABA Neurons

GABA (gamma-aminobutyric acid) receptor neurons are crucial to the cardiorespiratory system. These receptor neurons make cells in the amygdala less excitable. Because practices like breathing, yoga, and chanting all increase GABA, this provides another mechanism by which they influence cell excitability and reduce negative emotions. However, contradictory results were presented, so further evidence is needed to better understand this link.

Stress, Anxiety, and the Amygdala

Another way to look at breathing and emotions is to look at what happens during stressed states. In this condition, the activity of the amygdala is increased, and things like heart rate, blood pressure, and breathing rate all go up. They propose that this is due to “widespread depolarization” that changes cell membrane potential and makes cells more excitable.

This matters because slow deep breathing reverses this trend, decreasing amygdala activity and improving overall emotional states.

RAPs: “Respiration-Related Anxiety Potentials”

During inhalation, we create “waves of positive potentials”—called RAPs—in emotional areas of the brain. During faster breathing, these waves may actually “depolarize” these emotional brain regions, making them more excitable. By being in a sympathetic state, we breathe faster, increasing RAPs and ultimately destabilizing emotional areas of the brain.

Conversely, breathing slowly reduces the impact of RAPs and stabilizes emotional brain regions.



4. Why Do These Results Matter?

Given the above findings that link the breath to the ANS and amygdala, we see how voluntary slow deep breathing could be highly beneficial for controlling our emotional state. In fact, the authors even say, **“it is apparent that individuals possess the ability to alter emotional states using the voluntary control of breathing and mindset.”**

This is a genuinely life-changing statement. It provides you and me with self-efficacy, with the ability to take control of our emotional ups and downs using only our breath.

From a more mechanistic perspective, their results are significant because they say:

1. Stressful emotional states are associated with sympathetic activation and increased “cell excitability” in emotional regions of the brain.
2. Slow deep breathing activates the parasympathetic branch, increases cardiorespiratory harmony, and reduces “cell excitability” in emotional regions of the brain.

Together, this means slow breathing may physiologically prevent (or at least reduce) negative emotions from occurring.

1 Big Takeaway

Slow deep breathing is one of the simplest and most effective ways to reduce negative emotions. It quickly activates the parasympathetic nervous system, reducing the body’s physiological capability to experience negative emotions.

1 Practical Application

Regularly practice slow deep breathing to activate your parasympathetic nervous system and reduce your likelihood of experiencing negative emotions. Almost any method that gets your breathing down to less than 10 breaths a minute will be helpful. Techniques focusing on extended exhalations and/or short pauses after inhaling will likely be most beneficial.



They specifically mention Sudarshan Kriya yogic (SKY) breathing, which uses the 4-4-6-2 rate ([we recently covered here](#)), and ujjayi. SKY also uses short periods of fast breathing, which the authors hypothesize could be beneficial when combined with slow breathing. [See this Science 411](#) for a simple approach similar to the SKY sequence.

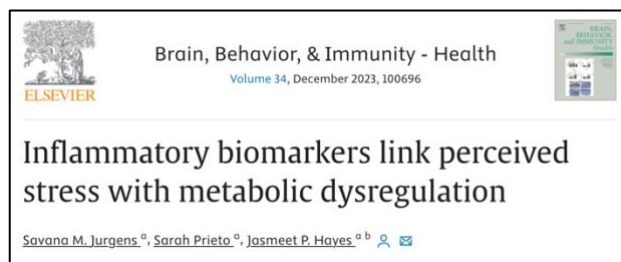
Lastly, note that consistency is key here. The breath-emotion link they're proposing ultimately comes back to rewiring our nervous systems. So, let's remember [Dr. Weil's advice](#) for breathing: **"You want to do this every day without fail because you are attempting to change rhythms in your nervous system, and it's the constancy of the input, it's the regularity of the input, that is going to produce these changes over time."**

Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Inflammatory Biomarkers Link Perceived Stress with Metabolic Dysregulation

411 Released: April 7, 2024



[Get the Paper Here](#)

Quick Summary

This 2023 analysis suggests (but doesn't directly prove) that higher perceived stress levels lead to chronic inflammation, which then leads to metabolic dysfunction. Stress-reduction techniques may, therefore, be an accessible and effective way to reduce inflammation and the risk of metabolic syndrome.

Article Citation

Jurgens SM, Prieto S, Hayes JP. Inflammatory biomarkers link perceived stress with metabolic dysregulation. Brain Behav Immun Health. 2023 Oct 17;34:100696. doi: 10.1016/j.bbih.2023.100696. PMID: 37928770; PMCID: PMC10623170.

Reading Time: 2 min 37 seconds

I hope the next 39'ish breaths are the most nourishing of your day.



4 Fundamentals

1. Essential Background Material

Perceived stress—the extent to which a person sees life experiences as stressful—adversely impacts health and contributes to metabolic syndrome. Metabolic syndrome refers to a cluster of risk factors, including insulin resistance, abnormal blood lipids, central adiposity, and high blood pressure, and is a predictor of health problems like heart disease and type-2 diabetes.

One way perceived stress may cause metabolic dysfunction is chronic inflammation via overactivation of the body's sympathetic nervous system. However, this indirect relationship (stress --> inflammation --> metabolic dysfunction) has not shown up consistently in research.

Therefore, this paper aimed to analyze if inflammation is a dominant pathway linking perceived stress to metabolic dysfunction. This relationship is critical because lifestyle changes can reduce perceived stress, potentially making it an accessible way to help metabolic syndrome.

2. What Did this Research Do?

Researchers used data from another project (the MIDUS project) in which participants had taken a perceived stress scale (PSS) questionnaire and had a blood draw and physical exam. For this analysis, they excluded people with diabetes or cardiovascular disease. **The final sample size was 648, with 52% female and a mean age of 52.**

Inflammation was assessed by five biomarkers: c-reactive protein, IL-6, E-selectin, fibrinogen, and ICAM-1. A composite score of all these was created to represent overall inflammation.

To assess metabolic function, they analyzed blood pressure, BMI and waist circumference, lipids (cholesterol and triglycerides), blood sugar, and blood insulin levels, representing the four metabolic syndrome factors mentioned above.

With this physiological data and the PSS stress scores, they used “structural equation modeling” and “mediation analysis” to test whether the link between perceived stress and metabolic syndrome can be explained by inflammation.



3 & 4. What Were the Major Findings and Why Do They Matter?

There was a small but significant direct effect of perceived stress on metabolic syndrome.

This means more perceived stress is associated with metabolic dysfunction.

Main Result: Higher perceived stress levels were associated with greater levels of inflammation, which were then associated with worse metabolic measures.

This suggests that inflammation is a key pathway through which perceived stress leads to metabolic dysfunction: Stress causes low-grade chronic inflammation, increasing metabolic syndrome risk.

Inflammation explained 61.5% of the connection between stress and metabolic dysfunction.

This means inflammation is a dominant pathway through which perceived stress leads to metabolic dysfunction. But other factors are also important (for example, stress may cause poor lifestyle choices [food, sleep] that increase the risk of metabolic dysfunction).

These results imply a relationship between stress and metabolic dysfunction through inflammation; however, causation cannot be inferred through this approach.

Structural equation modeling and mediation analysis are statistical approaches to map measured variables to underlying causes and then test whether one variable (in this case, inflammation) can help explain the relationship between two other variables (stress and metabolic dysfunction). It suggests the relationship exists but cannot establish direct causation.

1 Big Takeaway

This analysis suggests that higher perceived stress levels lead to chronic inflammation, which then leads to metabolic dysfunction. Stress-reduction techniques may, therefore, be an accessible and effective way to reduce inflammation and the risk of metabolic syndrome.



1 Practical Application

For Yourself:

Use slow breathing, mindfulness, or any relaxation technique you enjoy to lower your perceived stress and (potentially) improve metabolic health. [Here are a few you can try today.](#)

For the People You Work With:

This is a highly significant study for helping others. For example, implementing the PSS scale in your coaching or courses (it must be 4+ weeks long to work) is very simple and can be administered in a Google Form. [Here are the questions and how to grade the responses.](#) And here is [a Google Form I have used](#) in my own teaching that you can copy and modify as needed.

Moreover, if people improve their PSS, this study suggests that they might also reduce inflammation and improve overall metabolic health, giving the results more meaning.



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Citation

Kox M, van Eijk LT, Zwaag J, van den Wildenberg J, Sweep FC, van der Hoeven JG, Pickkers P. Voluntary activation of the sympathetic nervous system and attenuation of the innate immune response in humans. Proc Natl Acad Sci U S A. 2014 May 20;111(20):7379-84. doi: 10.1073/pnas.1322174111. Epub 2014 May 5. PMID: 24799686; PMCID: PMC4034215.

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December 27, 2022

Additional Note

I've called this study the "4-minute mile" of breathing because it showed that something previously thought "impossible" is possible.

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Voluntary activation of the sympathetic nervous system and attenuation of the innate immune response in humans

4 FUNDAMENTALS

1. Essential Background Material

Our immune systems are vital to our survival. But, in autoimmune diseases, overactivation of the innate immune response can be detrimental, causing problems such as chronic inflammation.

Somewhat unexpectedly, short-term, intense sympathetic nervous system activation (SNS; fight-or-flight branch) can reduce some of the harmful effects of inflammation. This seems counterintuitive—doesn't stress cause inflammation? Yes, chronic stress can. But short-term SNS activation can actually reduce it.

The problem is that activating the SNS and influencing our immune system is thought to be out of our conscious control. This means we can't just "turn it on at will" to reduce inflammation. But Wim Hof challenged this assumption.

As you might know, he proved that he could do precisely this using his method. Scientists assumed he was just an anomaly, but Wim insisted he wasn't. So they put it to the test. This study documents if ordinary people can learn Wim's techniques and achieve similar results in a randomized and controlled fashion.

2. What Did this Research Do?

Thirty healthy male volunteers were recruited for the study. Eighteen were



assigned to the Wim Hof group (I'll abbreviate it as WHM group), and 12 were in the control group. Note that only 12 WHM participants took part in the injection of the endotoxin described below. The others were there as backups.

The WHM Training

The WHM group trained with Wim and a few instructors in the method for just 4 days (*that's crazy*). The core training included the 3 pillars of Wim's method: breathing, cold, and meditation. They also did some of the other standard Wim Hof training: push-ups, yoga poses, horse stance, etc. Their training concluded with climbing a mountain wearing only shorts.

After the 4 days, they were told to practice the method on their own every day until the experiment, which occurred 5-9 days later.

The Endotoxin Experiment

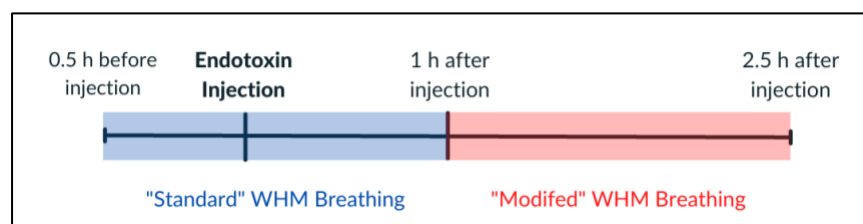
The experiment involved injecting all participants with an endotoxin that has been given to hundreds of people, most of which develop flu-like symptoms (fever, headache, nausea, etc.).

The control group simply got the endotoxin injection, and their symptoms were monitored and recorded.

The WHM group, however, performed WHM breathing before injection and during exposure. Here are the two exercises they performed:

- **Standard WHM Breathing:** 30 deep breaths → maximum breath hold after exhalation → inhale and hold for 10 seconds → repeat.
- **Modified WHM Breathing:** 30 deep breaths → inhale fully → hold for 10 seconds and tighten all muscles. Repeat. (This technique was not explained well, so this is my best guess based on other references.)

They started the standard WHM breathing 30 minutes before injection and continued it for 1.5 hours. Then, they began the modified WHM breathing for another 1.5 hours. **That's 3 total hours of WHM breathing.**



The researchers measured inflammatory cytokine levels, fever, blood gases, and flu-like symptoms.



3. What Were the Major Findings?

The WHM participants suppressed and/or modified most physiological and subjective symptoms of the endotoxin injection.

They significantly suppressed proinflammatory cytokines (51-57% reduction) and increased anti-inflammatory ones (194% increase).

The participants also had massive increases in adrenaline, which correlated with these changes in cytokines. The authors even state that the rise in adrenaline was more than that seen in bungee jumpers (Wim says this a lot on podcasts, and it's right here in the PNAS study).

The WHM group also reduced their fever and the number of symptoms they felt. At the peak, symptom scores reduced by ~56% compared to controls.

Lastly, during the breath retentions, WHM participants dropped their blood oxygen saturation down to about 50% (normal is 95% or higher).

4. Why Do These Results Matter?

This is the 4-minute mile of breathing.

For the first time, it is shown that ordinary people can use their breath to control their innate immune response and reduce acute inflammation.

Moreover, the results showed that participants could also reduce fever and subjective symptoms, meaning they felt better during the exposure, too.

Most critically, the administration of endotoxin is how some drugs (biologics) are first tested for autoimmune diseases. **Thus, Wim's method passed the same initial test that drugs must pass, using only the breath and mind.**

However, more studies are needed to assess whether Wim's approach can cause lasting reductions in chronic inflammation (spoiler alert: it can—learn more here).

1 BIG TAKEAWAY

With only 4 days of training, Wim Hof's method allows ordinary people to consciously influence their sympathetic nervous system and innate immune response, giving them the power to reduce inflammation and flu-like symptoms associated with the injection of endotoxin.



1 PRACTICAL APPLICATION

Although the “1 Practical Application” is my favorite section, this one is challenging.

First, you need to work with a certified Wim Hof instructor to ensure you perform his method safely.

Second, we don’t have endotoxin to inject ourselves with 😊. You also probably don’t have 3 h you can spend hyperventilating each day.

Third, the participants were trained in the complete method (breathing, cold, meditation) but only performed the breathing during the experiment. So, it’s unclear whether breathing is most critical, or if the benefits were due to the previous training in all three pillars.

With those caveats stated, these findings might suggest that using Wim’s approach could help you feel better when you have cold/flu symptoms (or perhaps prevent them altogether if caught early enough). Remember, they started the breathing and increased adrenaline before injection, which helped suppress the symptoms from the start.

Moreover, if you have an autoimmune disease, these results suggest that practicing Wim’s method may reduce some of the detrimental effects of inflammation associated with the overactivation of the immune response.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Laughter as Medicine: Laughing Reduces Cortisol by More than 30%

411 Released: September 15, 2023



[Get the Paper Here](#)

Article Citation

Kramer CK, Leitao CB. Laughter as medicine: A systematic review and meta-analysis of interventional studies evaluating the impact of spontaneous laughter on cortisol levels. PLoS One. 2023 May 23;18(5):e0286260. doi: 10.1371/journal.pone.0286260. PMID: 37220157; PMCID: PMC10204943.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2023 meta-analysis analyzed 8 studies and found that spontaneous laughter reduced cortisol by 31.9%, making it a potentially powerful tool for improving wellness.



4 Fundamentals

1. Essential Background Material

- Laughter is essential to human life and has been recognized as medicine for centuries.
- In fact, laughing is so critical to human life that the ability to laugh develops before the neural development of speech 🧠
- From a wellness perspective, laughter is effective for cardiovascular health, pain, diabetes, oncology, and rehabilitation.
- One way laughter may be helpful for so many conditions is through its effects on the neuroendocrine system, specifically the hypothalamic-pituitary-adrenocortical (HPA) axis, which releases stress hormones in response to physical or psychological stressors.
- Despite the anecdotal and scientific evidence supporting laughter as medicine, no reviews have examined how it impacts the HPA axis.
- To this end, this systematic review and meta-analysis analyzed how laughter affects cortisol, one of our primary stress hormones related to the neuroendocrine system.

2. What Did this Research Do?

This systematic review and meta-analysis collected studies published between 1950 and April 2022. Notably, their inclusion criteria were:

1. It must have been either a randomized-controlled trial or quasi-experimental, meaning that the participants may not have been randomly assigned or that the analysis was done on a pre-existing group of participants who had taken part in a laughter protocol.
 2. Compared to an intervention that didn't use laughter or a control group.
 3. Measured cortisol.
-



Overall, 8 studies with a total of 315 participants were included. The laughter interventions included watching a funny video (n=5), laughter therapy by an instructor (n=2), or self-administered laughter (n=1).

3 & 4. What Were the Major Findings and Why Do They Matter?

Pooled across all 8 studies, laughing reduced cortisol by a statistically significant 31.9%.

- Laughter has been shown to help many illnesses and promote better well-being. Thus, the above result is the first to provide an objective measure explaining these benefits, showing that laughter significantly reduces our body's primary stress hormone, cortisol.
- Laughter also promotes and communicates cooperation and positive feelings across age groups and cultural backgrounds, making this result potentially universally applicable for promoting health and wellness.

Even a single laughter session can reduce cortisol by an average of 36.7%.

- Five of the eight studies used a single laughter session ranging from 9 to 60 minutes. Across these five, laughing reduced cortisol by 15-63%. When analyzed together, they found the above-stated average of almost 37%. This is meaningful because it shows that we can significantly reduce stress almost immediately with laughter.

The method of inducing laughter did not affect these results, meaning that using a comedy video or participating in laughter therapy significantly reduced cortisol.

- They pooled together the studies by intervention type. They found that watching a funny video reduced cortisol by 36.7%, while laughter therapy reduced it by 18.9%. Both of these were statistically significant.
- This is meaningful because it shows that (similar to what we learned about breathing) the mode of delivery is less important than doing the act itself.

Although there wasn't much evidence of publication bias, the overall grade of the data was moderate/low, meaning the data itself was not entirely free of biases.

- This is important because it means that, although the results are meaningful, they do not meet the criteria for being used for clinical recommendations (they're not robust enough to give doctors laughter protocols they can prescribe to patients).

P.S. From the paper: "The author(s) received no specific funding for this work."



1 Big Takeaway

Across studies, spontaneous laughter reduces cortisol by over 30%, making it a potentially powerful tool for improving wellness.

1 Practical Application

I became interested in laughter for wellness about a year ago, so I'm going to provide three ways I've found helpful:

1. **Spotify:** You can just search "stand-up comedy" and hit shuffle. They're not all hilarious, but usually enough to make you laugh.
 2. **Comedy Videos:** Watch stand-up from comedians you enjoy or watch a funny movie.
 3. **Train Instagram (my favorite):** If you use social media, you can use the algorithm to your advantage. By spending most of my time looking at funny things on Instagram, it now only shows me funny videos and memes. Sometimes, I deliberately spend 10-15 on Instagram just to get a good laugh. This takes some time, but it is well worth it.
-



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Vagal Mediation of Low-Frequency Heart Rate Variability During Slow Yogic Breathing

411 Released: March 3, 2023

[Psychosom Med.](#) 2018 Jul/Aug;80(6):581-587. doi: 10.1097/PSY.0000000000000603.

Vagal Mediation of Low-Frequency Heart Rate Variability During Slow Yogic Breathing

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Affiliations + expand

PMID: 29771730 DOI: [10.1097/PSY.0000000000000603](#)

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Article Citation

Kromenacker BW, Sanova AA, Marcus FI, Allen JJB, Lane RD. Vagal Mediation of Low-Frequency Heart Rate Variability During Slow Yogic Breathing. *Psychosom Med.* 2018 Jul/Aug;80(6):581-587. doi: 10.1097/PSY.0000000000000603. PMID: 29771730.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

By separately blocking the parasympathetic and sympathetic nervous systems during slow breathing, a 2018 study in *Psychosomatic Medicine* conclusively found that observed increases in HRV during slow breathing are primarily due to increased vagal activity.



4 Fundamentals

1. Essential Background Material

- Slow yogic breathing (which I'll just call "slow breathing") has consistently been shown to reduce stress, anxiety, and depression and lower inflammation and blood pressure.
- These benefits are thought to be—at least partially—the result of better vagal tone (**more activity of the parasympathetic rest-and-digest nervous system [PNS]**).
- However, the way vagal tone is measured (via heart rate variability [HRV]) introduces uncertainty to this interpretation for two reasons:
 - i. **Low-frequency HRV (LF-HRV) usually increases significantly during slow breathing.** But the interpretation of LF-HRV is not concrete: some studies suggest that increases in LF-HRV could indicate *increased sympathetic (fight-or-flight) activity [SNS]*.
 - ii. A problem with LF-HRV measurements is that **the breathing rate itself interferes.** When LF-HRV increases, it's unclear how much is due to increases in vagal activity and how much is an artifact of the breathing rate itself.
- Vagal nerve stimulation is used for many health conditions (and slow breathing is often even called "vagal stimulation"). Thus, from a treatment perspective, it's vital to know if these slow breathing rates are, in fact, stimulating the vagus nerve.
- **This study used a neat approach to assess this directly.**

Nick's Sidenote: Even though this study was focused on LF-HRV, which may be of no interest to you, the methods they used were awesome and the results are super important to anyone interested in how slow breathing works.

2. What Did this Research Do?

This study had 6 participants (4 male, 2 female). The procedure was pretty intensive and required a smaller sample size. Still, this is a limitation, but the intervention produced substantial effects, making this a significant study despite the small sample.



Procedure

The participants rested for 10-20 minutes. Then, they were injected with one of three drugs:

1. Glycopyrrolate: Blocks activity of the parasympathetic nervous system
2. Esmolol: Blocks activity of the sympathetic nervous system
3. Saline: Placebo

Next, they performed 11 different slow breathing rates, ranging from 4 to 9 breaths a minute, using 0.5-step increments. Although the breathing rates varied, they used the same ratio of 4-4-6-2 (inhale 4, hold 4, exhale 6, hold 2), which is commonly used in Sudarshan Kriya Yoga.

They performed each breathing rate for 1 minute, followed by 4 minutes of normal breathing.

Example: They took glycopyrrolate and slowly breathed at five breaths a minute. HRV was measured to see if it went up despite the blocked PNS.

3. What Were the Major Findings?

Blocking the Sympathetic Nervous System

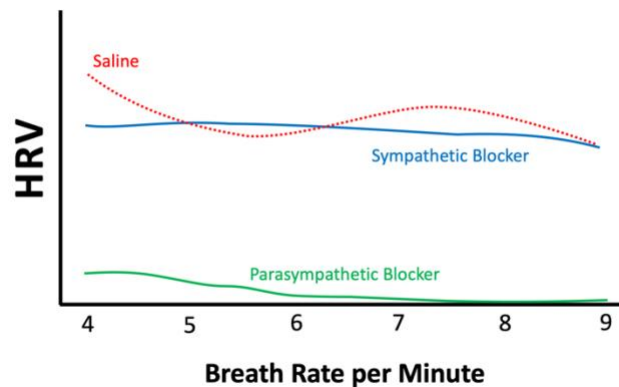
When they blocked the SNS, HRV was virtually unchanged compared to the placebo saline solution. This means the SNS plays a minimal role in HRV measurements during slow breathing.

Blocking the Parasympathetic Nervous System

HRV reduced significantly (basically to zero) when they blocked the PNS regardless of which breathing rate was used. This means the PNS (and thus vagal activity) is the primary reason HRV increases during slow breathing.

One Exception

At the slowest rate of 4 breaths a minute, there was a slight (but significant) increase in HRV even when the PNS was blocked. This indicated that the sympathetic nervous system contributes a small amount to HRV readings when breathing is less than 5 breaths a minute.



HRV plots during slow breathing at different rates and under different nervous system blockades. This is NOT REAL DATA. It's adapted from Figure 2 of this paper, Vagal Mediation of Low-Frequency Heart Rate Variability During Slow Yogic Breathing. Psychosom Med. 2018.

4. Why Do These Results Matter?

Together, these findings conclusively show that increases in HRV during slow breathing result from increased vagal activity, meaning slow breathing is indeed “vagal nerve stimulation.”

To reiterate the results, when the PNS was blocked, HRV basically went to zero. But when the SNS was blocked, HRV was basically unchanged. This means that when we see HRV increase during slow breathing, it's almost entirely due to increased vagal activity. That's super neat.

1 Big Takeaway

By separately blocking the parasympathetic and sympathetic nervous systems during slow breathing, this study conclusively found that observed increases in HRV during slow breathing are primarily due to increased vagal activity.



1 Practical Application

Practice slow breathing to stimulate your vagus nerve and increase the health of your cardiovascular and autonomic nervous systems.

More specifically, consider trying this 4-4-6-2 ratio to see how it makes you feel. I've been practicing it for 10 minutes before bed and find it very helpful for falling asleep.

You can easily program the 4-4-6-2 rate using the Breathe app:

[Get it for Apple.](#)

[Get it for Android.](#)



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Citation

Laborde S, Hosang T, Mosley E, Dosseville F. Influence of a 30-Day Slow-Paced Breathing Intervention Compared to Social Media Use on Subjective Sleep Quality and Cardiac Vagal Activity. J Clin Med. 2019 Feb 6;8(2):193. doi: 10.3390/jcm8020193. PMID: 30736268; PMCID: PMC6406675.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Influence of a 30-Day Slow-Paced Breathing Intervention Compared to Social Media Use on Subjective Sleep Quality and Cardiac Vagal Activity

4 FUNDAMENTALS

1. Essential Background Material

This study examined how 30 days of slow breathing before sleep affected subjective sleep quality and nighttime and morning vagal tone.

Good sleep is the foundation of health. And although I'm not a big fan of meticulously monitoring our sleep or getting bent out of shape over a few bad nights, I am a fan of *doing everything we can to set ourselves up for good sleep.*

And slow breathing before bed might be one such tool for this. Overarousal and overstimulation are two key drivers of poor sleep. **Slow breathing counters these by activating the parasympathetic (rest-and-digest) nervous system.**

Although the connection is straightforward, little research has been done on this topic, and only a few studies exist looking at the impact of nighttime slow breathing on sleep.

This study aimed to fill this gap by analyzing how slow breathing before bed impacted subjective sleep quality and, subsequently, how those changes were related to nighttime and morning vagal tone.



2. What Did this Research Do?

Sixty-four people (31 female; 33 male) between 18-29 years old took part in the study.

Before sleep, subjects in the breathing group performed **15 minutes of slow breathing at 6 breaths per minute using a 4.5-second inhale and 5.5-second exhale.** They used a nasal inhale and pursed lips exhale. They did this for 30 days and could not miss more than 3 sessions to be included in the study.

A control group looked at social media for 15 minutes before going to bed. I'm not going to consider this group—the focus here is on slow breathing itself, not on how slow breathing compares to social media.

To measure sleep quality, they used the Pittsburgh Sleep Quality Index (PSQI), which consists of 18 questions to assess different aspects of sleep.

To measure vagal tone, they used high-frequency HRV (HF-HRV). Recall from a previous [Laborde et al. paper](#) that RMSSD is typically the best marker of vagal tone. However, HF-HRV works here because the breathing rate was greater than 9 breaths/min when the measurements were taken (overnight and morning).

Measurements were taken one night at the start of the experiment (pre-test) and one night at the end of the experiment (post-test). Nighttime vagal tone was assessed during sleep, whereas morning vagal tone was measured first thing upon waking. These readings are comparable to what many people measure every day using Oura rings, Whoop straps, etc.

Importantly, no breathing was performed on the nights of the measurements. So, any observed changes in vagal tone indicate changes in resting autonomic nervous system activity.

3. What Were the Major Findings?

The take-home findings were that 15 minutes of slow breathing before bed:

- Significantly improved subjective sleep quality on the PSQI scale.
- Significantly increased nighttime vagal tone measured via HF-HRV.
- Increased (but not significantly) morning vagal tone.

Moreover, increases in vagal tone were correlated with changes in sleep quality: As vagal tone increased, sleep quality got better. This is associative, not causal, but still interesting nonetheless.



4. Why Do These Results Matter?

These results are meaningful because sleep is essential to basically every biological function, and here we learn that doing 15 minutes of slow breathing provides a free and effective way to improve sleep quality.

Moreover, the increase in morning and nighttime vagal tone is meaningful because vagal tone isn't just "vagal tone." It's better cognitive function, better social skills, better emotional and stress regulation, and better overall health.

1 BIG TAKEAWAY

Fifteen minutes of slow breathing (6 breaths per minute) before bed for 30 days improves sleep quality and increases vagal tone.

1 PRACTICAL APPLICATION

To implement the findings of this study in your life, aim to perform 15 minutes of slow breathing before bed. Use a slightly longer exhale than inhale, [which will aid in increasing vagal activity](#). This study used a 4.5-second inhale and 5.5-second exhale, but we can use 4:6 (or a variation that's comfortable for you) to make it easier on most breathing apps.

Moreover, my hunch is that basically any ratio focused on parasympathetic activation will work, so if you enjoy the 4-7-8 breath or another alternative, it's reasonable to assume you'll still achieve similar results.

Enjoy sleeping and feeling better with this simple nighttime routine.



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Citation

Laborde S, Iskra M, Zammit N, Borges U, You M, Sevoz-Couche C, Dosseville F. Slow-paced breathing: influence of Inhalation/Exhalation ratio and of respiratory pauses on cardiac vagal activity. Sustainability. 2021 Jul 12;13(14):7775.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Slow-Paced Breathing: Influence of Inhalation/Exhalation Ratio and of Respiratory Pauses on Cardiac Vagal Activity

4 FUNDAMENTALS

1. Essential Background Material

This study examined the impacts of inhalation:exhalation ratio and breathing pauses on vagal activity using heart rate variability (HRV) measurements.

As you probably know by now, slow breathing has consistently been shown to increase HRV. However, two aspects of it are still poorly understood:

1. Should the inhale and exhale be equal, or should the exhale be longer?
2. Does adding a pause after inhalation or exhalation increase HRV?

For number 1, we know that the parasympathetic nervous system controls exhalation. **Thus, extending the exhale should stimulate it more, resulting in more vagal activity.** However, study results have been mixed, which, according to these authors, might be due to misinterpretation of HRV measurements.

For number 2, it's hypothesized that a pause might increase vagal activity, as heart rate generally slows down when we pause our breath. However, again results have been mixed.

Therefore, this study aimed to fill these gaps by examining how the inhale:exhale ratio and respiratory pauses impact vagal activity as measured by HRV (specifically, RMSSD).



2. What Did this Research Do?

The authors recruited 64 athletes (27 female; 37 male). They had them perform 6 slow breathing exercises randomly for 5 minutes each. There was also a 5-minute washout period in between each exercise.

The 6 exercises they tested were:

1. Equal
 - a. 5-sec inhale; 5-sec exhale
 - b. 4.6-sec inhale; 0.4-sec pause; 4.6-sec exhale; 0.4-sec pause
2. Extended Exhale
 - a. 4.5-sec inhale; 5.5-sec exhale
 - b. 4.1-sec inhale; 0.4 sec pause; 5.1-sec exhale; 0.4-sec pause
3. Extended Inhale
 - a. 5.5-sec inhale; 4.5-sec exhale
 - b. 5.1-sec inhale; 0.4-sec pause; 4.1-sec exhale; 0.4-sec pause

The primary outcome measurement was HRV measured with RMSSD.

According to the researchers, RMSSD is the best indicator of vagal activity because it is not influenced by breathing rate. Thus, if RMSSD increases, this is due to an increase in vagal activity, not some kind of confounding interaction with the breathing rate (as can happen with LF-HRV or HF-HRV power).

This is what sets this study apart from others: it's the first to focus on RMSSD and thus the first to truly focus on vagal activity.

3. What Were the Major Findings?

There were two critical findings.

First, RMSSD (and thus vagal activity) increased most significantly during extended exhalation breathing. However, it is important to note that all the slow breathing practices increased RMSSD compared to baseline.

Second, adding slight pauses did not impact the results. However, the pauses might have been too short (only 0.4 seconds) to have any noticeable impacts.



4. Why Do These Results Matter?

This study confirms that extending the exhalation increases the activity of the vagus nerve. This made sense in theory; however, these results provide concrete evidence of increased vagal activity through RMSSD measurements.

1 BIG TAKEAWAY

Slow breathing with extended exhalations increases vagal activity more than an equal inhale:exhale ratio. Thus, we can extend our exhales to increase parasympathetic nervous system activity.

1 PRACTICAL APPLICATION

Use a slightly longer exhale than inhale for deeper relaxation during a slow breathing practice. **The most common approach is the “40:60 Rule”: 40% of your breath should be inhaling, and 60% should be exhaling.**

Thus, regardless of which rate you breathe at (6/5/4 breaths per minute), you can use the 40:60 rule to lengthen the exhalation and increase vagal activity.

And if you don't know which rate to start at, use this excellent advice from Leah Lagos, Psy.D., in [Heart Breath Mind](#): *“The standard rate that works for most individuals is to inhale for 4 seconds and exhale for 6 seconds with no pause in between.”*

Simple, easy, and highly effective.



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Citation

Laborde S, Allen MS, Borges U, Dosseville F, Hosang TJ, Iskra M, Mosley E, Salvotti C, Spolverato L, Zammit N, Javelle F. Effects of voluntary slow breathing on heart rate and heart rate variability: A systematic review and a meta-analysis. *Neurosci Biobehav Rev.* 2022 Jul;138:104711. doi: 10.1016/j.neubiorev.2022.104711. Epub 2022 May 24. PMID: 35623448.

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July 21, 2022

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Effects of voluntary slow breathing on heart rate and heart rate variability: A systematic review and a meta-analysis

4 FUNDAMENTALS

1. Essential Background Material

This systematic review and meta-analysis examined the effects of slow breathing on heart rate variability, a key marker of overall health and wellness.

Slow breathing has been used for millennia to reduce stress and improve physical and mental health. However, its underlying mechanisms are actually poorly understood.

In general, **we think most of the benefits come from stimulation of the vagus nerve**, the primary nerve of the parasympathetic (rest-and-digest) nervous system.

Heart rate variability (HRV) is a way to measure this vagal activity. Thus, if slow breathing works through the vagus nerve, it should be detectable via HRV.

And, of course, we know that HRV has been used in a ton of studies on slow breathing. However, this information has not been synthesized to examine how slow breathing impacts HRV at different time intervals across experiments.

This study filled that gap by generating a systematic review and meta-analysis on the impact of slow breathing on HRV at three time intervals.

1. During the practice (which they label DURING),
2. Immediately after one session (IM-AFTER1),
3. After a multi-session intervention (AFTER-INT).



2. What Did this Research Do?

The authors collected studies of voluntary slow breathing that measured HRV at one of the three time intervals mentioned above.

IMPORTANT

The introduction provided one of the best descriptions of the various HRV parameters I've read. It's too much to include, but here are the key takeaways: RMSSD and LF power can be used as a marker of vagal activity for the "DURING" measurements. For IM-AFTER1 (immediately after) and AFTER-INT (after multiple sessions), only RMSSD can be used. These latter two measurement times can be considered baseline vagal activity.

For each study, they calculated the risk of bias. They then used the Hedges' g score to evaluate the overall changes due to slow breathing.

According to the authors, if the Hedges' score is positive, the slow breathing group had higher HRV than the control group. Moreover, there are pre-defined g-value thresholds to see if the effects are "small," "medium," or "large."

223 studies were included in the analysis:

- 172 for DURING
- 16 for IM-AFTER1
- 49 for AFTER-INT

3. What Were the Major Findings?

Like almost all breathing studies, the risk of bias was high in most of the 223 included papers. Nonetheless, the results were encouraging.

For DURING:

- There was a "medium" effect size for RMSSD ($g = 0.53$)
- There was a "large" effect size for LF ($g = 1.49$)

This indicates that, across the 172 studies included for DURING, **slow breathing significantly increased cardiac vagal activity during the breathing practice.**



For IM-AFTER1:

- There was a “small” effect for RMSSD.

This indicates that, immediately after the practice, vagal activity is still increased, but not as significantly as during the exercise (this is expected).

For AFTER-INT:

- There was a “small” effect for RMSSD (but larger than for IM-AFTER1)
- The Hedges’ g score for AFTER-INT = 0.32 and IM-AFTER1 = 0.14.

This indicates that long-term practice increases baseline vagal activity.

4. Why Do These Results Matter?

Across all time points considered, slow breathing increased cardiac vagal tone. This supports the notion that the vagus nerve is the underlying mechanism behind the benefits of slow breathing.

Because the parasympathetic nervous system is critical in many chronic and acute diseases, **these results “encourage” the use of slow breathing in health-related settings. It’s cheap, easy, and basically has no adverse side effects.**

1 BIG TAKEAWAY

Slow breathing works through activation of the vagus nerve, and consistent practice of slow breathing improves overall cardiac vagal tone, which could be highly beneficial in many health conditions.

1 PRACTICAL APPLICATION

Because there were 223 studies with different protocols, we cannot provide explicit guidance from this paper. But, in general, slow breathing protocols are performed using a pace of 4 to 7 breaths per minute (6 bpm is most common) using a slightly longer exhale than inhale.

The most straightforward application is to perform slow breathing at 6 breaths per minute using a 4-second inhale and a 6-second exhale. Do this for at least 10 minutes daily to improve vagal tone both immediately and over the long run.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

The Influence of Slow-Paced Breathing on Executive Function

 FREE ACCESS

The Influence of Slow-Paced Breathing on Executive Function

S. Laborde , M. S. Allen, U. Borges, T. J. Hosang, P. Furley, E. Mosley, and F. Dosseville

Published Online: March 18, 2021 • <https://doi.org/10.1027/0269-8803/a000279>

[Get the Paper Here](#)

411 Released

February 17, 2023

Article Citation

Laborde, S., Allen, M. S., Borges, U., Hosang, T. J., Furley, P., Mosley, E., & Dosseville, F. (2021). The influence of slow-paced breathing on executive function. *Journal of Psychophysiology*.

Sharable: *Note: You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2022 study published in *J. Psychophysiology* found that 15-min of slow breathing:

- Improved attention control
- Improved working memory
- Increased cognitive flexibility

Take-home: slow breathing helps you plan, monitor, and execute your goals.



4 Fundamentals

1. Essential Background Material

- Executive function broadly refers to your ability to plan, monitor, and execute your goals.
- In general, executive function is comprised of the following:
 - **Inhibition:** The ability to resist impulses and distractions to do what you aim to do.
 - **Working Memory:** Working with information no longer directly in front of you—keeping information in your mind and working on it mentally.
 - **Cognitive Flexibility:** Being capable of changing your perspective, especially spatially or interpersonally.
- Theory suggests that your executive function is highly related to your vagal tone (measured via HRV RMSSD): **Higher vagal tone should lead to better executive function.**
- Slow breathing (< 10 breaths a minute) has consistently been shown to increase vagal tone, in both the short and long term.
- However, less is known about its specific ability to improve executive functioning.
- Thus, this study aimed to see if slow breathing could improve executive function and, if so, how those improvements were related to vagal tone.

2. What Did this Research Do?

This study had 78 participants (41 male, 37 female). They completed two different sessions, one week apart. Each session lasted a little over 1.5 hours.

During one session, the participants watched a neutral TV documentary (the control condition). During the other, they performed slow breathing (the experimental condition).

Here's a general description of the sessions:

1. Familiarization period: participants learned slow breathing or watched a TV documentary (11 minutes).



2. HRV measurement while resting (5 minutes)
3. Slow breathing or TV watching (17 minutes)
4. HRV measurement while resting (5 minutes)
5. Performed executive function tests (30 minutes)

Here's the Slow Breathing Exercise Used:

- Six breaths a minute using a 4.5-second inhale and 5.5-second exhale (slightly longer exhale used to increase vagal activity).
- Nasal inhale with pursed-lips exhale (they even mentioned the importance of nasal breathing for synchronizing brain wave activity 🧠).
- The 17-min breathing session (step #3 above) was performed like this: 5-min slow breathing → 1-min rest → 5-min slow breathing → 1-min rest → 5-min slow breathing

Measurements

Based on previous research, heart rate variability (measured using RMSSD) was used to quantify vagal tone. Executive function (inhibition, working memory, and cognitive flexibility) was assessed using clinically validated tests (listen to the audio version starting at the 5:00 min mark for more details).

3. What Were the Major Findings?

Overall changes in executive function:

- For the tests of inhibition skill, **their accuracy on the tests improved after slow breathing.**
 - However, the speed of their responses was not significantly different (it was slightly slower). In my opinion, this is exactly what we'd expect from slow breathing: you'd feel less rushed, think about the problem, and therefore be more accurate.
 - Working memory significantly improved following slow breathing compared to the control condition.
 - Cognitive flexibility significantly improved following slow breathing.
-



Relation to Vagal Tone

Mathematically, vagal tone did not explain the improved executive functioning.

During slow breathing, RMSSD (their measure of vagal tone) increased significantly. Then, after the breathing ended, RMSSD went back down to near-baseline levels (it was still higher but not statistically significant). This is a very normal response.

However, when they tried to create statistical relationships between the increases in RMSSD and the performance on the mental tests, none existed, meaning vagal tone couldn't explain the improved cognitive functioning.

4. Why Do These Results Matter?

Altogether, this study suggests that, by improving executive function, slow breathing may help us plan, monitor, and execute our goals.

Interestingly, however, they found that these improvements were not due to increases in vagal tone. But we should remember that using HRV RMSSD to measure vagal tone is still debated. In general, we know they're related, but saying that "RMSSD = vagal tone" may turn out to be untrue as we learn more about vagal tone.

Moreover, we know that slow-paced breathing also impacts brain chemistry, [altering the release of noradrenaline](#), which also improves focus. Nasal breathing also [organizes brain wave](#) activity across various regions of the brain. Thus, many factors probably explain these results, not just vagal tone.

Regardless of the precise mechanism, the most significant aspect of this study is that we learn that slow breathing improves executive function.

1 Big Takeaway

The short-term practice of slow breathing helps us resist distractions and impulses, improves our working memory, and enhances our cognitive flexibility. This means slow breathing enables you to plan, monitor, and execute your goals.



1 Practical Application

Practice slow breathing before performing any task in which you need your complete focus and want to resist impulses (such as the impulse to check your phone or email).

Moreover, although this research didn't address it, regularly practicing slow breathing may also improve your baseline executive functioning, increasing the likelihood of you being able to plan, monitor, and achieve your goals.



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Citation

Lehrer, P., Kaur, K.,
Sharma, A., Shah, K.,
Huseby, R., Bhavsar, J.,
& Zhang, Y. (2020).
Heart Rate Variability
Biofeedback Improves
Emotional and Physical
Health and
Performance: A
Systematic Review and
Meta Analysis. *Applied
Psychophysiology and
Biofeedback*, 45(3),
109–129.
<https://doi.org/10.1007/s10484-020-09466-z>.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Heart Rate Variability Biofeedback Improves Emotional and Physical Health and Performance: A Systematic Review and Meta Analysis

4 FUNDAMENTALS

1. Essential Background Material

This meta-analysis aimed to examine basically every issue that might be addressed by heart rate variability biofeedback (HRVB) based on previous research.

A hallmark of slow breathing is that it increases HRV. It does this by increasing respiratory sinus arrhythmia (RSA), which synchronizes your heart rate with your breathing. When they match, your heart rate increases while you inhale, and it decreases while you exhale.

With HRVB, a person's "ideal" breathing rate is determined—that is, one that maximizes HRV. And because increases in RSA and HRV are driven by increases in the calming parasympathetic branch of the nervous system, this can reduce harmful stress and increase overall resiliency. This has wide-reaching positive benefits, such as lowering blood pressure, lessening anxiety, and improved cardio-autonomic function. Because of this, HRVB has become extremely popular for treating various problems.

But here, they wanted to systematically examine which conditions and issues have benefited most from HRVB based on previously-published scientific literature.



2. What Did this Research Do?

To do this, the authors performed an extensive literature review to examine these benefits of HRVB from a broader statistical perspective. They included papers spanning a wide range of settings, measuring a wide range of outcomes.

Although HRVB sounds fancy (and it can be), many of the benefits are achieved by simply breathing at a rate of about 5-6 breaths per minute.

Therefore, this meta-analysis also included studies that used 6 breaths per minute without HRVB because it's likely that this is enough to get most of the benefits without fancy equipment.

This might be the most important aspect of this meta-analysis from a practical perspective.

After starting with more than 1,500 papers (!), they ended up with 58 studies having a total of 2,485 patients. This large number of papers is a testament to the power and popularity of HRVB and slow breathing.

3. What Were the Major Findings?

Their statistical analysis revealed that HRVB and slow breathing significantly improve many health and wellness aspects.

The **most significant benefits** were for:

- Athletic performance
- Artistic performance
- Depression
- Gastrointestinal problems
- Anxiety and anger
- Respiratory disorders
- Systolic blood pressure
- Pain

Smaller but still meaningful benefits were found for:

- Self-reported stress
- Quality of life
- Diastolic blood pressure
- PTSD



- General energy
- Sleep

Interestingly, I would have expected several items on the second list to be on the first. But that's why meta-analyses like this are so important :) Also, note that measures like "self-reported stress" are harder to quantify. The authors even mention that these findings might result from how the questionnaires were given.

In any case, the overall results of this meta-analysis are quite exceptional: **HRVB and slow breathing both have wide-ranging benefits for overall health and wellness.**

4. Why Do These Results Matter?

These results are significant because they show that, across studies and disciplines, slow breathing is a valuable tool for improving a variety of outcomes. Just imagine it helping you with a few things on the lists above. **Even minor improvements on 4 or 5 different things could be life-changing. Now we see that a broad range of studies suggest it's possible.**

1 BIG TAKEAWAY

Slow breathing basically helps everything, with very rare side effects. This study supports almost any positive statement about slow breathing, as the results show many benefits across studies and disciplines.

1 PRACTICAL APPLICATION

Make slow breathing at around 5-6 breaths per minute part of your daily routine to improve many aspects of your physiological, emotional, and creative health. Although this study didn't provide precise guidance on how long these slow breathing sessions must be, we know that at least 10 minutes a day is the bare minimum for improvements.

For a deeper application, consider buying an HRV monitor (I use the [Elite HRV CorSense®](#), which cost me \$165) and going through a full HRVB protocol.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Respiratory therapy for the treatment of anxiety: meta-analytic review and regression

411 Released: July 3, 2024



Quick Summary

Interventions targeting dysfunctional breathing (for example, diaphragmatic breathing, slow breathing, and HRV biofeedback) significantly improve anxiety both immediately and over the long term, providing effects comparable to the gold-standard treatment of cognitive behavioral therapy.

[Click Here to Get the Paper](#)

Article Citation

Leyro TM, Versella MV, Yang MJ, Brinkman HR, Hoyt DL, Lehrer P. Respiratory therapy for the treatment of anxiety: Meta-analytic review and regression. Clin Psychol Rev. 2021 Mar;84:101980. doi: 10.1016/j.cpr.2021.101980. Epub 2021 Jan 24. PMID: 33540222; PMCID: PMC8302658.

Reading Time: 2 min 48 seconds

I hope the next 42'ish breaths are the most nourishing of your day.



4 Fundamentals

1. Essential Background Material

It's estimated that over 11% of people worldwide suffer from anxiety and its related disorders. The “gold-standard” treatment of cognitive behavioral therapy (CBT) offers small to medium effect sizes when measured statistically, suggesting that other therapies are needed.

Breathing is a natural process to target because hyperventilation (breathing more than our metabolic need) is a hallmark feature of anxiety, and a feeling of breathlessness often accompanies anxiety and its related disorders. Moreover, hyperventilation may result in the excess use of accessory breathing muscles in the chest and shoulders, causing excess sympathetic activation and exacerbating feelings of breathlessness. Lastly, due to the Bohr effect, overbreathing will reduce blood flow and oxygen delivery, especially to the brain, causing cognitive impairment, which may enhance feelings of anxiety.

Conscious breathing techniques, on the other hand, can counteract these adverse effects by activating the relaxation response, inducing a calmer body and mind.

Altogether, interventions targeting breathing may be effective, given the close link between breathing and anxiety. However, few papers have comprehensively reviewed the evidence to see if breathing is clinically useful in the treatment of anxiety, which is the goal of this paper.

2. What Did this Research Do?

This meta-analysis searched for studies using a breathing intervention, having a control group, and measuring at least one anxiety index. If a study included multiple measures of anxiety, each was assigned an equal weight, and the overall changes in all measurements were aggregated. Overall, they ended up finding 39 studies with around 3,000 participants that fit these criteria.

The studies consisted primarily of diaphragmatic breathing (n=12) and heart rate variability (HRV) biofeedback (n=16), with the remaining studies being a mix of slow breathing, inspiratory muscle training, CO2 biofeedback, and others.



3 & 4. What Were the Major Findings and Why Do They Matter?

There was a high risk of bias in most of the studies.

Thirty-five of the 39 studies had a “high” risk of bias, and the other four had “some concerns.” Their analysis revealed this was primarily due to failure to use an appropriate control condition. (As an aside, breathing studies often suffer from quality issues.)

Breathing interventions significantly reduced anxiety, showing a medium effect size.

This is meaningful because, as they noted in the introduction, the gold-standard therapy of CBT has a small to medium effect size. Here, we see that breathing has slightly stronger effects. However, let’s keep in mind that the studies had a high risk of bias.

These significant improvements persisted over time.

Eight studies had multiple follow-ups and showed a medium to large effect size, indicating that the benefits for anxiety persisted over time.

Results were not significant for panic and post-traumatic stress symptoms.

They isolated specific symptoms and found that, for panic, the improvements from breathing were large, but they did not reach statistical significance. For post-traumatic stress, the results were small and insignificant.

There were no statistically significant differences between biofeedback and non-biofeedback studies.

Because of the diverse set of breathing interventions used, they couldn’t isolate which techniques were most effective. However, they could delineate between interventions using biofeedback (n=19) and those that did not (n=20). The results for biofeedback interventions were medium to large, whereas those for all other interventions were medium. So, they were slightly better for biofeedback. However, these differences didn’t reach statistical significance, meaning biofeedback is not necessary to get the benefits of breathing for anxiety.

Treatment duration and intensity were not statistically significant.



Studies ranged from 1 to 20 weeks in duration and included between 1 and 28 different breathing sessions. Their analysis revealed that neither of these factors mattered statistically, meaning that the longer studies or the ones with more sessions didn't show any statistical improvement over the shorter, less intense ones.

1 Big Takeaway

Interventions targeting dysfunctional breathing (for example, diaphragmatic breathing, slow breathing, and HRV biofeedback) significantly improve anxiety both immediately and over the long term, providing effects comparable to the gold-standard treatment of cognitive behavioral therapy.

1 Practical Application

Combining these results with those from [HHPF's more recent paper](#), we can choose any slow-paced breathing technique we enjoy and practice for at least five minutes daily to see meaningful improvements in anxiety symptoms.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Effect of Breathing Exercises on Oxidative Stress Biomarkers in Humans: A Systematic Review and Meta-analysis

411 Released: September 29, 2023

 frontiers | Frontiers in Medicine

Effect of breathing exercises on oxidative stress biomarkers in humans: A systematic review and meta-analysis

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Zhong-guang Sun^{1*}

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Article Citation

Li TT, Wang HY, Zhang H, Zhang PP, Zhang MC, Feng HY, Duan XY, Liu WB, Wang XW, Sun ZG. Effect of breathing exercises on oxidative stress biomarkers in humans: A systematic review and meta-analysis. *Front Med (Lausanne)*. 2023 Apr 5;10:1121036. doi: 10.3389/fmed.2023.1121036. PMID: 37122322; PMCID: PMC10132211.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2023 meta-analysis examined 10 studies and found that breathing exercises of all kinds significantly reduce biomarkers of oxidative stress while increasing the body's antioxidant capacity.



4 Fundamentals

1. Essential Background Material

- Oxidative stress refers to an excess production of free radicals (reactive oxygen species) compared to antioxidant defenses.
- Oxidative stress causes inflammation, vascular damage, and cell death (to name just a few of its damaging consequences).
- Biomarkers of oxidative stress generally increase with disease or chronic inflammation and then decrease as the condition is resolved.
- Excess oxidative stress is basically associated with every chronic condition. It is thought to be a primary driver of complications in many diseases (such as diabetes).
- Breathing exercises of all kinds have been shown to improve oxidative stress in healthy adults and people with diabetes, hypertension, and COPD. However, small sample sizes and inconsistent findings have made it challenging to accurately state whether they're good for oxidative stress.
- This systematic review and meta-analysis analyzed how breathing exercises impact oxidative stress biomarkers to assess if they are clinically applicable.

2. What Did this Research Do?

This review collected studies that used the following biomarkers of oxidative stress:

1. **Malondialdehyde (MDA):** Result of lipid peroxidation by excess oxidative stress. Higher levels indicate higher oxidative stress.
 2. **Superoxide dismutase (SOD):** Antioxidant that protects the body from free radical damage. Higher levels indicate greater protection from oxidative stress.
 3. **Glutathione (GSH):** Important antioxidant that helps protect cells and vital organs from oxidative damage.
 4. **Nitric oxide (NO):** Disease- and condition-specific marker of oxidative stress. Needs to be interpreted carefully as high/low values mean different things in different conditions.
-



5. **Vitamin C:** Antioxidant. It must be interpreted carefully because it can be obtained externally.
6. **Total antioxidant capacity (TAC):** Marker of overall antioxidant status.

Overall, 10 studies with a total of 519 participants were included. The breathing exercise interventions included slow and fast breathing (n=1), respiratory muscle trainer (n=1), diaphragmatic breathing (n=1), pranayama (n=4), straight leg raising breathing (n=1), and inspiratory muscle training (n=2).

The interventions ranged from 20 days to 3 months and included healthy participants (n=2) and people with diabetes (n=3), COPD (n=3), and hypertension (n=2).

3 & 4. What Were the Major Findings and Why Do They Matter?

MDA decreased significantly in 8 studies that measured it (with a total of 443 participants).

- Since MDA is a byproduct of oxidative stress, this is meaningful because it provides quantitative evidence that breathing exercises directly lower oxidative stress.

SOD increased significantly in 7 studies that measured it (with a total of 433 participants).

- This is meaningful because it shows that breathing exercises increase the body's antioxidant capacity.
- Caveat: The authors performed a sensitivity analysis and found that removing one study made the changes insignificant, making this finding less robust.

GSH increased significantly in 7 studies that measured it (with a total of 383 participants).

- This also suggests that breathing exercises increase the body's antioxidant capacity. This is specifically meaningful because GSH protects cells from oxidative damage.

Nitric oxide, vitamin C, and total antioxidant capacity levels did not change significantly.

- Note that these were not measured in as many studies (NO in 4 studies with 111 people total, vitamin C in 3 studies with 220 people total, and TAC in 2 studies with 50 people total). Thus, it's harder to draw conclusions about these findings.

The limited number of studies made it impractical to analyze the results by breathing type (e.g., fast vs. slow vs. resistance) and condition type (e.g., diabetes vs. hypertension).



- This is important. This means that this meta-analysis basically says, “Breathing exercises seem to help oxidative stress.” But, there’s not enough data to put together guidelines or best practices from the published studies alone.

1 Big Takeaway

Breathing exercises of all kinds significantly reduce biomarkers of oxidative stress while increasing the body’s antioxidant capacity, which may benefit both healthy and disease states.

1 Practical Application

Unfortunately, this review used an assorted sample of studies, making it impossible to create a simple, practical application directly from the results.

However, from the studies included, it seems plausible that just about any breathing exercise you enjoy doing will be helpful.

Further, I have not read all the studies. But, from what I’ve seen, [slow breathing](#) and [inspiratory muscle training](#) have fairly robust results, so those seem like natural places to start getting the benefits of breathing for reducing oxidative stress.



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Citation

Lundberg JO,
Settergren G, Gelinder
S, Lundberg JM, Alving
K, Weitzberg E.
Inhalation of nasally
derived nitric oxide
modulates pulmonary
function in humans.
Acta Physiol Scand.
1996 Dec;158(4):343-7.
doi: 10.1046/j.1365-
201X.1996.557321000.
x. PMID: 8971255.

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Additional Note

**When you see in books
or social media posts
that nasal breathing
increase oxygenation
by 10-18%, this is “the
paper” it comes from.**

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Inhalation of nasally derived nitric oxide modulates pulmonary function in humans

4 FUNDAMENTALS

1. Essential Background Material

Nitric oxide (NO) has many beneficial properties. For instance, it's anti-microbial, anti-viral, and a potent vasodilator (it improves blood flow).

Interestingly, NO is continuously produced in the paranasal sinuses. Because of this, **NO is carried through the airways and into the lungs with each nasal inhale we take.** However, for this same reason, mouth breathing does not bring as much NO into the airways and lungs.

Thus, since nasal breathing harnesses NO, the authors of this study wondered *if nose breathing might improve blood flow in the lungs, thereby leading to better oxygenation than mouth breathing? And if so, is it purely due to NO, or are other factors playing a role?*

They did several neat experiments to test these hypotheses.

2. What Did this Research Do?

First Experiment: Nose vs. Mouth in Healthy Subjects

They had eight healthy participants perform two different breathing protocols:

- **Nasal Breathing:** Inhale through the nose and exhale through the mouth (**exhaling through the mouth does not wash out residual NO in the nasal**



airways, maximizing NO delivery to the lungs on the next inhale).

- **Mouth Breathing:** Inhale through the mouth and exhale through the nose (this minimizes NO delivery because it bypasses the nasal airways during inhalation and washes NO out of the airways during exhalation).

They repeated 5-min segments of each breathing protocol (5-min mouth breathing, 5-min nose breathing, 5-min mouth breathing, etc.) **while transcutaneous oxygen tension was measured.**

Sub-Experiment 1: They had two people perform mouth breathing with added NO to see if it could reproduce some of the same nasal-breathing results.

Sub-Experiment 2: They also had two participants perform mouth breathing with added moisture to see if humidification might explain the differences.

Second Experiment: Intubated Mechanically Ventilated Participants

In patients who were on long-term mechanical ventilation (and therefore receiving no nasally-derived air), they did something simple yet profound. They extracted air from the patients' noses and added it to the inspiratory air of the ventilator. Arterial oxygenation was measured in these participants.

Sub-Experiment 3: They also added just exogenous NO to the inspiratory air of the ventilator to see if that affected arterial oxygenation similarly to nasal air.

There was another experiment with coronary bypass grafting patients, which I am leaving off for simplicity—the results presented next are the most critical.

3. What Were the Major Findings?

- **In six of the eight healthy participants, nasal breathing increased oxygenation by an average of 10% compared to mouth breathing.**
- Mouth breathing with exogenous NO increased oxygenation by 6% in one participant and 23% in another compared to mouth breathing without NO; mouth breathing with added moisture *did not* impact oxygenation.
- **Adding nasal air to long-term ventilated patients increased arterial oxygenation by an average of 18%.**
- Adding exogenous NO to long-term ventilated patients increased arterial oxygenation by an average of 20%.

4. Why Do These Results Matter?



Nose vs. Mouth Breathing

The 10% increase in transcutaneous oxygen tension is meaningful because this is a measurement of tissue oxygenation, which is a function of blood oxygen levels, blood flow, and oxygen release. **Thus, it shows that nasal breathing increases the amount of oxygen actually getting to the tissues by 10%.** That's crazy.

However, note that there were only eight participants, and nasal breathing did not increase oxygenation in two (the authors were unsure why). So, we'd like a much bigger sample size, but we have to take what we can get.

Lastly, the additional sub-experiments suggest that NO is the primary mechanism through which nasal breathing increases oxygenation, not humidification.

Intubated Mechanically-Ventilated Patients

These results are perhaps even more crucial. Here's why. The structure of the nose and nasal airways does all kinds of amazing things for our breath. But the intubated mechanically ventilated patients bypass all of this.

Yet, adding nasally-derived air still increased oxygenation by 18%, and adding NO alone increased it by 20%. **This suggests that, of all the nose's features, the production of NO might be the most significant factor for body oxygenation.**

1 BIG TAKEAWAY

Nasal breathing increases tissue oxygenation more than mouth breathing (an average of 10%) due to inhalation of nasally-derived nitric oxide.

1 PRACTICAL APPLICATION

This one is pretty straightforward: Breathe primarily through your nose.

Another practical takeaway is that we can use nose-in/mouth-out to maximize NO delivery to the lungs. Thus, you might start or end each breathing exercise with a few breaths (or a few minutes) like this to optimize oxygenation.

Lastly, in my opinion, these results might also support the standard advice to breathe in nasally and out orally during exercise. This would maximize NO delivery and enhance oxygenation. But, it could also [dehydrate us faster](#), so we just have to be careful not to overdo it.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

The Effect of Slow Breathing in Regulating Anxiety

Article Published: March 2025

411 Released: March 23, 2025

Quick Summary

This 2025 study in Nature Scientific Reports found that just 30 seconds of slow breathing reduced self-reported anticipatory anxiety while lowering heart rate and slowing brainwaves, suggesting it may help prepare the body and brain for upcoming stress.

Article Citation

Luo Q, Li X, Zhao J, Jiang Q, Wei D. The effect of slow breathing in regulating anxiety. Sci Rep. 2025 Mar 11;15(1):8417. doi: 10.1038/s41598-025-92017-5. PMID: 40069274; PMCID: PMC11897343.

[Link to Get the Paper](#)

4 Fundamentals



1. Essential Background Material

Anxiety disorders are highly prevalent and often characterized by persistent worry and physiological symptoms like increased heart rate, shortness of breath, and sweating. One of the largest contributors to anxiety is uncertainty. As Elissa Epel, Ph.D., puts it: **“Chronic uncertainty is the most ubiquitous form of chronic stress.”** This is type of anxiety (often called anticipatory anxiety) is problematic because it causes distress before any actual threat occurs.

To deal with anticipatory anxiety, breathing may be a particularly helpful tool. As you know, breathing is directly linked to our nervous system, brain activity, and emotional state. In fact, an elevated breathing rate is an indicator of anxiety, but unlike most physiological responses, breathing is under voluntarily control. This suggests we might be able to use our breath not only for detecting but also for mitigating anticipatory anxiety.

This study explored how both slow and fast breathing affected subjective and physiological measures of anticipatory anxiety. It's a valuable contribution because, while many studies have explored breathing for anxiety, few have looked at how breathing affects anticipatory anxiety in uncertain situations, something we all deal with in everyday life. Let's look at what they did.

2. What Did this Research Do?

This study included 25 female participants between the ages of 18 and 26. The study procedure first began with either slow or fast breathing for 30 seconds. The slow breathing condition consisted of inhales and exhales lasting 6 seconds (5 breaths a minute). The fast breathing condition consisted of 2-second inhales and exhales, or 15 breaths a minute. (Note: 15 breaths a minute is within the range of normal resting breathing; thus a more accurate name might have been “controlled breathing” instead of “fast breathing”.)

After the breathing, a visual clue was shown on a screen to introduce either certainty or uncertainty. The certain cue indicated that they would be shown a negative image. The uncertain cue indicated that the upcoming image could be positive, negative, or neutral, adding an element of ambiguity.

Next, the image showed up on the screen for three seconds and the participants then rated their emotional response using valence and arousal. Valence measured how negative an image felt (0 = not negative at all, 6 = very negative). Arousal measured emotional intensity (0 = not feeling at all, 6 = very strong).



They performed this procedure—breathing --> cue --> image --> rate valence --> rate arousal—for 60 different images.

The researchers recorded heart rate, heart rate variability (HRV), and brainwave activity (EEG) throughout the procedure to physiologically assess how breathing and anticipatory anxiety affected the body and brain.

3 & 4. What Were the Major Findings and Why Do They Matter?

Slow breathing reduced the emotional impact of negative images.

- After the slow breathing task, participants rated images lower on the valence scale (meaning they found them less negative than after fast breathing)
- Their arousal scores were also lower, indicating that slow breathing dampened their emotional response to the images.

Slow breathing helped regulate the anxiety associated with uncertainty.

- Perhaps unsurprisingly, in the uncertain condition (when they didn't know what type of image would appear), the participants rated images as more negative and experienced higher arousal levels.
- Slow breathing dampened these effects, lowering valence and arousal in uncertain situations. This is meaningful for real life because uncertainty is a major cause of stress.

Slow breathing led to a lower heart rate.

- Slow breathing also improved heart rate variability (HRV), but the only statistically significant cardiovascular effect was a reduction in heart rate compared to fast breathing.

Slow breathing altered overall brain activity.

- Slow (but not fast) breathing increased power in the delta, theta, alpha, and beta bands.
- This could reflect greater brainwave synchrony and overall brain coherence during slow compared to fast breathing.

Slow and fast breathing had different effects on the brain during anticipation.

- During fast breathing, when participants were uncertain about the upcoming image, beta brainwave activity increased, which is a sign of heightened anxiety or arousal.
-



- Conversely, during slow breathing, beta activity decreased in the same situation, suggesting that it had a calming effect.
- This suggests that slow breathing may be able to help reduce anticipatory anxiety associated with uncertain scenarios (public speaking, social settings, etc.).

1 Big Takeaway

Slow breathing reduced self-reported valence and arousal alongside physiological markers of anxiety (heart rate and brainwaves) during uncertain situations, suggesting that it may help prepare your body and brain for upcoming stress.

1 Practical Application

Practice slow breathing before a stressful or uncertain event to help calm your physiological and emotional responses. As this study shows, even just 30 seconds is enough to make a meaningful impact on how you respond mentally and physically.



[DOWNLOAD
PAPER](#)

Citation

Magnon V, Dutheil F, Vallet GT. Benefits from one session of deep and slow breathing on vagal tone and anxiety in young and older adults. Sci Rep. 2021 Sep 29;11(1):19267. doi: 10.1038/s41598-021-98736-9. PMID: 34588511; PMCID: PMC8481564.

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May 4, 2022

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Benefits from one session of deep and slow breathing on vagal tone and anxiety in young and older adults

4 FUNDAMENTALS

1. Essential Background Material

This study examined the effects of one 5-minute slow breathing session (6 breaths/min) on anxiety and heart rate variability (HRV) in young and older adults.

Anxiety is widespread. It contributes to poor mental health and is associated with increased all-cause mortality. Moreover, anxiety might pose a more significant threat as we get older because our bodies might lose the resiliency to adapt to it. Thus, we must develop effective treatments for anxiety, especially in the older adult population.

It is established that slow deep breathing (usually around 6 breaths/min) can reduce anxiety and increase HRV, a physiological measure of resiliency. However, it's unclear if these same effects will hold true in an older population (surprisingly, no one has tested this explicitly).

Thus, the goal of this paper was to use slow breathing in younger and older adults and measure how it impacted psychological anxiety (via a questionnaire) and physiological anxiety (using HRV).



2. What Did this Research Do?

They recruited 25 younger adults with a mean age of ~20 years old and 22 older adults with a mean age of ~66 years old (a sample of at least 20 was needed to determine significance).

This paper was part of a more extensive study, so there seemed to be many tasks going on. However, for this paper, here are the three main components:

1. Take a state anxiety questionnaire (pre-test period)
2. Slow deep breathing session (5 min)
3. Retake state anxiety questionnaire (post-test period)

The slow breathing exercise consisted of a 4-second inhale, and a 6-second exhale (6 breaths per minute). Because they were focused on anxiety, they specifically used a longer exhale to maximize relaxation and promote a shift into a parasympathetic dominant state.

The questionnaire used was called Spielberger's State Anxiety Inventory, which consisted of 20 questions assessing anxiety.

HRV was measured continuously throughout the protocol. They measured differences between the "pre-test period" and "post-test period" to determine the significance of the breathing protocol. They focused on high frequency power (HF-HRV) and RMSSD (see sidebar).

3. What Were the Major Findings?

State anxiety was significantly lower after slow breathing in both the younger and older adults.

HF-HRV power increased (meaning physiological anxiety decreased) and was negatively correlated with perceived anxiety. **Thus, the greater the increase in HF-HRV, the lower the participant's subjective anxiety.**

Somewhat surprisingly, **HF-HRV increased more in older adults than younger ones.** This was surprising because the authors hypothesized the younger adults would have greater autonomic flexibility and would therefore be able to increase HRV more robustly.

NOTE: They make a strong argument that HF-HRV is a good measure of vagal tone when using breathing with slow, extended exhalation. I've read other studies saying HF-HRV is not a measure of vagal tone, so there doesn't seem to be agreement on what these HRV frequency-domain measurements really mean. In any case, this was published in Nature, and it's almost certain a reviewer brought it up. So, I am going to accept their justification for calling HF-HRV "vagal tone" since they're the experts getting published in Nature, not me.



4. Why Do These Results Matter?

These results provide three significant outcomes.

First, they show that older adults' vagal tone is persevered (perhaps even strengthened). **This suggests that deep breathing exercises might be particularly helpful in this population (but really for everyone) to improve aging and reduce anxiety.**

Second, they show that HF-HRV is negatively associated with perceived anxiety. Thus, measuring the HF-HRV of yourself or your clients might be a reliable way to assess physiological anxiety.

Third, HF-HRV is potentially a good indicator of vagal tone and overall physiological resiliency. The different HRV metrics still confuse me, and I'm not entirely sold on their arguments (see above sidebar). But, they make a good case for using HF-HRV in the paper. So, it seems like one that's worth measuring if you have the equipment.

1 BIG TAKEAWAY

Slow breathing at 6 breaths per minute (4-sec in/6-sec out) reduces subjective and physiological anxiety. **This holds true in younger and older adults and could be a valuable practice for promoting healthy aging.**

1 PRACTICAL APPLICATION

Perform slow breathing with a slightly longer exhalation (4/6, 5/7, etc.) anytime you experience anxiety to help with symptoms. Moreover, use this practice regularly to improve vagal tone, enhance physiological resiliency, and promote healthy aging.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Modulating Heart Rate Oscillation Affects Plasma Amyloid Beta and Tau Levels in Younger and Older Adults

411 Released: May 12, 2023

[nature](#) > [scientific reports](#) > [articles](#) > [article](#)

Article | [Open Access](#) | Published: 09 March 2023

Modulating heart rate oscillation affects plasma amyloid beta and tau levels in younger and older adults

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Article Citation

Min J, Rouanet J, Martini AC, Nashiro K, Yoo HJ, Porat S, Cho C, Wan J, Cole SW, Head E, Nation DA, Thayer JF, Mather M. Modulating heart rate oscillation affects plasma amyloid beta and tau levels in younger and older adults. Sci Rep. 2023 Mar 9;13(1):3967. doi: 10.1038/s41598-023-30167-0. PMID: 36894565; PMCID: PMC9998394.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

This is the first study to show that a behavioral intervention may reduce the risk of Alzheimer's. They found that slow breathing reduces levels of A β , a key Alzheimer's biomarker, in younger and older adults, suggesting that it may reduce the risk of Alzheimer's and promote healthy brain aging.



4 Fundamentals

1. Essential Background Material

- One reason that rates of Alzheimer's disease might increase with age is that, in general, the activity of the sympathetic nervous system increases with age. In contrast, the activity of the parasympathetic branch decreases as we age.
- This age-related nervous system dysfunction may impact Alzheimer's by modulating how peptides and proteins are produced and removed in the body and brain.
- Specifically, amyloid- β ($A\beta$) peptides and tau proteins, both of which are critical in Alzheimer's risk, may increase as sympathetic nervous system activity increases.
- Therefore, the authors reasonably hypothesized that activities that increase parasympathetic activity might reduce $A\beta$ and tau concentrations, thus reducing Alzheimer's risk.
- Because slow breathing is a noninvasive way to stimulate the parasympathetic nervous system, they hypothesized that it could potentially reduce $A\beta$ and tau levels.

2. What Did this Research Do?

There were 108 participants in this study, subdivided into older ($N=54$; an average age of about 66) and younger ($N=54$; an average age of about 23) groups to separately test the effects of the breathing exercise on $A\beta$ and tau levels in these different populations.

The participants were randomly separated into two groups. One group used HRV biofeedback **to maximize heart rate oscillations using slow breathing** (which I'll call the "breathing group"). This group performed a standard one-day HRV biofeedback procedure, testing breathing rates from $\sim 4.5 - 6.5$ breaths a minute. Whichever rate maximized their HRV was considered their resonant frequency, and they were told to breathe at this rate for the duration of the study. They used an equal inhale-to-exhale ratio.

The other group (I'll call them the "non-breathing group") minimized HRV through various methods, such as listening to music or visualization of different scenes.



During the 4-week study period, both groups did 20-40 minutes of daily practice with a heart rate monitor clipped to their ear while watching a screen (in my opinion, this is an essential part of the upcoming results).

The primary measurements here are $A\beta_{40}$, $A\beta_{42}$, pTau-181, and tTau, as they appear to be most relevant to determining the risk of Alzheimer's.

However, this was part of a larger, longer-lasting study. So, in addition to $A\beta$ and tau, there were also cognitive tests, blood tests, emotional tests, MRI scans, and the participants were told to wear a Whoop strap for the duration of the experiment to monitor sleep.

3 & 4. What Were the Major Findings and Why Do They Matter?

(Because this paper is outside our usual scope, I'm testing out combining these two sections to make it easier to understand. The **major result will be bolded**, and the "Why It Matters" will come immediately after.)

The breathing group significantly lowered both $A\beta_{40}$ and $A\beta_{42}$ peptides. In contrast, they significantly increased in the non-breathing group. These findings held true when analyzed separately for younger and older adults.

- *This is the first study to show that a behavioral intervention can lower $A\beta$ peptides.* Because these peptides are a significant risk factor for Alzheimer's disease, this provides evidence that slow breathing could help reduce the risk of Alzheimer's while also promoting healthy $A\beta$ levels throughout adulthood. **This is groundbreaking.**

There were no significant changes in tTau or pTau-181. Ratios of the different measurements ($A\beta_{42}/A\beta_{40}$ and pTau-181/tTau) did change for the different age groups, but altogether, there were no conclusive findings.

- These ratios are important for determining if slow breathing reduces the production of AB and tau or increases the clearance of these critical markers. From their discussion, it seems that slow breathing has the potential to do both (reduce production and increase clearance). But, from a mechanistic standpoint, this is something future research will have to examine more thoroughly.

There were no significant changes in "negative affect" (negative emotions and moods) for either group.



- This was a surprising yet glossed-over finding of the study. Shouldn't slow breathing help them feel more relaxed, leading to better emotional health? To understand this unexpected finding, I think looking at the study protocol is crucial. Participants were asked to use an ear clip and stare at a computer screen the whole time. Additionally, they were wearing Whoop straps, and the younger adults took saliva samples every morning. If anything, the entire process probably stressed them more than it relaxed them. Thus, these findings on emotional health don't surprise me, but this is just my speculation.

1 Big Takeaway

Slow breathing reduces levels of amyloid- β , a key biomarker of Alzheimer's risk, in younger and older adults, suggesting that it may reduce the risk of Alzheimer's and promote healthy brain aging.

1 Practical Application

Practice slow breathing at a comfortable rate between 4.5-6.5 breaths a minute for 20 minutes daily to potentially reduce your risk of Alzheimer's and promote healthy brain aging.

In this study, they used an equal inhale-to-exhale ratio. However, using a slightly longer exhalation would likely be ok too (if that's what you prefer) because some research suggests this approach leads to greater activation of the parasympathetic nervous system.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

The “Embreathment” Illusion Highlights the Role of Breathing in Corporeal Awareness

411 Released: March 31, 2023

> [J Neurophysiol](#). 2020 Jan 1;123(1):420–427. doi: 10.1152/jn.00617.2019. Epub 2019 Dec 4.

The "embreathment" illusion highlights the role of breathing in corporeal awareness

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Affiliations + expand

PMID: 31800367 PMCID: [PMC6985859](#) DOI: [10.1152/jn.00617.2019](#)

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Article Citation

Monti A, Porciello G, Tieri G, Aglioti SM. The "embreathment" illusion highlights the role of breathing in corporeal awareness. *J Neurophysiol*. 2020 Jan 1;123(1):420–427. doi: 10.1152/jn.00617.2019. Epub 2019 Dec 4. PMID: 31800367; PMCID: PMC6985859.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

By mapping peoples’ real breathing rate and amplitude onto virtual avatars, this study found that—even without conscious awareness—breathing is essential to our feeling of body ownership and, most significantly, to our sense of body agency.



4 Fundamentals

1. Essential Background Material

- Corporeal awareness is a form of self-consciousness. It refers to being aware that you own a body, that you control that body, and that that body occupies a position in space.
- The factors determining this “sense of embodiment” are diverse (and likely different from person to person). But breathing seems like a potentially significant factor in corporeal awareness because it is intimately tied to our physiology, brains, and emotions.
- Moreover, because breathing is both automatic and under voluntary control, it may be especially vital to our sense of embodiment and feeling of body agency.
- This study assessed how much breathing contributes to corporeal awareness through a unique virtual reality experiment.

2. What Did this Research Do?

[Watch a YouTube Clip of the Experiment Here](#)

They brought 32 male volunteers into a lab, had them put on an immersive virtual reality headset and breathing monitor, and then ran them through several experiments. **Note that they did not know the research was about breathing.**

They called their virtual reality setup the “embreathment” illusion.

What is the “Embreathment” Illusion?

The “embreathment” illusion refers to the real-time mapping of a person’s breathing rhythm and amplitude onto a virtual avatar. (It’s also one of the best words I’ve seen in a science paper.)



Embreathment Illusion Experiments

Each participant went through eight different conditions ranging from “most congruent” to “least congruent”:

- **Most Congruent:** A human-like avatar breathed at the same rate and amplitude as the person and was viewed from the first-person perspective.
- **Least Congruent:** A wooden avatar, viewed from the third-person, breathed out-of-phase with the person. For instance, the avatar inhaled when the person exhaled.

The other six conditions were in between these extremes. Participants experienced each situation for two minutes.

Assessments

Before the virtual reality experiments, participants answered the Multidimensional Assessment of Interoceptive Awareness (MAIA) questionnaire **to assess interoceptive awareness**.

After each two-minute virtual reality segment, participants answered a five-question survey assessing how much they felt that: (1) the virtual body was theirs (ownership), (2), they controlled the movements of the virtual body (agency), and (3) they were in the same place as the virtual body (location).

Then, after the experiments were over, they performed two more tests to assess interoception:

1. **Heartbeat Counting Task:** They were asked to count their heartbeats over a specified time interval.
2. **“Pneumoception” Task:** They listened to 26 breathing soundtracks and had to label whether they were their own or not (how amazing is this?!)

3. What Were the Major Findings?

For these results, keep in mind that the participants had no clue that breathing was being modified. They were just looking in the goggles and answering the questions.

- The participants felt like the virtual body was more their own **when the avatar breathed in phase** with their actual breathing.
 - They also felt more body agency—that they were in control of the avatar’s body movements—when its breathing matched their own. In fact, **breathing was more important than body perspective and visual appearance for having a sense of agency**.
-



To put this in perspective, this means switching from a human avatar to a wooden block avatar was less important than breathing in phase or out of phase. Pretty crazy.

- The participants' actual breathing changed based on how in phase it was with the avatar's breathing. For example, the amplitude of each breath increased slightly when in phase with the avatar and decreased slightly when out of phase with the avatar.
- These impacts depended on how interoceptive each person was—**the more interoceptive they were, the less susceptible they were to the experimental manipulations.** (Note: recall that interoception was measured by the heartbeat counting and pneumoception tests.)

4. Why Do These Results Matter?

This study shows that breathing is a fundamental physiological variable providing a sense of embodiment. It suggests that breathing may play a key role in our feelings of self-consciousness 🤖.

Critically, these results show that breathing increased body agency. Meaning when they saw the avatar's breathing move with their breathing (even though it wasn't conscious awareness), it gave them a sense of control over the avatar's actions.

I believe these results help explain why conscious breathing exercises also increase self-agency in **our real bodies**. Anecdotally, almost everyone who practices breathing feels a greater sense of body agency.

Here, we learn why. If breathing increases our sense of agency even without conscious awareness, **it just makes sense that deliberately controlling your breathing would provide an even greater sense of agency.**

Finally, these results are significant because they highlight the role of interoception with breathing and embodiment. **Participants who scored higher on the interoceptive tasks were less likely to be influenced by the different scenarios.** This suggests that more interoception may lead to higher levels of corporeal awareness and, thus, less susceptibility to bodily illusions.

I believe these results are especially significant because breath practices build interoception, simply because you become aware of your breathing and body. So, the more you practice, the more interoception you'll develop, and thus the greater your sense of corporeal awareness will be. Stated simply, you will know yourself and be less susceptible to outside influences. But this is just my speculation.



1 Big Takeaway

Even without conscious awareness, breathing is essential to your feeling of body ownership and, most significantly, to your sense of body agency.

1 Practical Application

Breath Awareness + Breath Practices

Because this study found that breathing uniquely contributes to our feeling of body ownership and agency, practicing breath awareness may increase these aspects of corporeal awareness and interoception, reducing your susceptibility to bodily illusions.

An easy practice you can do anytime is to stop and notice where you feel the sensations of breathing in your body, in whatever body position you are currently occupying. It's that simple.

Moreover, my subjective interpretation of these results suggests that voluntary control of breathing may increase our sense of agency even more. Thus, regularly taking time **to perform any breathing practice you enjoy** could increase your sense of self-agency, which may have positive ripple effects on all aspects of your health and well-being.



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PAPER](#)

Citation

Pagaduan J, Wu SS, Kameneva T, Lambert E. Acute effects of resonance frequency breathing on cardiovascular regulation. *Physiol Rep.* 2019 Nov;7(22):e14295. doi: 10.14814/phy2.14295. PMID: 31782265; PMCID: PMC6882954.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Acute effects of resonance frequency breathing on cardiovascular regulation

4 FUNDAMENTALS

1. Essential Background Material

This study examined differences in autonomic nervous system and cardiovascular outcomes from slow breathing at a person's resonant frequency (RF) compared to one breath above their RF ($RF + 1$).

RF breathing is defined as the breathing rate that synchronizes heart rate oscillations with the breath and maximizes heart rate variability (HRV).

(Note: Most peoples' RF is between 4.0-6.5 breaths/min, with 5.5 breaths/min being most common. However, there's a big range, so I'll use 3-7 breaths/min throughout the 411 to account for this.)

It is theorized that breathing at our RF will amplify the benefits of our slow breathing practice.

However, in practice, it is unclear if this theory holds true. Countless studies have shown positive results for both RF breathing and "plain old slow breathing" (usually a set pace of 3-7 breaths a minute).

So, it begs the question: **How important is breathing at our precise RF? And is breathing at a comfortable pace between 3-7 breaths/min good enough?**

This study aimed to help address these questions by examining the differences between breathing at RF and $RF + 1$.



2. What Did this Research Do?

Ten healthy participants took part in the study (**an incredibly small sample size...but we have to take what we can get with breathing studies**).

On the first day, participants were taught to breathe slowly in a semi-supine position using a nasal inhale and pursed-lips exhale. RF was defined as the breathing rate that maximized low-frequency HRV (LF-HRV).

(Nick's personal note: This is one problem I have with "resonance." There's no perfect characterization of it. Typically, it's the rate that maximizes LF-HRV or the mix-max heart rate difference, along with other factors. Even though there is [a published protocol](#) for finding RF, it's pretty complicated & subjective.)

On the second day, the volunteers came in and performed the following protocol:

- 10-min of baseline measurements
- 10-min of RF (or RF + 1) breathing
- 10-min rest
- 10-min of RF (or RF + 1) breathing

The RF or RF + 1 breathing was randomized.

Key measurements were:

- HRV
- Muscle sympathetic nerve activity (MSNA)
- Blood pressure
- Baroreflex function

3. What Were the Major Findings?

Overall, there were no significant differences between RF and RF + 1 breathing. Both slow breathing methods lowered blood pressure, reduced sympathetic nervous system activity, and increased HRV.

Here are some of the specifics:

- RF breathing lowered blood pressure slightly more (-4.6 mmHg) than RF + 1 (-3.1 mmHg), but it did not reach significance.
- Sympathetic nervous system activity was similarly reduced for both RF and RF + 1 as measured by MSNA burst frequency and phase.
- Results suggested that a session of at least 10 min is needed to see changes in nervous system activity via MSNA measurements.



- Both breathing approaches improved baroreflex efficiency, but neither affected the “baroreflex function” (there seemed to be some unresolved problems with measuring this).

And just to reiterate, a significant limitation of these results is that there were only 10 participants.

4. Why Do These Results Matter?

Determining (and breathing at) your precise RF can be complicated, and it costs money to buy the necessary equipment. Moreover, a recent study [published in Nature](#) (411 coming soon) showed that our RF can frequently change.

Here, we learn that even if we breathe close to our RF, we get the same cardio-autonomic benefits. **Thus, this paper provides evidence that simply breathing at a comfortable rate between 3-7 breaths a minute is “good enough.”**

Lastly, it’s important to acknowledge that this isn’t the only study to analyze RF versus non-RF slow breathing. Some have found that you’ll get most of the benefits if you breathe “pretty close” to your RF. Others have found that RF breathing does provide more benefits. So, only time will tell, but slow breathing between 3-7 breaths/min still gives us many benefits without any extra equipment.

1 BIG TAKEAWAY

There are no significant cardio-autonomic differences between RF and RF + 1 breathing. This suggests that slow breathing at a comfortable pace, between 3-7 breaths per minute, is sufficient to get most of the benefits.

1 PRACTICAL APPLICATION

Practice slow breathing at any pace between 3 and 7 breaths a minute **that is most comfortable for you.** Use a nasal inhale and pursed-lips exhale and practice for at least 10 min per day to get the most benefits for your nervous system.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Respiratory Feedback in the Generation of Emotion

411 Released: August 4, 2023

Original Articles

Respiratory feedback in the generation of emotion

Pierre Philippot, Gaëtane Chapelle & Sylvie Blairy

Pages 605-627 | Published online: 09 Sep 2010

Download citation <https://doi.org/10.1080/02699930143000392>

[Get the Paper Here](#)

Article Citation

Philippot P, Chapelle G, Blairy S. Respiratory feedback in the generation of emotion. *Cognition & Emotion*. 2002 Aug 1;16(5):605-27.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

This groundbreaking study (21 years ago) found that emotional states elicit similar breathing patterns across people and, remarkably, breathing in specific ways can induce the corresponding emotional state. Meaning we can change our breath to change how we feel.



4 Fundamentals

1. Essential Background Material

- Emotions relate to our appraisal of a situation, facial expressions, physiological changes, and to our subjective feelings.
- One aspect of emotions that is unclear is whether physiological changes occur in response to emotions or if they actually trigger emotions or some combination of the two.
- Along these lines, the relationship between breathing and emotions is of particular interest because breathing is a physiological function that is both voluntary and involuntary. It is well-known to reduce (or increase) subjective feelings of anxiety, hinting that it's directly related to our emotional state.
- Moreover, before this study, research found generalized results showing that specific breathing patterns emerge under different emotional states. However, it is unclear if breathing itself can change our emotional state.
- This study analyzed whether certain breathing patterns are related to specific emotional states and whether breathing patterns themselves can induce emotional states without a person consciously knowing that's the goal of the breathing pattern.

2. What Did this Research Do?

Study Part 1

The first part of the study had 23 participants generate emotional states and then describe what happened to their breathing via a questionnaire. Specifically, **they were asked to generate joy, anger, fear, and sadness** by modifying their breathing and using memories or imagination.

Once they felt they achieved the desired emotional state, they were asked to answer a questionnaire describing how the following parameters of their breathing changed: frequency, amplitude, pauses, sighs, tremors, regularity, and tension in their chests.



Study Part 2

This part aimed to see if breathing patterns could elicit specific emotional states.

Specifically, 26 new participants were instructed to breathe in the joy, anger, fear, or sadness breathing patterns identified in Part 1 of the study for two minutes. Then, they answered a questionnaire about bodily sensations, with information about emotional states. They also wore an elastic tube around their chests so the researchers could ensure they followed the breathing directions correctly.

Crucially, to ensure the participants were unaware of the study's real goals, they were told that it was assessing how breathing impacted cardiovascular characteristics. Thus, any changes in emotional state will result from the breathing pattern itself and not the person's expectations that their emotional state should change.

3 & 4. What Were the Major Findings and Why Do They Matter?

Study Part 1

In Part 1, participants were able to elicit the emotional and feelings states using their breathing and memories/imagination.

- The responses to the questionnaire showed that they induced the emotional states with significant intensity. This is critical and obvious: to find breathing patterns associated with specific emotional states, we first must ensure they elicit the emotional state.

Across participants, similar breathing patterns emerged for each emotional state.

- This is significant because it shows that similar breathing patterns emerge across people during the same emotional states. Specifically, here were some of the commonalities:
 - **Joy:** Slower, deeper, through the nose, with a smooth rhythm and a relaxed ribcage.
 - **Anger:** Faster and slightly deeper nose breathing, with more irregularity, more tremors, and a tense ribcage.
 - **Fear:** Faster and more upper chest breathing with normal amplitude. More irregular, more tremors, and a tense ribcage.
 - **Sadness:** Approximately normal amplitude and pace, a slightly tense ribcage, and more frequent sighs.
-



- These results were then used as instructions in Part 2 of the study.

Study Part 2

The breathing patterns discovered in Study 1 elicited the related emotional states.

- This is the most significant finding of this study. It shows for the first time that breathing and emotions have a two-way street: we can breathe in certain ways to elicit specific emotions.

Some breathing patterns elicited the related emotional states better than others.

- Specifically, the joy and anger breathing patterns elicited the most similar corresponding feelings. The fear and sadness breathing patterns evoked mixed emotions, some matching expectations, and others not. It's not 100% clear why this happened. But, it appeared to be a combination of individual differences and slight differences in how well the participants followed the breathing instructions.

Overall, this is the first study to show that breathing patterns alone can change our emotional state and even induce specific emotions if the correct instructions are given.

- Just restating this because it's the most significant finding of the paper. Emma Seppälä, Ph.D., summarized these results perfectly: *"The finding that we can change how we feel by using our breath is revolutionary."*

1 Big Takeaway

Emotional states elicit similar breathing patterns across people, and remarkably, breathing in specific ways can induce the corresponding emotional state. Meaning we can change our breath to change how we feel.

1 Practical Application

Try one of the four breathing patterns discovered in Part 1 of the study for two minutes and see how you feel. And more broadly, the next time you're hit with an emotional storm, instead of trying to change your thoughts to change how you feel, consider changing your breath to change how you feel 🙏



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Ultra-Slow Mechanical Stimulation of Olfactory Epithelium Modulates Consciousness by Slowing Cerebral Rhythms in Humans

411 Released: September 16, 2024

Quick Summary

Mechanical nasal airflow stimulation led to significant increases in theta and delta brainwaves, which are typically associated with states of relaxation and creativity. It also induced an altered state of consciousness. This indicates that nasal airflow alone, independent of any attentional or postural factors, is sufficient to slow brainwave activity and alter our mental states.

Article Citation

Piarulli A, Zaccaro A, Laurino M, Menicucci D, De Vito A, Bruschini L, Berrettini S, Bergamasco M, Laureys S, Gemignani A. Ultra-slow mechanical stimulation of olfactory epithelium modulates consciousness by slowing cerebral rhythms in humans. Sci Rep. 2018 Apr 26;8(1):6581. doi: 10.1038/s41598-018-24924-9. PMID: 29700421; PMCID: PMC5919905.

[Link to Get the Paper](#)



4 Fundamentals

1. Essential Background Material

Research consistently shows that meditation increases and synchronizes theta brainwaves, frequencies which are associated with states like sleep, creativity, memory consolidation, and relaxation. While the exact cause of this shift in brainwaves is unclear, meditation often involves slower breathing, suggesting that nasal respiration may play a role.

For example, both human and animal studies show that nasal breathing influences activity across the whole brain, especially during inhalation. The olfactory epithelium, stimulated by incoming air, transmits signals to other brain regions. This effect may potentially affect meditation outcomes.

However, isolating these effects is difficult since meditation engages multiple brain regions through focused attention. Moreover, postures and mindsets used in meditation may impact results, making it challenging to determine the precise influence of slow nasal breathing.

To address these limitations, this study used a unique approach and found some rather remarkable results. Let's dive into what they did.

2. What Did this Research Do?

The study involved 12 participants (6 female, 6 male). Odorless air was mechanically delivered to their noses through a nasal cannula while lying down, with the rest of the nose blocked to prevent additional breathing. Thus, the participants actually breathed through their mouths during the experiment.

The air was delivered for 8 seconds, followed by 12 seconds of no air, mimicking a breathing rate of 3 breaths per minute. This nasal stimulation lasted 15 minutes, with pre- and post-measurement and rest periods of 7 minutes (~30 minutes of total experiment time).

Participants also took part in a sham control session with the same setup but no air delivery. Both sessions were randomly assigned, one week apart. EEG monitored brain activity, and a belt tracked respiration. Afterward, participants completed questionnaires assessing executive function, hippocampal function, and consciousness.



3 & 4. What Were the Major Findings and Why Do They Matter?

There was no change in actual breathing during both conditions.

- The participants' breathing remained consistent before, during, and after both nasal stimulation and the sham control. This means there were no reflexive breathing changes, cognitive efforts, or postural adjustments that could interfere with the results. **This is what makes this study so unique: they separated nasal stimulation from the act of conscious breathing, providing a clearer understanding of how airflow can influence the brain.**

Nasal stimulation led to widespread increases in theta and delta brainwaves.

- This was not the case for the sham control. This indicates that nasal stimulation—without conscious control or breathing reflexes—can effectively slow down brainwaves.

Nasal stimulation alone increases brainwave connectivity.

- They analyzed how information was flowing between different brain hubs and found that slow nasal stimulation enhanced information flow between brain regions in the delta and theta bands. Both unidirectional and bidirectional increases were observed, indicating stronger brain connectivity.

Certain brain hubs may transmit more information to others after nasal stimulation, acting as a focused a “source of information” for the brain.

- They performed an analysis to see if nasal stimulation made certain brain regions more dominant than others when it comes to information flow. While there were no overall significant changes, certain brain hubs did show trends toward significance, indicating a potential shift in importance to those regions. However, these results will require further research to be more conclusive.

Nasal stimulation alone led to the feeling of being in an altered state of consciousness.

- Even without changes in breathing or focused attention, participants reported that the nasal stimulation led to **altered perceptions of self and time, an inward-focused attention, and a diminished ability to control thoughts.** This suggests that nasal stimulation can impact consciousness independently of other components, such as activation of the parasympathetic nervous system.
-



1 Big Takeaway

Slow mechanical nasal airflow stimulation (independent of breathing reflexes, cognitive effort, and posture) significantly increases theta and delta brainwaves while increasing brain connectivity, resulting in an altered state of consciousness. This indicates that nasal airflow alone is sufficient to slow brainwave activity and alter our mental states.

1 Practical Application

While we usually focus on slow breathing at ~5-6 breaths per minute, this study suggests **trying ultra-slow breathing at around 3 breaths per minute**. So, if you can do it comfortably, try sitting or lying down and breathing with an 8-second inhale and a 10-12 second exhale for 5-15 minutes. This practice can noticeably slow your mind and provide a calming effect almost immediately.

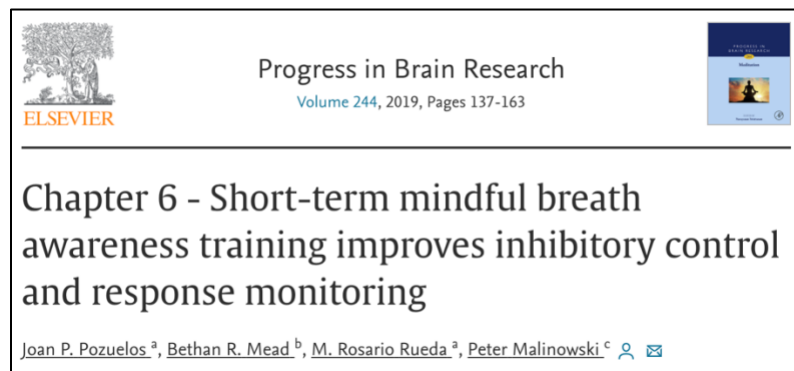
P.S. Keep in mind that consciously controlling our breath might influence the brainwave response. However, [a more recent study](#) confirmed that ultra-slow nasal breathing does slow brainwaves and enhance brainwave connectivity. So, it seems reasonable to think that deliberately practicing this technique will produce similar outcomes.

Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Mindful Breath Awareness Improves Impulse Control and Brain Dynamics

411 Released: August 18, 2023



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Article Citation

Pozuelos JP, Mead BR, Rueda MR, Malinowski P. Short-term mindful breath awareness training improves inhibitory control and response monitoring. Prog Brain Res. 2019;244:137-163. doi: 10.1016/bs.pbr.2018.10.019. Epub 2019 Jan 17. PMID: 30732835.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2019 study in *Progress in Brain Research* found that 17 minutes of mindful breath awareness five days a week for three weeks improved impulse control and brain dynamics related to the metacognitive processes of conflict and error monitoring.



4 Fundamentals

1. Essential Background Material

- Mindfulness meditation and mindfulness-based interventions generally have positive impacts on metacognition.
 - Metacognition (or meta-awareness) refers to the ability to control behavioral responses and the general awareness of one's thought patterns. Two critical metacognitive skills are:
 - **Inhibitory Control:** Control of our habitual reactions and urges, so we can do what is more appropriate in any situation.
 - **Response Monitoring:** Ability to monitor your actions and progress toward a predefined goal.
- An issue is that most studies on meditation use multiple components, such as group discussions and yoga, which make it challenging to determine the precise effects of meditation on metacognition.
- To understand the specific effects of meditation, it's critical that studies be performed on one meditation exercise rather than a mindfulness-based intervention with multiple non-meditative components.
- One attractive practice for such studies is mindful breath awareness because it captures the essence of all meditative techniques: focusing on one task (breathing), noticing when your mind wanders (meta-awareness), returning to the breath sensations, and assuming an open, non-judgmental attitude.
- Therefore, this study examined if the regular practice of mindful breath awareness improves metacognition, specifically inhibitory control and response monitoring.

2. What Did this Research Do?

The study had 21 participants that were divided into meditation and waitlist control groups.

The intervention (meditation or waitlist) lasted 3 weeks. The waitlist participants were instructed to continue normal activities during the three weeks. The meditation group received the following intervention:



- A 2-hour introduction to mindful breath awareness.
- Two 1-hour follow-up trainings (once per week).
- Daily practice of at least 10 min, 5 days a week.

The meditation instructions were as follows:

1. Sit in an upright but comfortable position.
2. Focus on the sensations of the breath at the nostrils or the abdomen region.
3. Do not change the breath; just notice it.
4. When the mind wanders, re-orient back to the breath.
5. Do not elaborate on or judge mind wandering; just notice it and re-orient to the breath.

Measuring Inhibitory Control and Response Monitoring

The participants received a Go/No-Go task, which quickly flashed a traffic light (for 500 ms). The participants had to click the mouse if it was green or withhold their response if it was red.

During the test, EEG was recorded to monitor brain activity. This data was used to assess specific brain signatures of metacognition, which have been defined in previous studies.

3 & 4. What Were the Major Findings and Why Do They Matter?

On average, participants meditated for approximately 17 minutes five times a week.

- That's a reasonable amount of time and much less than is typically dedicated to more extensive mindfulness-based interventions with multiple components.

The mindful group significantly reduced impulsive responses to the Go/No-Go stoplight test. However, no overall improvements in accuracy were discovered.

- This is meaningful because a major goal of mindful training is to reduce our impulsivity (i.e., to build better inhibitory control). This skill is critical to many health conditions (like addictions) and emotional regulation. Thus, although the accuracy did not change significantly, reducing impulsivity is an essential finding showing that the simple act of breath awareness does promote one intended outcome of meditation.

The mindful breath awareness practice improved conflict monitoring processes.



- EEG signatures revealed increased “centro-parietal N2 amplitude,” a brain indicator of better conflict monitoring. Conflict monitoring refers to noticing mental activity that conflicts with your goals. For example, during the mindful breathing exercise, conflict monitoring is noticing when your mind is thinking thoughts not associated with the meditation instructions. This awareness is transferrable to any goal you set for yourself.

Meditation increased EEG signatures of response inhibition, but they did not reach statistical significance.

- Previous research has shown that the “frontal P3 EEG signature” is linked to response inhibition. Although there were changes in the meditation group in the right direction, they did not reach significance. The authors were unsure why these effects were not greater, but it could be related to many complex interacting factors.

The meditation group increased signatures of error-related negativity (ERN), a robust way of assessing metacognition.

- ERN is a brain signature that occurs after a negative response (i.e., you realize you clicked the mouse when you shouldn’t have). It thus serves as a marker of self-monitoring during a given task (here, the Go/No-Go test). This finding was significant because they also showed that it was strongly correlated with the time spent meditating, meaning people who meditated more showed greater ERN, suggesting this is a direct result of meditation practice.

1 Big Takeaway

Practicing mindful breath awareness for 17 minutes, five days a week, for three weeks improved impulse control and brain dynamics related to the metacognitive processes of conflict and error monitoring, even in a task outside the meditation practice itself.

1 Practical Application

Add a form of mindful breath awareness to any contemplative practice (e.g., a breathwork practice, yoga, or walking) to improve impulsivity and brain dynamics related to metacognition, enhancing your ability to stay focused on goals even outside of your formal practice.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Nasal Nitric Oxide and Regulation of Human Pulmonary Blood Flow in the Upright Position

411 Released: June 9, 2023

> J Appl Physiol (1985). 2010 Jan;108(1):181-8. doi: 10.1152/jappphysiol.00285.2009.
Epub 2009 Oct 29.

Nasal nitric oxide and regulation of human pulmonary blood flow in the upright position

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Affiliations + expand

PMID: 19875719 DOI: 10.1152/jappphysiol.00285.2009

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Article Citation

Sánchez Crespo, A., Hallberg, J., Lundberg, J. O., Lindahl, S. G., Jacobsson, H., Weitzberg, E., & Nyrén, S. (2010). Nasal nitric oxide and regulation of human pulmonary blood flow in the upright position. Journal of applied physiology, 108(1), 181-188.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2010 study found that nasal breathing redistributes blood flow from the bottom to the top of the lungs and from the back to the front due to nitric oxide. This suggests that nasal nitric oxide may be an evolutionary adaption to counteract gravity and allow us to walk upright.



4 Fundamentals

1. Essential Background Material

- Blood flow in the lungs is critical for gas exchange (more blood flow = better gas exchange) and for fighting infections (areas of the lung with less blood flow are more susceptible to infection).
- In general, gravity makes blood flow greater in the lower portions of the lungs.
- However, blood flow can also be quite heterogeneous in human lungs, indicating that other factors besides gravity play a role.
- The authors of this study hypothesized that inhalation of nasally-derived nitric oxide (NO), a potent vasodilator, may be essential to controlling lung blood flow distribution.
- Moreover, because nasal NO is found mainly in upright primates, they hypothesized that it may have been an evolutionary adaptation to counteract the effects of gravity in the upright position.

2. What Did this Research Do?

This study used two radiotracers to map blood flow in the lungs.

The experiment had 14 participants that were broken into three groups:

- **Control Group (N=3):** This group was used to ensure the two tracers worked as intended.
 - **Nasal Group (N=7):** This group sat upright and breathed through their mouths for 20 min. Then, they were injected with a tracer and switched to nose-in/mouth-out breathing for 10 min. At the end of the 10 minutes, they were injected with a second tracer.
 - **Mouth NO Group (N=4):** This group did the same protocol as the Nasal Group. However, instead of switching to nose breathing, they continued mouth breathing, but with exogenous NO added at levels close to those produced by the paranasal sinuses. *This experiment was performed to see if NO explained blood flow changes between nasal and mouth breathing or if it was some other mechanism associated with the nose.*
-



3 & 4. What Were the Major Findings and Why Do They Matter?

Nasal breathing, and mouth breathing with added NO, both redistributed blood flow throughout the lungs in similar magnitudes.

- This is a critical finding because it shows that *NO causes blood flow redistribution*, not some other factor associated with nasal breathing.

Total lung blood flow shifted from the bottom third to the top third of the lung, resulting in a 24% net increase in blood flow in this upper region. (*Blood flow in the middle third of the lung remained basically unchanged.*)

- This indicates that NO specifically redistributes blood flow to counteract gravity, moving more from the bottom to the top of the lung. *This suggests (but doesn't prove) that it may indeed be an evolutionary adaptation to gravity.*

Blood flow also shifted from the back of the lungs toward to center and front of the lungs, resulting in a 7% net increase in blood flow to each of these regions.

- This is meaningful because it shows that NO doesn't just redistribute blood flow from bottom-to-top, but also from back-to-front.

Altogether, these results help explain how nasal breathing improves oxygenation, protects against infection, and counteracts the effects of gravity.

- **Oxygenation:** We've learned that both nasal breathing and mouth breathing with NO increase oxygenation by 10-18%. This study's results provide one reason why: blood flow is redistributed in the lungs, improving gas exchange.
 - **Fighting Infection:** Lung infections occur more frequently in regions of the lungs that receive less blood flow. Thus, nasal NO may have been an evolutionary adaptation to redistribute lung blood flow to help protect all areas of the lungs.
 - **Counteracting Gravity:** In their words: "*In this work, we have demonstrated that nasal breathing counteracts the effects of gravity on pulmonary blood flow in the upright position by redistribution of blood to the nondependent lung regions.*" 🙌
-



1 Big Takeaway

Due to nitric oxide, nasal breathing redistributes blood flow from the bottom to the top of the lungs and from the back to the front. This suggests that nasal nitric oxide may have been an evolutionary adaption to counteract gravity and allow us to walk upright.

1 Practical Application

This one is straightforward: breathe through your nose as much as possible (and perhaps practice nose-in/mouth-out now and then) to maximize NO delivery, improve blood flow distribution in the lungs, and potentially reduce your risk of lung infections.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Mindful Breath Awareness Meditation Facilitates Efficiency Gains in Brain Networks: A Steady-State Visually Evoked Potentials Study

411 Released: July 21, 2023

[nature](#) > [scientific reports](#) > [articles](#) > [article](#)

Article | [Open Access](#) | [Published: 12 September 2018](#)

Mindful breath awareness meditation facilitates efficiency gains in brain networks: A steady-state visually evoked potentials study

[Benjamin Schöne](#), [Thomas Gruber](#), [Sebastian Graetz](#), [Martin Bernhof](#) & [Peter Malinowski](#) 

[Scientific Reports](#) **8**, Article number: 13687 (2018) | [Cite this article](#)

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Article Citation

Schöne B, Gruber T, Graetz S, Bernhof M, Malinowski P. Mindful breath awareness meditation facilitates efficiency gains in brain networks: A steady-state visually evoked potentials study. Sci Rep. 2018 Sep 12;8(1):13687. doi: 10.1038/s41598-018-32046-5. PMID: 30209327; PMCID: PMC6135840.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2018 study in Nat Sci Rep found that the brief but consistent practice of mindful breath awareness by itself improves our attention, our ability to resist distractions, and the efficiency of neural networks responsible for these processes.



4 Fundamentals

1. Essential Background Material

- Mindfulness and mindfulness-based interventions improve attention, working memory, and meta-awareness. These enhancements are generally associated with better emotional outcomes and improved well-being.
- However, most studies employ interventions with several modalities, such as multiple forms of meditation, group discussions, and yoga. This makes it challenging to determine the precise role of mindfulness (and the type of mindfulness) in the observed benefits.
- To make mindfulness more accessible, it's important that we move away from generalized statements about mindfulness interventions in general and into more practical statements on which type of meditation to do and how long to do it.
- One accessible form of meditation is mindful breath awareness. This practice is of particular interest because it's simple and draws on the essential components of mindfulness: focusing on one task (breath sensations), noticing when your mind wanders, returning to the breath sensations, and assuming an open, non-judgmental attitude.
- This study aimed to assess if the practice of mindful breath awareness (by itself) altered sustained attention and visual working memory and if these changes were associated with changes in brain patterns.

2. What Did this Research Do?

Thirty-four participants without meditation experience were randomly assigned into meditation (N=17) or progressive muscle relaxation (N=17) groups:

Meditation Practice: Mindful breath awareness, focusing on the sensations of the breath either at the tip of the nostrils or in the abdomen. Participants were instructed not to change their breath, just observe it. When their minds wandered, they were instructed to become aware of the distraction and return to breathing sensations. Lastly, they were instructed to assume an accepting and non-judgmental attitude.



Progressive Muscle Relaxation: Deliberate tensing and relaxing of muscles in a pre-defined sequence. Participants focused on the difference between the two states (tension and relaxation), helping to decrease muscle tension and stress.

The study lasted 8 weeks. Over that period, participants met with an instructor 6 times for 1.5-hour sessions. They were asked to perform at least one home practice a week (a very minimal commitment, in my opinion).

Cognitive and Brain Assessments

MOT

Cognitive changes were measured through the multiple objects tracking (MOT) task. In this task, dots are placed on a screen. Then, 2 to 5 dots flash a different color, indicating you should follow them. The dots then move around, and, at the end, they flash one dot, and you must identify whether it was a dot you were supposed to be tracking.

This task is a good measure of sustained attention and visual working memory under distracting conditions.

SSVEP

EEG data were used to analyze steady-state visual evoked potentials (SSVEP). The amplitude of the SSVEP is interpreted as **“the amount of neural resources devoted to the perception of an exogenous stimulus.”** So, if the SSVEP goes down for a given task, it suggests the efficiency of those neural networks has improved.

3 & 4. What Were the Major Findings and Why Do They Matter?

Participants meditated for about 10 minutes 3-4 times a week.

- More precisely, they meditated for an average of 41.71 minutes over 3.86 weekly sessions. That’s an incredibly small amount of meditation.

The meditation group significantly improved their scores on the MOT task (the dot-tracking task), while the progressive muscle relaxation group did not.



- This is extremely meaningful because it shows that the short, consistent practice of mindful breath awareness by itself can improve attention and working memory in the face of distractors (a superpower in today's world).

The SSVEP amplitudes were significantly smaller after training in the breath awareness group but not in the progressive muscle relaxation group.

- This, in combination with the improved MOT, is noteworthy because it suggests that the breath awareness practice *improved neural efficiency*. It also trained participants to ignore distractions (or separate from them quickly), keeping only relevant information in visual short-term memory. Again, this is a superpower in today's world.

There were no positive psychological outcomes from either practice (the meditation group actually got a little worse).

- Although not the main point of the study, questionnaires revealed that the negative affect of the meditation group went up slightly. They provided several arguments for why this might happen. To me, the most compelling was that in traditional meditative practices, attention is first developed, which is the foundation for improving mental health. Thus, given that the participants didn't meditate much, this could explain the lack of positive results here: They just needed more time with mindfulness.

1 Big Takeaway

The brief but consistent practice of mindful breath awareness improves our attention, our ability to resist distractions, and the efficiency of neural networks responsible for these processes.

1 Practical Application

Here's a simple mindful breath awareness based on the paper:

1. Sit comfortably with a straight back (or lay down, as long as you stay awake).
 2. Focus on the sensations of your breathing at the tip of your nose or in your abdomen.
 3. When you notice your mind has wandered, non-judgmentally say, "oh well," and return to your breath sensations.
 4. Continue for ~10 minutes daily.
-



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

The Effect of Slow-Paced Breathing on Cardiovascular and Emotion Functions: A Meta-Analysis and Systematic Review

411 Released: March 9, 2024



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Article Citation

Shao, R., Man, I.S. and Lee, T., 2024. The Effect of Slow-Paced Breathing on Cardiovascular and Emotion Functions: A Meta-Analysis and Systematic Review. *Mindfulness*, pp.1-18.

Quick Summary

This 2024 meta-analysis published in *Mindfulness* found that slow-paced breathing, at least in the short term, has positive effects on the cardiovascular system, and these physiological changes are associated with a modest reduction in negative emotions.

Reading Time: 2 min 47 seconds

I hope the next 41'ish breaths are the most nourishing of your day.



4 Fundamentals

1. Essential Background Material

- Slow breathing at around 6 breaths a minute is a highly effective and commonly used way to reduce stress ([see this meta-analysis for more](#)).
- Because stress is related to the cardiovascular system via the hypothalamus-pituitary-adrenal (HPA) axis, this reduction in stress provides a pathway for slow breathing to benefit heart health as well.
- In general, research shows that slow breathing does help cardiovascular health by increasing heart rate variability (HRV) and baroreflex sensitivity while reducing blood pressure.
- However, no meta-analyses have directly analyzed the association between the cardiovascular effects of slow breathing (such as those listed above) and the emotional outcomes (such as reduced stress).
- This paper aimed to provide an updated review that brought together slow breathing's benefits for cardiovascular and emotional health while looking at how the two are associated in both the short and long term.

2. What Did this Research Do?

Thirty-one studies with a total of 1133 participants were included in this analysis.

The studies were selected if they included slow-paced breathing and cardiovascular outcomes. Moreover, they also found a subset of papers that included both cardiovascular and emotional outcomes.

The slow breathing intervention of each study needed to last at least 5 minutes, and the breathing needed to be performed at a rate less than 10 breaths a minute.

The outcomes examined were HRV, heart rate, blood pressure, and negative emotions. The analysis focused on changes occurring during and immediately after the practice compared to the pre-practice measurements.



Lastly, they examined the long-term effects of slow breathing, too. However, not enough studies were available to perform a meta-analysis (only two studies had post-intervention follow-ups). Thus, they could only perform a systematic review of long-term outcomes.

3 & 4. What Were the Major Findings and Why Do They Matter?

Slow breathing led to a small but significant reduction in heart rate immediately after the practice, but not during.

- In general, this represents a beneficial effect on cardiovascular health and an activation of the relaxation response.

Slow breathing significantly improved time-domain HRV measurements (RMSSD and SDNN) during and after the practice.

- This generally indicates a beneficial outcome for the cardiovascular system and greater activity of the calming parasympathetic nervous system. However, [as we've discussed](#), the exact meaning of all the different HRV parameters is not straightforward, and studies consistently contradict each other.
- With that said, there seems to be a correlation between RMSSD and better cardiovascular health, making this finding meaningful nonetheless.

Low-frequency HRV went up significantly during the practice but not after. No other frequency-domain HRV index showed consistent changes.

- This is meaningful because [LF-HRV is directly associated](#) with activating the parasympathetic nervous system. So, this result confirms that slow breathing does activate the parasympathetic nervous system, at least during the practice.

Systolic (but not diastolic) blood pressure was significantly lower immediately after the practice and showed long-term improvements, too.

- Two studies examined long-term effects on blood pressure. It wasn't enough for a meta-analysis, but they both did show improvements. However, these studies used prehypertensive patients, which may have skewed the results.

Interestingly, for negative emotions, there was only a marginal improvement.



- I would have guessed this would have been the most significant improvement. Nonetheless, it did show enhancements, but they were only marginal.

For the 7 studies that included emotional and physiological results, a correlation analysis showed that reductions in heart rate and increases in HRV were positively correlated (correlation is not causation, though) with improvements in emotion, but this did not reach significance.

- This is meaningful because it does show that the physiological changes from slow breathing are at least associated with the emotional benefits. More studies with both measurements are likely needed to get more concrete findings.

1 Big Takeaway

Slow-paced breathing, at least in the short term, has positive effects on the cardiovascular system, and these physiological changes are associated with a modest reduction in negative emotions.

1 Practical Application

Pick almost any breathing rate less than 10 breaths a minute that is comfortable for you. Focus on conscious, nasal, diaphragmatic breathing for at least 5 minutes to quickly improve emotional and cardiovascular health. Most guided practices on the [practice page](#) can help you with this.

[DOWNLOAD
PAPER](#)

Citation

Tonelli AR, Aulak KS, Ahmed MK, Hausladen A, Abuhalmeh B, Casa CJ, Rogers SC, Timm D, Doctor A, Gaston B, Dweik RA. A pilot study on the kinetics of metabolites and microvascular cutaneous effects of nitric oxide inhalation in healthy volunteers. PLoS One. 2019 Aug 30;14(8):e0221777. doi: 10.1371/journal.pone.0221777. PMID: 31469867; PMCID: PMC6716644.

411 Created On:
April 22, 2022

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

A pilot study on the kinetics of metabolites and microvascular cutaneous effects of nitric oxide inhalation in healthy volunteers

4 FUNDAMENTALS

1. Essential Background Material

Inhaled nitric oxide (NO) has many body-wide impacts. For example, it can protect against brain injury and cause vasodilation and increased blood flow in distant regions. However, it has remained unclear how inhaled NO exerts these effects.

In general, inhaled NO is believed to react away quickly after reaching the lungs. Thus, the systemic benefits of NO could simply be due to improved gas exchange in the lungs, which then has positive downstream impacts. However, another explanation could be that metabolites of NO transport it around the body, where bioactive NO can then reform under specific conditions (for instance, in areas of low oxygen).

Interestingly, despite its widespread clinical use, there have been very few studies testing this hypothesis to truly discover how inhaled NO exerts its systemic effects. This paper sought to fill that gap.



2. What Did this Research Do?

The researchers recruited 15 healthy volunteers. They had them inhale NO at concentrations of 40 ppm (the maximum produced in the paranasal sinuses is on the order of ~20 ppm, but typically much less). They inhaled the added NO for 15 minutes. Blood samples were collected before inhalation, at the end of the 15 minutes of inhalation, and then at 5, 15, and 30 minutes post-inhalation.

3. What Were the Major Findings?

The results were quite striking. **They found that NO inhalation significantly increased circulating levels of SNO-Hb and nitrite.** This is important because SNO-Hb plays a significant role in whole-body oxygenation. For example, a 2015 PNAS study discovered that SNO-Hb “senses” areas of low oxygen, and then releases bioactive NO to increase blood flow and oxygenation. This discovery led to breathing be considered as a three-gas system involving oxygen, carbon dioxide, and NO. All of this is to say that if inhaling NO increases SNO-Hb, it could be playing a critical role in whole-body oxygenation.

They also observed increases in circulating nitrite. This is important because, like SNO-Hb, nitrite can also [release bioactive NO in regions of hypoxia](#). However, nitrite can do this independent of the hemoglobin, thus providing a “back-up mechanism” for increasing blood flow in regions of low oxygen.

Lastly, another interesting finding from this study was that nitrite levels were most significant at the 5-min post inhalation mark. In contrast, SNO-Hb continued rising throughout the 30 minutes. This led the authors to believe that the lungs might be acting as an SNO-Hb reservoir, releasing it “as needed.”

4. Why Do These Results Matter?

When we breathe through our nose, we carry NO into the lungs (although not at concentrations as high as those studied here). Based on these findings, we can now be reasonably confident this NO enters the bloodstream and is carried as SNO-Hb and nitrite. **Thus, breathing through your nose might not just improve gas exchange in the lungs. It might also help make sure oxygen gets delivered where it is needed most throughout the body.**

Additionally, their finding that SNO-Hb levels continued increasing after NO inhalation is intriguing. It might support the idea that nose breathing provides a baseline level of NO that keeps SNO-Hb in its normal range. Then, when excess



NO is inhaled, the body stores that "just in case." This is speculative, but interesting to think about.

1 BIG TAKEAWAY

For the first time, it has been shown that metabolites of inhaled nitric oxide enter the blood stream and are carried systemically throughout the body in forms that can convert back to bioactive NO. This provides a mechanism for how inhaled NO has distant effects.

1 PRACTICAL APPLICATION

This is one study, and it's relatively new. So, we'll need more to confirm/deny that NO inhalation consistently increases SNO-Hb and nitrite across different populations. **In the meantime, the one application here is to simply breathe through our noses as much as possible. It may just be the key to whole-body oxygenation.**



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PAPER](#)

Citation

Wilson T, Kelly KL,
Baker SE. Review: Can
yoga breathing
exercises improve
glycemic response and
insulin sensitivity. J
Yoga Phys Ther.
2017;7:270.

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April 24, 2022

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

**Review: Can yoga breathing exercises
improve glycemic response and insulin
sensitivity?**

4 FUNDAMENTALS

1. Essential Background Material

This review examined how yogic breathing (pranayama) might help as a non-pharmacological intervention for improving insulin sensitivity and blood sugar control.

(Note that this was not a systematic review or meta-analysis. It's more of a general overview article.)

Yoga, in general, has been shown to improve diabetes control. However, yoga involves many aspects other than just breathing (like physical movement). But not everyone can perform physical activity, especially people with chronic diseases.

Encouragingly, a study published in 2013 showed that a relaxation breathing technique lowered blood glucose in healthy college-aged participants. This suggests that pranayama alone could benefit insulin sensitivity and blood sugar control.

Here, they wanted to examine this proposition further and see if yogic breathing by itself could benefit insulin sensitivity and blood sugar control. This would be helpful since breathing exercises can generally be performed safely by everyone.



2. What Did this Research Do?

This wasn't a traditional paper with a "Methods" section. Instead, it was broken into three subsections, and relevant information was cited for each one. Here are the areas they examined:

- Glucose, Insulin Sensitivity, and Stress
- Effect of Pranayama on Insulin Sensitivity and Glucose
- Sleep Apnea and Diabetes

3. What Were the Major Findings?

Glucose, Insulin Sensitivity, and Stress

No surprise here: stress and an overactive sympathetic nervous system are two key areas that pranayama techniques help with. Interestingly, the liver generates glucose via gluconeogenesis, and when the sympathetic nervous system is activated, it increases this process. We shift out of a sympathetic state and into a more parasympathetic dominant state by breathing slowly. **Thus, slow yogic breathing that promotes relaxation might reduce the amount of glucose produced by the liver and help reduce our blood sugar.**

Insulin Sensitivity and Glucose Control

Despite their intention to focus solely on pranayama, they referred to several studies that involved pranayama and yoga poses. Although these studies showed improvements in insulin sensitivity, there was no way to tease out the importance of breathing versus physical exercises.

However, one study (their own from 2013, mentioned above) only used breathing. Here, it was shown that slow breathing for 30 min reduced the blood sugar spike associated with an oral glucose tolerance test. **This provides some evidence that pranayama, independent of physical poses, can benefit blood sugar and insulin sensitivity.**

Sleep Apnea and Diabetes

There were no concrete results here, just speculation. The authors suggest that pranayama breathing techniques might reduce the severity of sleep apnea. This, in turn, would then help improve insulin sensitivity and fasting blood sugar levels (since sleep apnea harms these significantly). It seems like a reasonable assumption, in my opinion, but there were no studies to back it up.



4. Why Do These Results Matter?

Although the evidence was limited, this overview suggests that breathing exercises could be a beneficial and cost-effective add-on for people suffering from conditions that involve insulin resistance and impaired glucose control. These pranayama breathing exercises could be particularly helpful in regions of low socioeconomic status.

1 BIG TAKEAWAY

Yogic breathing practices that focus on slow breathing and parasympathetic activation might reduce the body's endogenous production of blood sugar and help improve insulin sensitivity.

1 PRACTICAL APPLICATION

Use slow breathing (almost any rate and ratio that promotes relaxation will do) to potentially improve insulin sensitivity and reduce blood sugar.

In particular, the study that found impressive improvements in blood sugar following an oral glucose tolerance test used the following exercise:

1. Inhale 2 Seconds, Exhale 1 Second
2. Inhale 2 Seconds, Exhale 2 Seconds
3. Inhale 2 Seconds, Exhale 3 Seconds
4. Inhale 2 Seconds, Exhale 4 Seconds
5. ... (Exhale keeps getting longer) ...
6. Inhale 2 Seconds, Exhale 10 Seconds

You can count in your head (I sometimes use this to help me fall asleep) or use the [Breathwrk app](#), which has this breath pre-programmed for you. On the app, it's called the "Relaxation" breath, and it can be found in the "Calming" section (not to be confused with "Deep Relaxation," which is an entirely different exercise on the app).

Give it a shot and see how you feel.



[DOWNLOAD
PAPER](#)

Citation

Yadav G, Mutha PK.
Deep Breathing
Practice Facilitates
Retention of Newly
Learned Motor Skills.
Sci Rep. 2016 Nov
14;6:37069. doi:
10.1038/srep37069.
PMID: 27841345;
PMCID: PMC5107920.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Deep Breathing Practice Facilitates Retention of Newly Learned Motor Skills

4 FUNDAMENTALS

1. Essential Background Material

This study aimed to assess whether controlled alternate nostril breathing could improve the learning of motor skills. Wikipedia tells us that “A *motor skill* is a function that involves specific movements of the body's muscles to perform a certain task. ... In order to perform this skill, the body's nervous system, muscles, and brain have to all work together.” So, if breathing can help with these most fundamental of skills, that would be meaningful.

And we know that controlled breathing does do many incredible things related to the brain and memory. For instance, slow deep breathing has been shown to increase pain threshold to hot and cold exposure. Nasal breathing has also been shown to synchronize brainwave activity and improve response times compared to mouth breathing. Alternate and uni-nostril breathing have improved verbal and spatial performance, suggesting a strong influence over cognitive functioning.

However, it is unclear if these benefits of breathing will translate to learning motor skills. But, the authors of this paper hypothesized they might because learning and retaining motor skills requires the activation of many different brain regions that are also activated by controlled breathing. And, as we will see, they were remarkably correct.



2. What Did this Research Do?

The researchers found 40 participants (31 male, 9 female) aged 18 to 27 to take part in the study. They had them draw a circle on an electronic tablet (sort of like signing your name on an electronic pad). However, they could not actually see their hand; they could only see the drawing as it appeared on the screen.

And importantly, they had bounds within which they had to keep their circle. If they went outside of those bounds, errors began accumulating. They also had to draw the circle within 2.1 seconds. So, time and spatial constraints to this task provided a quantifiable error.

They drew the circle 100 times to minimize their error. Then, they rested for 30 minutes. The control group (n=14) sat quietly during this 30-min period. The breathing group performed 30 minutes of alternate nostril breathing (ANB) at a pace of about 7.5 breaths per minute using a 2-second inhale, 2-second hold, and 4-second exhale.

Then, both groups were tested again on their ability to draw the circle. They were also tested again 24 hours later to see how the breathing affected long-term motor skill retention.

3. What Were the Major Findings?

This is where it gets insane :)

Imagine you do something 100 times. You feel like you have it. Then, you rest for 30 minutes and do it again, only to find out you are almost the same as when you first started—frustrating, right? That's what happened to the control group.

However, the ANB group basically didn't lose any of their skill during the 30-minute rest. Their results looked almost identical to where they left off before the 30-min break. **That is, they learned the motor skill better due to the breathing.**

But it gets even crazier. At the 24-hour retest, the breathing group looked just like they did the day before, whereas the control again looked like they had never taken the test before.

Overall, these results showed that the ANB group significantly outperformed the control group. They retained the motor skill better after the 30-min break, and they also improved long-term motor skill retention 1 day later, with no additional breathing.



One 30-minute session. One day of improved learning and retention.

4. Why Do These Results Matter?

I think the importance of these results is immediately apparent. This research showed that performing ANB after learning a motor skill improves both immediate and long-term retention of that skill. Thus, a simple breathing exercise (**and only one session**) could be used to improve short and long-term learning of motor skills.

These results could be immediately applicable in sports training. Moreover, ANB could also potentially be used in people relearning motor skills after an injury.

1 BIG TAKEAWAY

We can use slow alternate nostril breathing immediately after learning a skill to improve that skill's short and long-term retention. **More broadly and simply stated, this study supports using ANB after a task for enhanced learning.**

1 PRACTICAL APPLICATION

Perform slow alternate nostril breathing immediately after learning a new skill if you're looking for a quick and effective way to improve your retention of that skill.

Using a breathing app, use the following settings (or just count in your head like they did in this study):

- Inhale: 2 seconds
- Hold: 2 seconds
- Exhale: 4 seconds

Or, use any variation that maintains that ratio that is comfortable for you (e.g., 3:3:6; 4:4:8; etc.).



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Citation

Yau KK, Loke AY. Effects of diaphragmatic deep breathing exercises on prehypertensive or hypertensive adults: A literature review. Complement Ther Clin Pract. 2021 May;43:101315. doi: 10.1016/j.ctcp.2021.101315. Epub 2021 Jan 26. PMID: 33530033.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

Effects of diaphragmatic deep breathing exercises on prehypertensive or hypertensive adults: A literature review

4 FUNDAMENTALS

1. Essential Background Material

Hypertension and prehypertension (high blood pressure) are present in about 1 in 4 people globally. They both increase the likelihood of cardiovascular disease, making it critical that people control their blood pressure.

These conditions are characterized by increased sympathetic nervous system activity and decreased parasympathetic nervous system activity, especially during stressful situations ([see Thought #2 here for more](#)).

Slow deep breathing (SDB) has been shown to [reduce blood pressure](#) and improve nervous system health by reducing sympathetic activity. Moreover, [SDB reduces psychological stress and anxiety](#), critical drivers of the excess sympathetic activity causing high blood pressure.

So, SDB is a proven treatment option for controlling high blood pressure.

*The uniqueness of this literature review was that it synthesized findings to provide practical guidance for voluntary and device-guided SDB, **specifically for people who already have hypertension and prehypertension.***

2. What Did this Research Do?



This review collected studies published since 2010 examining the effects of SDB (either voluntary or device-guided) in hypertensive and prehypertensive adults.

Overall, 13 studies were included in the review. Seven used voluntary SDB, where participants were taught to breathe slowly by counting in their heads. The other six used device-guided SDB, where a device provided the breathing cues.

Five studies had people with prehypertension with systolic blood pressure (SBP) from 120 – 139 mmHg and/or diastolic blood pressure (DBP) from 80-98 mmHg who were not using medication. The other 8 studies had people with stage I or II hypertension (SBP 140-160 and DBP 90-100) using medications.

All 13 studies differed in SDB protocol, length, and duration. For example, some had a breathing practice twice weekly, and others were twice daily. Significantly, however, all of them were rated “moderate to good” based on the “Quality Assessment Tool for Studies with Diverse Designs (QATSDD).”

3. What Were the Major Findings?

- **All 13 studies showed significant reductions in blood pressure in hypertensive and prehypertensive individuals.**
- Although less consistent, SDB also showed a general trend in lowering resting heart rate (only five studies measured resting HR, and 3 showed improvements).
- Three studies measured HRV and found significant improvements.
- Three studies included measurements of mean arterial pressure (MAP). Two found significant reductions in MAP.
- Only one study measured anxiety and depression scores. It showed a significant reduction in anxiety but not depression.
- One study measured quality of life and did not find significant changes.
- **Overall, they found that practicing slow breathing between 6 to <10 breaths a minute for at least 10 minutes a day can significantly improve blood pressure if practiced consistently for weeks or months.**

4. Why Do These Results Matter?

The finding across all 13 studies that slow breathing improves blood pressure in hypertensive and prehypertensive patients shows that **this simple intervention can potentially decrease the likelihood of cardiovascular complications in this**



susceptible population. The results also confirm that SDB reduces sympathetic activity and improves autonomic function in this population.

Moreover, the finding that MAP was reduced in two studies is meaningful. *A reduction in MAP indicates better blood flow to major organs during every cardiac cycle.* Better blood flow to vital organs further supports the notion that SDB can reduce the risk of cardiovascular-related problems in this population.

Lastly, these results are significant because they show that no device is necessary to get the benefits. This means we already have all the tools we need to reduce our blood pressure. It's cost-free and effective.

1 BIG TAKEAWAY

Slow breathing between 6 to <10 breaths a minute, with or without a device, for at least 10 minutes a day for 4+ weeks significantly reduces blood pressure, improves blood flow to vital organs, and improves autonomic dysfunction in people with hypertension or prehypertension.

1 PRACTICAL APPLICATION

Usually, such heterogeneity among studies is undesirable if we want to generalize the results. However, in this case, it's even more potent because we learn that we can use a device, count in our heads, **or use basically any method that gets our breathing to less than 10 breaths a minute** to get positive outcomes.

(Note: I put 6 to <10 breaths/min in the 1 Big Takeaway because that's what the study showed. But it seems likely that any comfortable slow breathing rate between 3-10 breaths/min will work.)

Thus, to apply this study's findings, perform slow breathing at a comfortable pace of less than 10 breaths/min for at least 10 minutes a day for ~4 weeks.

If you're unsure where to start, 6 breaths/min is usually the best place: Inhale for 4 seconds, exhale for 6. Count in your head or [use this app](#).

Enjoy improving your cardiovascular and autonomic health with this simple practice.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Immediate Impact of Yogic Breathing on Pulsatile Cerebrospinal Fluid Dynamics

411 Released: June 23, 2023



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Article Citation

Yildiz S, Grinstead J, Hildebrand A, Oshinski J, Rooney WD, Lim MM, Oken B. Immediate impact of yogic breathing on pulsatile cerebrospinal fluid dynamics. Sci Rep. 2022 Jun 28;12(1):10894. doi: 10.1038/s41598-022-15034-8. PMID: 35764793; PMCID: PMC9240010.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2022 study published in *Nature Scientific Reports* found that slow yogic breathing increases the flow of cerebrospinal fluid toward the head by 16-28%, which may help explain how slow breathing techniques improve brain health.



4 Fundamentals

1. Essential Background Material

- Cerebrospinal fluid (CSF) is a clear fluid with many vital roles for the brain and spinal cord, such as providing cushioning, delivering nutrients & hormones to the brain, and removing waste from the brain.
- Therefore, CSF plays a critical role in the health of the central nervous system and in brain diseases like Alzheimer's.
- For these reasons, understanding what drives and modifies CSF movement is crucial for potentially developing non-pharmacological interventions to improve brain health.
- Previous studies have shown that breathing directly impacts CSF flow and that the rate and depth of breathing are particularly important to CSF dynamics.
- Because of these effects, the authors of this paper hypothesized that the benefits of slow, deep, yogic breathing may be partially explained by changes in CSF flow (in addition to parasympathetic nervous system activity).

2. What Did this Research Do?

This study analyzed **the immediate impacts** of yogic breathing on CSF flow.

The experiment involved 18 participants. They had them learn four yogic breathing techniques:

1. **Slow Breathing:** Breathing slowly at a comfortable count between 3-5 seconds for the inhale and 3-5 seconds for the exhale, with no instruction on breathing depth.
 2. **Deep Abdominal Breathing:** Same optional slow breathing count as above, but with a focus on expanding the abdomen during inhalation and relaxing it during exhalation.
 3. **Deep Diaphragmatic Breathing:** Same optional slow breathing, but with a focus on expanding the ribs out to the side during inhalation while relaxing them on exhalation.
-



4. **Deep Chest Breathing:** Same optional slow breathing, but focusing on expanding the chest to the sides, front, and back during inhalation and relaxing during exhalation.

Note that numbers 2-4 make the components of the “yogic complete breath,” which involves expanding the abdomen, then the ribs, and then finally the chest, all in one breath. They isolated each part to see how different breathing depths impacted CSF dynamics.

For the experiment, the participants lay down in an MRI scanner and breathed spontaneously for baseline measurements. Then, they did one minute of each of the above breathing techniques, with a 30-60 second rest in between. They did each breathing technique twice:

1. Slow breathing (1-min)
2. Rest
3. Slow breathing (1-min)
4. Rest
5. Deep abdominal breathing (1-min)
6. Rest
7. Deep abdominal breathing (1-min)
8. Rest
9. And so on.

The results examined how CSF changed in each breathing condition compared to the spontaneous breathing measurements.

3 & 4. What Were the Major Findings and Why Do They Matter?

All four yogic breathing conditions increased the flow of CSF toward the head by 16-28%. Specifically, deep abdominal breathing increased it the most by 28%, deep diaphragmatic breathing by 23%, and slow breathing by 22%. Deep chest breathing increased it by 16%, which was not statistically significant.

- Because CSF is essential for brain waste clearance, these results suggest that this may be one way slow breathing techniques improve brain health. However, more studies are needed to directly study CSF with brain waste removal during yogic breathing.

Despite these significant increases from breathing, the pulsation of CSF was primarily driven by the heart rhythm. However, the 28% increase seen during deep abdominal breathing made the contribution from breathing as significant as from the heart.



- This separation of the heart and breath components is significant because it shows that the depth of breathing plays a critical role in CSF dynamics. Although all breathing conditions increased breathing's contribution to CSF flow, **only deep abdominal breathing** increased it to be comparable with the magnitude of the heart's influence on CSF. **Thus, the depth of breathing is critical to influencing CSF dynamics.**

Overall, deep abdominal breathing had the most significant impact on CSF flow.

- The authors hypothesized this is due to greater diaphragm activation and more significant changes in intrathoracic and intraabdominal pressures during deep abdominal breathing.

CSF moved toward the brain during inhalation and back down during exhalation.

- This is just a neat finding, which many previous studies have also shown. It relates back to intraabdominal pressure changes during the respiratory cycle.

1 Big Takeaway

Slow yogic breathing increases the flow of cerebrospinal fluid toward the head by 16-28%, with the most significant changes occurring during deep abdominal breathing. These findings may help explain how slow breathing techniques improve brain health.

1 Practical Application

This one is straightforward: Practice slow breathing (particularly slow abdominal breathing) at a comfortable rate of less than 10 breaths a minute to increase CSF flow and potentially improve brain health. You may also try out the full yogic breath if you feel so inspired :)



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Citation

Zaccaro A, Piarulli A, Laurino M, Garbella E, Menicucci D, Neri B, Gemignani A. How Breath-Control Can Change Your Life: A Systematic Review on Psycho-Physiological Correlates of Slow Breathing. Front Hum Neurosci. 2018 Sep 7;12:353. doi: 10.3389/fnhum.2018.00353. PMID: 30245619; PMCID: PMC6137615.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

How Breath-Control Can Change Your Life: A Systematic Review on Psycho-Physiological Correlates of Slow Breathing

4 FUNDAMENTALS

1. Essential Background Material

(Note: I think this paper wins “Best Title Ever” award, lol. Ok, on with the 411.)

There has been a lot of research on meditation and its underlying physiology and associated cognitive benefits. It’s also been used extensively in preclinical and clinical conditions.

Simultaneously, we know that breathing is intertwined with meditative practices, often serving as an anchor for our focus. Moreover, ancient contemplative practices understood that the breath and mind are closely linked. For example, the Hatha Yoga Pradipika states, "**When the Breath wanders, the mind is unsteady, but when the Breath is still, so is the mind still.**"

Modern science also recognizes many health benefits of breath control (specifically slow breathing), such as reduced stress and anxiety. However, the breath-mind connection was largely overlooked when this paper was published. Most studies had focused on how meditation affects the mind rather than pure breathing practices.

Thus, unraveling the effects of breathing versus other meditative practices remained unanswered.



This review aimed to fill that gap by synthesizing the scientific literature on slow breathing and psychological/behavioral outcomes. They were trying to answer the following question: **What physiological changes are common among pure slow breathing studies showing improvements in psychological results (like less stress & anxiety)?**

2. What Did this Research Do?

They found studies that used slow breathing (<10 breaths/min) without other mental components (such as breath counting or breath awareness). They only included studies with healthy individuals to not muddle the results with condition-specific results.

Their focus was on studies that had physiological measurements accompanying behavioral outputs. For example, if someone felt calmer, there needed to be physiological measurements that might explain that improvement in mood. The physiological measurements they focused on were brainwave activity, heart rate variability (HRV), respiratory sinus arrhythmia (RSA), and cardio-respiratory synchronization.

After using some rigorous criteria for their literature search (see the paper if you're interested), **they ended up with 15 studies that met the inclusion criteria.**

3. What Were the Major Findings?

As it is with science, there was nuance and many contradictory results. However, a few common themes did emerge.

First, slow breathing was associated with increases in HRV, particularly in the low frequency (LF) band. Second, it was also associated with increases in RSA. **These two results indicate increased activity in the parasympathetic (rest-and-digest) branch of the nervous system.** Moreover, slow breathing was also associated with increases in alpha brainwave activity (brainwaves thought to be associated with “flow”) and decreases in theta brainwave activity.

Critically, these common *physiological changes* observed during/after slow breathing were associated with improved *psychological and behavioral outcomes*. For example, slow breathing generally increased relaxation while reducing anxiety, depression, and anger.



Lastly, they examined the importance of the nose. They reviewed studies showing that nasal breathing directly correlates with brain activity. For instance, certain areas of the brain show oscillations synchronized with nasal (but not mouth) breathing.

Based on the evidence, the authors hypothesize that the nose is the link between slow breathing, brain, autonomic functioning, and positive emotional outcomes.

4. Why Do These Results Matter?

This review provides the physiological mechanisms of slow breathing that improve behavioral outcomes. Thus, it shows that slow breathing (independent of meditation, yoga, etc.) can be a powerful way of changing our physiology to promote positive psychological outcomes.

Moreover, these results are significant because they give us breathing nerds a grounded basis for explaining **how** slow breathing helps: it increases parasympathetic action (which helps with relaxation, anxiety, etc.) and enhances alpha brainwave activity (which facilitates focus, attention, etc.).

1 BIG TAKEAWAY

Altogether, slow breathing consistently increases HRV, RSA, and alpha brainwave activity. These *physiological changes* are associated with better *psychological and behavioral outcomes*.

1 PRACTICAL APPLICATION

From a practical perspective, most studies showing improved HRV and RSA used breathing rates of about 5-6 breaths/min (although some were as high as 9 breaths/min).

The studies showing increased alpha brainwave activity used slower rates of 3-4 breaths/min (although one used a 10 breath/min rate).

Lastly, the studies used different inhale-to-exhale ratios: either an equal or slightly longer exhale.



So, to apply these findings in your life, grab your favorite breathing app and set it to a comfortable pace for you, between 3-10 breaths/min. Use an equal ratio or a slightly longer exhale. Practice regularly to increase HRV, RSA, and alpha brainwave activity, and ultimately improve your overall mood.



Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Neural Correlates of Non-ordinary States of Consciousness in Pranayama Practitioners: The Role of Slow Nasal Breathing

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ORIGINAL RESEARCH article

Front. Syst. Neurosci., 21 March 2022

Volume 16 - 2022 | <https://doi.org/10.3389/fnsys.2022.803904>

Neural Correlates of Non-ordinary States of Consciousness in Pranayama Practitioners: The Role of Slow Nasal Breathing

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Article Citation

Zaccaro A, Piarulli A, Melosini L, Menicucci D, Gemignani A. Neural Correlates of Non-ordinary States of Consciousness in Pranayama Practitioners: The Role of Slow Nasal Breathing. Front Syst Neurosci. 2022 Mar 21;16:803904. doi: 10.3389/fnsys.2022.803904. PMID: 35387390; PMCID: PMC8977447.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2022 study found that slow nasal breathing has significantly different effects on the brain than slow mouth breathing, resulting in less physical and psychological tension, less anxiety, more joy, and a relaxed yet fully aware altered state of consciousness. Nasal stimulation should be considered alongside vagal stimulation as a primary mechanism behind the benefits of slow breathing.



4 Fundamentals

1. Essential Background Material

- Slow breathing is the cornerstone of many contemplative and ancient breathing practices, such as meditation and pranayama.
- Modern research repeatedly shows that **slow breathing improves psychological outcomes** (like lowering stress and increasing relaxation) **and physiological outcomes** (like better blood pressure and heart rate variability).
- Most research assumes that these benefits result from **increased vagal activity** (increased activity of the parasympathetic nervous system).
- Although this mechanism is essential, the authors hypothesized that the nose plays a critical role in slow breathing's benefits because it organizes oscillations across broad brain regions, especially in emotional areas.
- Moreover, they had recently discovered that simply **stimulating the nose with odorless puffs of air** at a slow pace can induce non-ordinary states of consciousness comparable to those reached by meditators.
- Thus, it appears that two primary factors are at play during slow breathing: **vagal activity (bottom-up) and nasal stimulation (top-down)**. This study aimed to disentangle these two pathways by comparing slow nasal breathing (which provides both vagal and nasal activation) to slow mouth breathing (which only provides vagal activation).

2. What Did this Research Do?

The researchers recruited twelve participants with extensive meditation experience (the average lifetime hours of formal practice was 1,688).

The participants performed 15 minutes of box breathing using a 6:6:6:6 count (6 seconds for each breathing phase), coming out to 2.5 breaths a minute. The practice was performed while lying down comfortably.



For one session, they performed slow nasal breathing (SNB); for the second session (occurring one week later), they performed slow mouth breathing (SMB). The SNB and SMB sessions were randomized among participants.

In an exceptional display of insight, the researchers recognized that mouth breathing is unnatural for experienced meditators. So, they had them train in SMB for 30 minutes a day for one week before the experiment to get them comfortable with the practice 🧘

They measured breathing and heart rate variability (HRV) throughout the experiment. They measured brain activity via an EEG for 5 minutes just before and for 5 minutes immediately after the breathing practice.

Lastly, they gave the participants two questionnaires for assessing non-ordinary states of consciousness and stress and anxiety before and after the breathing session:

1. Phenomenology of consciousness inventory (PCI)
2. State-trait anxiety inventory Y short form (STAI-6)

3. What Were the Major Findings?

- They found that **SNB and SMB activated the parasympathetic nervous system to the same extent** (as measured by HRV). HRV was slightly higher during nasal breathing, but it did not reach statistical significance.
 - Both SNB and SMB **increased power at slower brain wave frequencies**, but the enhancement was greater in SNB, and it occurred in different brain regions.
 - There was heightened connectivity at slow and fast brain wave frequencies during SNB versus SMB.
 - There was an increased coupling between slower brain waves (theta) and faster brain waves (high-beta) after SNB compared to SMB.
 - The post-SNB state **was accompanied by higher positive emotions, especially joy**, compared to baseline and post-SMB.
 - After SNB, participants felt they were in an altered state of consciousness more than with SMB: they reported higher scores of “altered awareness” and “altered experience.”
-



- They experienced **lower physical and psychological tension after SNB than SMB**, but the differences did not reach statistical significance.
- They also showed **significantly lower anxiety** levels after SNB compared to SMB.

4. Why Do These Results Matter?

This is the first study (to my knowledge) to **separate the effects of vagal vs. nasal stimulation** during slow breathing by comparing slow nasal breathing to slow mouth breathing.

Critically, they found that both practices activated the parasympathetic nervous system to the same extent. Thus, if vagal nerve stimulation was the primary mechanism, these practices should have shown similar outcomes. But they didn't...

They showed that nasal breathing had significantly different effects on brain waves and brain wave connectivity throughout the brain. Most importantly, **participants felt less physical and psychological tension, less anxiety, more joy, and a greater perception of being in an altered state of consciousness during SNB than SMB.**

Notably, questionnaires revealed that the altered states of consciousness did not reduce their rational thinking or will. This indicated that SNB put them into (as the researchers put it) **“a deeply relaxed but fully aware non-ordinary state of consciousness comparable to meditation.”** This was not achieved with slow mouth breathing.

1 Big Takeaway

Slow nasal breathing has significantly different effects on the brain than slow mouth breathing, which results in less physical and psychological tension, less anxiety, more joy, and a relaxed yet fully aware altered state of consciousness. Nasal stimulation should be considered alongside vagal stimulation as a primary mechanism behind the benefits of slow breathing.



1 Practical Application

You probably don't need to hear this by now, but make sure you're using your nose while performing slow breathing to get the maximum benefits for your body, brain, and mental and emotional health.

And if you're up for it, try out the super slow pace used in this study (2.5 breaths a minute) and see if it leaves you in a relaxed yet alert altered state of consciousness.

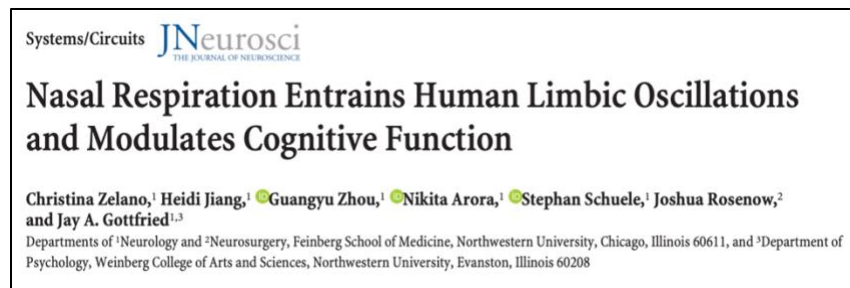


Science 411

4 fundamentals, 1 big takeaway, & 1 practical application

Nasal Respiration Entrain Human Limbic Oscillations and Modulates Cognitive Function

411 Released: July 7, 2023



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Article Citation

Zelano C, Jiang H, Zhou G, Arora N, Schuele S, Rosenow J, Gottfried JA. Nasal Respiration Entrain Human Limbic Oscillations and Modulates Cognitive Function. J Neurosci. 2016 Dec 7;36(49):12448-12467. doi: 10.1523/JNEUROSCI.2586-16.2016. PMID: 27927961; PMCID: PMC5148230.

Sharable: *You can share anything you want from these 411s—the more sharing the better. But this is my attempt to condense the findings into a bite-sized wisdom nugget:*

A 2016 study found that nasal breathing synchronizes brain oscillations in the piriform cortex, amygdala, and hippocampus, the latter two regions being critical to emotions, memory, and behavior. This coherence leads to improved cognitive function compared to mouth breathing.



4 Fundamentals

1. Essential Background Material

- Nasal breathing affects brain oscillations in the piriform cortex, a brain region associated with the olfactory system and sense of smell (thus, it makes intuitive sense that nasal breathing would impact this area).
- Interestingly, though, the piriform cortex is also linked with “limbic” brain areas, those associated with memory, emotion, and behavior.
- We know that memories, emotions, and behaviors can alter our breathing rate and depth. However, this close link between the olfactory and limbic brain regions might also provide a pathway for breathing (specifically nasal breathing) to influence memory, emotion, and behavior.
- To assess this possibility, this study used intracranial EEG and several cognitive tests to (1) examine the influence of breathing on brain oscillations in the piriform cortex and limbic areas and (2) to see if nasal breathing impacted cognitive function.

2-4. What this Research Did, the Results, and Why They Matter

This study had three separate experiments (it was also one of the longest “Materials and Methods” sections I’ve ever read 😊).

To make it easier to follow, I’m going to abandon the normal 411 format and just go through each experiment, its results, and its implications individually.

Experiment 1: Intracranial EEG

To get the time and spatial resolution needed to examine how breath oscillations impact brain oscillations, they recorded intracranial EEG in patients with temporal lobe epilepsy. Getting



intracranial EEG is very invasive, so they piggybacked on operations these patients were already undergoing to get this data. This is the most unique aspect of the study.

The participants sat down, kept their eyes open, and breathed normally for 15 minutes of measurements (there were no breathing exercises used in this study—just normal breathing).

Experiment 1 Results and Why They Matter

Oscillations in the piriform cortex are in harmony with nasal breathing, especially nasal inhalations.

- They found that oscillations in the piriform cortex were significantly correlated with natural nasal breathing, especially during inhalations. Although these results were somewhat expected, this was the first time iEEG had been used to definitely show that nasal breathing does influence oscillations in this brain region.

Nasal breathing influenced brain oscillations in emotional regions of the brain too.

- They also found that natural nasal breathing entrained brain oscillations in the amygdala and hippocampus, with the most pronounced impacts during inhalation. These results are meaningful because these are emotional regions of the brain. Thus, these results show that nasal breathing synchronizes oscillations over widespread brain areas.

These effects did not occur during mouth breathing.

- Three of the subjects were measured during both nasal and mouth breathing. They found that brain oscillations did not synchronize with respiration during mouth breathing like during nasal breathing. Thus, nasal breathing is critical to synchronizing brain oscillations.

Experiment 2: Emotional Recognition Task

Given the above results, the researchers wanted to see if nasal breathing, especially nasal inhales, affected emotional recognition.

Sixty-two participants took part in this phase of the study. They were shown faces with fearful or surprised expressions and asked to identify them as quickly as possible.

Here were the breathing conditions for this task:

- **Nasal Breathing (N = 21)**
 - **Mouth Breathing (N = 17)**
-



- **Nasal Breathing with Mouth Open (N = 24):** In the mouth breathing group, the researchers were concerned that keeping the mouth open (which is generally unnatural) may have been contaminating the results. So, they used this condition (nasal breathing with an open mouth) as an attentional control.

Experiment 2 Results and Why They Matter

Participants identified surprised/fearful faster during nasal breathing vs. mouth breathing.

- However, there were no statistically significant differences in overall accuracy comparing nasal vs. mouth breathing. It was simply that they were slightly faster during nasal breathing, which is still meaningful.

Participants identified fearful faces faster during nasal inhalation than exhalation.

- This inhale/exhale difference was not seen during mouth breathing. This is noteworthy because it shows that the iEEG results, which displayed more brain synchronization during nasal inhales, translate into real-world changes in emotional processing.

Participants who breathed nasally but with open mouths still identified fearful faces faster during inhalation than exhalation.

- This result confirms that nasal stimulation is critical to these inhale/exhale differences.

Experiment 3: Visual Object Memory Task

The limbic brain areas that nasal breathing impacted in Experiment 1 are also associated with memory. Therefore, the researchers used a memory task to analyze if nasal breathing also affected this aspect of cognition.

In this experiment, 33 participants viewed 180 images over a 15-min period. Then, they had a 20-min break. Next, they were presented with 180 old images (the ones they saw in the original session) and 180 new images (ones they hadn't seen yet) in random order. They had to identify if they had seen the images or not.

The same breathing conditions were used:

- Nasal Breathing (N=11)
 - Mouth Breathing (N=11)
 - Nasal Breathing with Mouth Open (N=11)
-



Experiment 3 Results and Why They Matter

In the nasal breathing group, memory accuracy was better during inhalation than exhalation.

- The participants had better memory if they inhaled through their nose when they encountered the picture the first time or saw it the second time. This was not true for mouth breathing.

However, there were no overall differences in accuracy between nasal and mouth breathing.

- Thus, although the previous result is neat, it may not be meaningful from a practical perspective since overall accuracy was not changed.

1 Big Takeaway

Nasal breathing—especially nasal inhalation—synchronizes brain oscillations in the piriform cortex, amygdala, and hippocampus, the latter two regions being critical to emotions, memory, and behavior. This coherence leads to improved cognitive function compared to mouth breathing.

1 Practical Application

This one is straightforward: Breathe through your nose as much as possible, especially when learning or in emotional situations, to improve brain coherence and cognitive functioning.



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Citation

Zwaag J, Naaktgeboren R, van Herwaarden AE, Pickkers P, Kox M. The Effects of Cold Exposure Training and a Breathing Exercise on the Inflammatory Response in Humans: A Pilot Study. Psychosom Med. 2022 May 1;84(4):457-467. doi: 10.1097/PSY.0000000000001065. Epub 2022 Feb 23. PMID: 35213875; PMCID: PMC9071023.

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Additional Note:

This was one paper with two studies. Although the studies complement each other, I'm dividing it into two 411s to make it easier to digest and because the results of both studies are extremely significant to WHM practitioners.

Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

The Effects of Cold Exposure Training and a Breathing Exercise on the Inflammatory Response in Humans: A Pilot Study **PART ONE: Breathing Exercises Study**

4 FUNDAMENTALS

1. Essential Background Material

[The PNAS Wim Hof study](#) showed for the first time that participants were able to willfully influence their innate immune response and reduce inflammation associated with endotoxin injection.

However, during the injection, two different WHM breathing exercises were used (see original [Science 411](#) for more). Moreover, the participants were trained by Wim Hof himself, which could introduce a placebo effect.

From practical and clinical perspectives, it's vital to know which breathing exercise was responsible for the anti-inflammatory effects and if the same results can be achieved by an independent trainer (not Wim Hof). Moreover, it's necessary to know how much training is needed to get the results.

That's what this study aimed to discover. The three questions it answers are:

1. Which WHM breathing exercise is most effective for reducing inflammation?
2. How long do people need to train to achieve the anti-inflammatory effects?
3. Can the same results be obtained by an independent trainer (not Wim Hof)?

2. What Did this Research Do?



Forty male participants were randomized into four different groups:

- **Intensive Training by Wim Hof (n=10):** Four days of training with Wim for 2 hours every morning.
- **Intensive Training by Independent Instructor (n=10):** Four days of training with an independent instructor who was part of the research team.
- **Short Training by Wim Hof (n=10):** Two hours of training with Wim.
- **Short Training by Independent Instructor (n=10):** Two hours of training with the independent instructor.

The 2-hour short training occurred on the day of the experiment. Note that this was only breath training, not cold and meditation.

On Experiment Day

In the morning, half of the participants were randomized to perform the standard Wim Hof breathing (30 deep breaths, hold after exhale, inhale and hold for 10 seconds, repeat) for 1.5 hours.

The other half performed the modified Wim Hof breathing (30 deep breaths, inhale and hold for 10 seconds while squeezing all muscles) for 1.5 hours.

Both groups rested for 1.5 hours. Then, in the afternoon, they performed the other breathing exercise for 1.5 hours.

Primary Outcome

The researchers were most interested in adrenaline release during the breathing, as this is what they found to be linked with the anti-inflammatory effects of Wim's method in the original PNAS study. It was measured every 30 min during the breathing sessions.

3. What Were the Major Findings?

- In the morning, both breathing exercises initially spiked adrenaline to similar levels.
- However, the standard WHM breathing achieved more sustained elevated adrenaline levels at the 1.5 h mark than the modified breathing.
- **In the afternoon, neither breathing exercise significantly increased adrenaline.**
- Adrenaline levels were not significantly different between the participants trained by Wim or the independent instructor.



- Adrenaline levels were not significantly different between the 4-day and 2-hour training groups.

4. Why Do These Results Matter?

The original PNAS study found that the initial spike in adrenaline associated with WHM breathing was the source of the anti-inflammatory effects.

Here, we learn that to get that adrenaline spike (and thus the anti-inflammatory effects):

- We can use the standard or modified WHM breathing (this is critical for people who cannot tolerate hypoxia).
- We can train for just two hours instead of four days.
- We can use an independent trainer (not Wim Hof himself).
- We should practice in the morning only.

1 BIG TAKEAWAY

If performed in the morning, the anti-inflammatory effects of Wim Hof breathing can be achieved with or without breath holds in just 2 hours of instruction from an independent trainer.

1 PRACTICAL APPLICATION

NOTE: To use the method, it's critical that you work with a certified Wim Hof instructor or obtain a physician's approval to ensure you do not have any contraindications.

These results suggest that we should practice the WHM in the morning to maximize the anti-inflammatory effects.

Even more importantly, they also show us that we can use WHM breathing without hypoxia if we have any condition preventing us from doing long breath holds.

Most practically, [the first page of this PDF](#) gives the precise instructions used in this study.



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Citation

Zwaag J, Naaktgeboren R, van Herwaarden AE, Pickkers P, Kox M. The Effects of Cold Exposure Training and a Breathing Exercise on the Inflammatory Response in Humans: A Pilot Study. Psychosom Med. 2022 May 1;84(4):457-467. doi: 10.1097/PSY.0000000000001065. Epub 2022 Feb 23. PMID: 35213875; PMCID: PMC9071023.

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Science 411

4 fundamentals, 1 big takeaway, and 1 practical application

The Effects of Cold Exposure Training and a Breathing Exercise on the Inflammatory Response in Humans: A Pilot Study **PART TWO: Human Endotoxemia Study**

4 FUNDAMENTALS

1. Essential Background Material

In [Part One of this study](#), we learned four critical pieces of information:

1. The anti-inflammatory effects of Wim Hof breathing can be achieved with or without breath holds.
2. They can be achieved using an independent instructor.
3. They can be achieved with only 2 hours of training.
4. They can only be achieved if practiced in the morning, not afternoon.

Those results are essential from practical and clinical perspectives. However, two critical questions remain.

First, will these results hold up when participants are injected with endotoxin? If so, this would be yet another monumental finding because it would mean people only need 2 hours of training from an independent instructor in one breathing method to reduce inflammation.

Second, cold therapy is a central component of Wim's method, and it was used in the [original PNAS endotoxin study](#). But, it's unclear how much it contributed to the anti-inflammatory response. So, is cold therapy needed, or is breathing enough?

That's what Part Two of this study aimed to find.



2. What Did this Research Do?

Forty-eight participants enrolled in the study. They were divided into four groups with 12 people each:

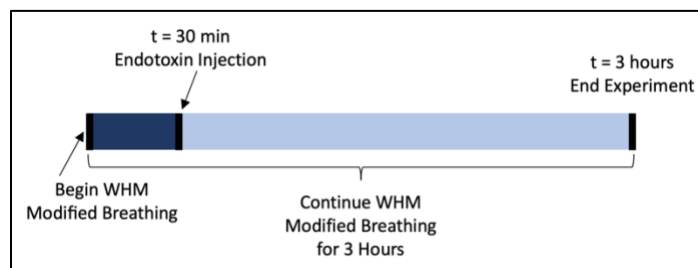
1. **Cold Only:** Intensive 4-day cold exposure training led by an independent trainer. No deliberate breath training.
2. **Breathing Only:** One 2-hour instruction of the WHM modified breathing technique. They were told not to practice this at home. They chose this one based on Part One, where both WHM breathing exercises produced similar initial spikes in adrenaline. The logic was that using the method without hypoxia would have broader applicability.
3. **Cold + Breathing:** Both above training programs: 4 days of cold training and one 2-hour breath training session.
4. **Control:** No training.

Endotoxin Administration Day

The *Breathing Only* and *Cold + Breathing* groups started the modified WHM breathing 30 minutes before injection. They continued it for 2.5 hours, for a total of 3 hours of hyperventilation. This is the same protocol used in the original PNAS study, except that only one breathing exercise is used here.

The *Cold Only* and *Control* groups did not perform any breathing exercises.

Like the original PNAS study, adrenaline, blood gases, inflammatory cytokines, and flu-like symptom scores were all recorded throughout the experiment.



3. What Were the Major Findings?



The table below summarizes the most important findings. The changes are comparisons with controls. So, if it says that inflammation improved, that means it improved compared to the control group.

Summary

	Inflammation	Flu-like Symptoms
Cold Only	No change*	Improved the most
Breathing Only	Improved	Worsened
Breathing + Cold	Improved the most	Improved

* One inflammatory cytokine, MCP-1, was lower than the control group. This one was not measured in the original PNAS study.

4. Why Do These Results Matter?

These results are meaningful because they show that as little as 2 hours of Wim Hof breathing without breath holds can reduce acute inflammation due to endotoxin injection.

Moreover, they show that cold exposure alone does not reduce inflammation. However, the combination of cold and breathing has a synergistic effect, reducing inflammation more than breathing alone while also reducing flu-like symptoms.

We should note that the anti-inflammatory response was not as profound as in the original PNAS endotoxin experiment. There's no way of knowing why. But, it could be that the WHM breathing with hypoxia, which has a more sustained release of adrenaline, works better when participants are exposed to endotoxin.

Lastly, note that this study did not measure baseline inflammation. So, cold therapy could potentially help there. These results just show that it didn't have a significant effect when participants were exposed to intense short-term inflammation associated with the endotoxin.

1 BIG TAKEAWAY

Breathing is the primary pillar of Wim's method that reduces acute inflammation during endotoxin exposure. However, cold training boosts these anti-inflammatory effects while also reducing flu-like symptoms.



1 PRACTICAL APPLICATION

NOTE: To use the method, it's critical that you work with a certified Wim Hof instructor or obtain a physician's approval to ensure you do not have any contraindications.

These results show that Wim Hof breathing by itself can reduce inflammation. So, if you're completely against the cold, this suggests you can still use Wim's breathing to get anti-inflammatory effects.

However, the results also show that using both cold and breathing has a synergistic effect of reducing inflammation more than breathing or cold alone.

What's more, the results suggest that regular cold exposure may help with your ability to reduce flu-like symptoms if you become sick.

So overall, incorporating cold might be beneficial if you have an autoimmune disease or any condition that leads to chronic inflammation and increased susceptibility to illness. See the [Book 411 on The Wim Hof Method](#) for Wim's basic cold exposure guidelines.